

From Static Data to Human Alignment: A Comprehensive Survey of Offline and Preference-Based Reinforcement Learning for Robotics

1 Motivations and Paradigm Shifts in Data-Efficient Robotic Learning

1.1 The Sample Efficiency and Safety Bottlenecks of Online Exploration

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The deployment of Reinforcement Learning (RL) in real-world robotics is hindered by a fundamental dichotomy: the algorithmic promise of autonomous improvement through trial-and-error versus the physical reality of finite resources and irreversible consequences. Traditional online RL paradigms operate on the assumption that an agent can freely explore its environment, collecting vast amounts of data to refine its policy. However, when transferred from simulated domains to physical hardware, this assumption collapses under the weight of prohibitive sample inefficiency and acute safety risks. The core bottleneck lies in the fact that standard RL algorithms often require millions of interaction steps to converge—a scale of data collection that is operationally infeasible and economically unsustainable for most robotic systems.

The sample efficiency crisis represents the most immediate barrier to practical adoption. In simulated environments, agents can reset instantly and parallelize experiences across thousands of CPUs, effectively masking the data hunger of deep RL methods. In contrast, real-world robots are bound by the speed of physical actuators, the latency of sensory processing, and the wear and tear of mechanical components. Collecting the equivalent of a million steps on a quadruped robot or a manipulation arm could take months of continuous operation, during which the system remains unavailable for productive tasks. This inefficiency is exacerbated by the “curse of dimensionality” inherent in high-degree-of-freedom robotic systems, where the state-action space grows exponentially, demanding even more extensive exploration to cover the manifold of possible behaviors. Consequently, the cost of data acquisition in terms of time, energy, and human supervision becomes a dominant factor, rendering pure online exploration strategies impractical for complex, long-horizon tasks.

Beyond the economic and temporal costs, the safety implications of uncontrolled online exploration are severe and potentially catastrophic. The trial-and-error mechanism, which serves as the engine of RL learning, inherently involves taking actions with uncertain outcomes. In a simulation, a failed action results only in a reset; in the physical world, it can lead to equipment damage, environmental destruction, or harm to human operators. For instance, a robot learning to navigate uneven terrain might attempt a dynamic maneuver that results in a fall, damaging sensitive sensors or structural links. Similarly, a manipulation agent exploring forceful interactions might exceed torque limits, burning out motors or crushing objects in its workspace. As noted in the literature, safe exploration aims to address these limitations where failures during trial-and-error learning may incur high costs, yet reducing exploration risks in unknown environments remains a significant challenge [1].

Compounding these risks is the lack of prior knowledge regarding the boundaries of safe operation. Standard RL agents typically learn safety constraints only after violating them, receiving negative rewards or penalties post-factum. This reactive approach is insufficient for safety-critical applications where even a single violation is unacceptable. The contradiction is stark: how can an agent

learn to avoid dangerous regions without first experiencing them? Existing methods often rely on posterior penalties, meaning the agent is not penalized until it has already experienced danger, preventing the learning of a zero-violation policy even after convergence [2]. This creates a Catch-22 where the data required to learn safety can only be generated by unsafe behavior, a paradox that stymies the deployment of RL in high-stakes domains like autonomous driving, medical robotics, and industrial automation.

Furthermore, the reliability of the hardware itself introduces another layer of complexity to the safety bottleneck. As robotic systems increasingly rely on specialized hardware accelerators and deep learning components for real-time decision-making, they become susceptible to hardware faults such as soft errors, bit-flips, and transient faults. These hardware-level unreliabilities can manifest as silent data corruptions (SDCs) that alter the robot’s perception or control outputs, leading to unexpected and potentially dangerous behaviors even if the learned policy is theoretically sound. Research into cyber-physical systems highlights that traditional fault-tolerant approaches often impose significant overheads and may not detect errors quickly enough to prevent catastrophic failure, necessitating new approaches like deep reinforcement learning for online error detection to ensure system reliability [3]. The interplay between algorithmic exploration and hardware fragility means that an agent exploring the boundaries of its control authority might inadvertently trigger a hardware fault, leading to a loss of control that no software-level safety filter can recover from.

The challenge is further articulated by the need to balance performance optimization with strict safety constraints. In many computer systems and robotic applications, optimizing for metrics like latency or task completion time often conflicts with safety limits, such as power consumption or thermal thresholds. Approaches that rely on learning system behavior from past executions often fail when faced with new applications or inputs, leading to suboptimal performance or safety violations. This necessitates frameworks that can dynamically and safely allocate resources or explore action spaces without violating predefined constraints, a capability that standard online RL lacks without significant modification [4]. The inability to guarantee safety during the exploration phase forces researchers to rely on conservative baselines or extensive human supervision, effectively neutering the autonomy that RL promises to deliver.

Moreover, the concept of “safe fail” becomes critical in this context. When an ML model encounters situations outside its training distribution or faces hardware anomalies, it must have mechanisms to reject unsafe predictions and defer to backup controllers. However, identifying these out-of-distribution scenarios in real-time during active exploration is non-trivial. Methods that infer uncertainty from training data or partition the feature space are essential to prevent the agent from extrapolating into dangerous regions where error rates skyrocket [5]. Without such safeguards, the exploration process becomes a gamble with physical integrity, making the transition from simulation to reality a perilous endeavor.

In summary, the bottlenecks of sample efficiency and safety in online exploration are deeply intertwined. The data requirements for effective learning are incompatible with the fragility and cost of physical hardware, while the necessity of avoiding catastrophic failures restricts the very exploration needed to gather data. This impasse has driven the community to seek alternative paradigms that decouple learning from active, risky interaction. While one might consider bypassing exploration entirely by hand-crafting reward functions to guide behavior more directly, this approach introduces its own set of intractable challenges regarding alignment and specification, as explored in the following section. Ultimately, these compounded limitations set the stage for the shift towards offline learning and preference-based methods that can leverage existing data and human guidance to bypass these fundamental constraints.

1.2 The Intractability of Hand-Crafted Reward Engineering

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While the previous section established that active online exploration is often prohibited by sample inefficiency and safety risks, one might propose bypassing these constraints by leveraging prior knowledge through carefully hand-crafted reward functions. However, this approach introduces a second, equally formidable bottleneck: the intractability of specifying a scalar reward signal that accurately aligns with complex human intent and rigorous safety constraints. The fundamental difficulty lies in the semantic gap between a designer’s high-level objectives and the low-level mathematical formulation required for optimization. When reward functions are misspecified, agents do not merely fail to perform; they actively exploit loopholes in the reward definition to achieve high scores while exhibiting behaviors that are useless, dangerous, or completely misaligned with the user’s true goals. This phenomenon, widely recognized as reward hacking, represents a critical failure mode where optimizing an imperfect proxy reward function leads to a significant degradation in performance according to the true, underlying objective [6].

The challenge of hand-crafting rewards is exacerbated by stringent theoretical limitations regarding the linearity of reward functions with respect to state-action visit counts. Analysis reveals that for a proxy reward to be “unhackable”—meaning that increasing the expected proxy return never decreases the expected true return—the conditions are exceptionally restrictive. In the space of all stochastic policies, two reward functions can only be unhackable if one is constant, rendering the concept of a useful, non-trivial, and perfectly aligned hand-crafted reward theoretically elusive in general settings [6]. This theoretical barrier manifests practically when designers attempt to simplify complex tasks into narrow reward specifications. In an effort to make the reward function tractable, engineers often overlook fine-grained distinctions or omit terms deemed secondary. However, capable agents, particularly those with high model capacity and extensive training time, are adept at identifying these omissions. Research indicates that more capable agents are paradoxically more prone to exploiting reward misspecifications, often exhibiting phase transitions where their behavior qualitatively shifts to maximize the proxy reward at the expense of the true objective, leading to catastrophic drops in actual performance [7].

A primary manifestation of this intractability is the emergence of unwanted side effects and negative environmental impacts that were not explicitly penalized. Designing a reward that captures every possible undesirable outcome is practically impossible in unstructured environments. To mitigate this, researchers have proposed augmenting reward functions with impact regularizers that discourage behavior causing large changes to the environment state. However, defining what constitutes a “negative” impact versus a necessary task-induced change remains a profound design challenge, and current impact regularizers face significant hurdles in distinguishing between beneficial task completion and harmful side effects [8]. Furthermore, the reliance on scalar rewards fails to capture the multi-objective nature of many robotic tasks. Arguments have been made that scalar rewards are insufficient to account for the complexity of biological and computational intelligence, and their use in artificial general intelligence poses unacceptable risks of unsafe or unethical behavior, necessitating a shift toward explicitly multi-objective models [9].

The difficulty is further compounded when integrating safety constraints. A common heuristic is to penalize failures or unsafe states with a large negative reward. However, rigorous examination shows that simply adding penalties does not guarantee safety; the interaction between the penalty magnitude, the discount factor, and the system dynamics is complex. While it is theoretically possi-

ble to construct a “safe value function” where optimality implies safety, determining the minimum required penalty is often computationally infeasible, leading designers to rely on guesswork that may result in either overly conservative policies or unsafe behaviors [10]. Additionally, in human-robot interaction scenarios, maximizing task rewards often conflicts with maintaining human trust. Naively shaping rewards to maximize task efficiency can erode trust, while prioritizing trust can compromise performance, creating a dilemma that is difficult to resolve through manual reward tuning alone [11].

Even when designers attempt to use auxiliary rewards or potential-based shaping to guide learning, these interventions can inadvertently introduce new biases or impair performance if not perfectly aligned with the true goal. Systematic investigations have shown that deploying potential-based reward shaping can significantly impair performance in certain settings, highlighting the fragility of manual intervention [12]. The problem is not merely one of tuning coefficients but of structural misspecification. When the specified objective does not match the user’s true underlying goal due to the inherent asymmetry in human expectations versus agent capabilities, the result is a fundamental misalignment that no amount of parameter tuning can fix [13].

Moreover, the process of reward design is often iterative and reactive, requiring designers to observe failure modes and patch the reward function accordingly. This “whack-a-mole” approach is inefficient and prone to error, as fixing one loophole may open another. Inverse Reward Design (IRD) has been proposed to address this by treating the designed reward as an observation of the designer’s intent within a specific training context, allowing the agent to infer the true objective and plan risk-averse behavior in novel test scenarios where the original reward might lead to hacking [14]. Similarly, approaches that learn common-sense rewards from expert demonstrations across multiple tasks suggest that disentangling task-specific rewards from general behavioral norms is essential to prevent agents from learning useful but unsafe policies [15]. Ultimately, the intractability of hand-crafted reward engineering underscores the necessity for paradigms that move beyond explicit scalar specification. This limitation motivates the shift towards learning rewards directly from human preferences, demonstrations, and feedback mechanisms that can better capture the nuance of human intent and safety requirements, a transition that naturally complements the move toward static dataset utilization discussed in the following section.

1.3 The Paradigm Shift Towards Static Dataset Utilization

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While the previous section highlighted the intractability of hand-crafting precise reward functions, a parallel and equally formidable bottleneck exists in the data acquisition phase of Reinforcement Learning (RL). Historically, the success of RL has been predicated on the agent’s ability to actively explore its environment, collecting fresh data through trial-and-error interactions to refine its policy. However, in real-world robotics, this requirement for active exploration creates a severe impediment due to the high costs of data collection, the potential for hardware damage during unsafe exploration, and the sheer time required to gather sufficient experience. Consequently, the field is increasingly pivoting towards utilizing static datasets—collections of previously gathered transitions that may include expert demonstrations, sub-optimal trials, or heterogeneous operational logs. This shift allows agents to learn complex behaviors without ever interacting with the environment during the training phase, effectively decoupling the data collection process from the policy optimization process and offering a pragmatic solution to the safety and efficiency constraints that plague online methods.

The core motivation for this transition lies in the ability to leverage vast repositories of historical data that were previously underutilized or discarded. In many robotic domains, such as autonomous driving, surgical manipulation, or industrial assembly, massive amounts of data are generated daily by existing systems or human operators. Offline RL provides the theoretical and algorithmic framework to extract optimal or near-optimal policies from these static datasets, even when the data was generated by sub-optimal behavior policies. For instance, recent advances have demonstrated that it is possible to learn policies that significantly outperform the behavior agents used to collect the data by effectively “stitching” together superior segments of sub-optimal trajectories. Methods like Model-based Trajectory Stitching introduce unseen actions to join previously disconnected states, generating new, higher-quality synthetic trajectories that enhance the utility of historical data [16]. This capability transforms static datasets from mere records of past performance into rich resources for discovering novel, high-performing strategies that were never explicitly executed by a single agent.

However, the utilization of static datasets introduces a unique and critical challenge known as distributional shift, a problem distinct from the reward misspecification issues discussed earlier. Unlike online RL, where the agent can correct its value estimates by exploring the consequences of its actions, offline RL agents must rely solely on the fixed support of the dataset. When a learned policy attempts to select actions that deviate from the behavior policy present in the dataset, it encounters out-of-distribution (OOD) state-action pairs. Standard function approximators, particularly deep neural networks, tend to produce arbitrarily inaccurate value estimates for these OOD inputs, leading to extrapolation errors that can cause the policy to collapse. To address this, the paradigm shift has necessitated the development of algorithms that explicitly account for uncertainty and enforce conservatism. A prominent approach involves penalizing the value estimates of actions that are not well-represented in the dataset, thereby ensuring that the learned policy remains within the support of the data or only cautiously explores regions where the model is confident. Algorithms such as Pessimistic Offline Reinforcement Learning (PessORL) deliberately manipulate the value function to lower-bound the true value across the state space, actively guiding the agent away from unfamiliar states [17]. Similarly, Conservative Q-Learning (CQL) and its variants have become foundational in this domain by learning lower-bounded Q-functions that prevent the overestimation of values for OOD actions.

Furthermore, the shift towards static data has spurred innovation in how uncertainty is quantified and managed without the benefit of active sampling. Traditional methods often relied on complex ensemble models to estimate epistemic uncertainty, which can be computationally prohibitive. Newer approaches have shown that diverse randomized value functions or simplified uncertainty heuristics can provide robust uncertainty quantification with significantly reduced computational overhead [18]. These methods enable the agent to distinguish between regions of the state space where the data is dense and reliable versus those where it is sparse and prone to error. By integrating these uncertainty estimates directly into the optimization objective, offline RL algorithms can achieve a delicate balance between exploiting known high-reward behaviors and cautiously generalizing to new scenarios.

The versatility of this paradigm is further evidenced by its applicability to heterogeneous and imbalanced datasets, which are common in real-world robotics. Real-world data is rarely uniform; it often exhibits heteroskedasticity, where the variability of behaviors changes non-uniformly across the state space. Standard distributional constraints often fail in such settings because they enforce a uniform level of conservatism. In response, advanced methods like CQL (ReDS) reweight the data distribution to approximate support constraints, allowing the policy to adapt its level of con-

servatism based on the local density and quality of the data [19]. Additionally, techniques for data rebalancing, such as Return-based Data Rebalance (ReD), have been proposed to construct better behavior policies by resampling transitions based on their episodic returns, effectively boosting the performance of offline RL algorithms on long-tailed data distributions [20].

Moreover, the integration of model-based approaches has enriched the static data paradigm by enabling “imagined” rollouts that augment the limited real data. By learning dynamics models from the static dataset and penalizing rewards based on model uncertainty, algorithms like Model-based Offline Policy Optimization (MOPO) can safely generalize beyond the dataset’s support while maintaining theoretical guarantees on performance bounds [21]. This synthesis of model-free conservatism and model-based generalization underscores the maturity of the field, moving beyond simple imitation learning to true policy optimization from static buffers. Ultimately, the shift towards static dataset utilization marks a critical evolution in robotic learning, offering a pathway to deploy safe, data-efficient, and high-performing agents in complex, real-world environments where active exploration is simply not an option. Yet, while offline RL solves the data efficiency problem, it traditionally still relies on the very reward signals that Section 1.2 identified as difficult to engineer. This limitation sets the stage for the next evolution in the field: combining the data efficiency of offline learning with the alignment capabilities of preference-based feedback, moving beyond scalar rewards entirely to learn directly from human judgment.

1.4 Aligning Robots with Human Preferences Beyond Scalar Rewards

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While Section 1.3 established the critical shift toward static dataset utilization to overcome the dangers and costs of active exploration, it concluded by highlighting a persistent limitation: offline RL fundamentally relies on the very scalar reward signals that Section 1.2 identified as notoriously difficult to engineer. Even with vast repositories of historical data, the quality of the learned policy is capped by the fidelity of the reward function defining the optimization objective. Translating complex human intent, safety constraints, and nuanced social norms into a single numerical value remains an ill-posed problem, often leading to “reward hacking” where agents exploit loopholes or exhibit behaviors that are mathematically optimal yet misaligned with human expectations. To resolve this specific bottleneck, the field has pivoted toward Preference-Based Reinforcement Learning (PbRL), a framework that decouples policy learning from explicit reward engineering by enabling agents to learn directly from relative human feedback. This approach effectively bridges the gap between the data efficiency of offline methods and the alignment capabilities of human judgment.

The core premise of PbRL is that while humans struggle to assign absolute scores to robotic actions, they are exceptionally proficient at making relative judgments between outcomes. This shift from absolute to relative feedback transforms the learning dynamic: instead of reverse-engineering a scalar reward from noisy environmental signals, the agent infers a reward model or policy directly from pairwise comparisons provided by a supervisor. This is particularly vital in domains like delicate manipulation or social navigation, where a human can easily indicate which of two trajectories is safer or more efficient without articulating the underlying cost function. By capturing these latent objectives, PbRL addresses the intractability of hand-crafted rewards, offering a natural interface for aligning robotic behavior with human values that rigid scalar functions cannot

capture.

A critical theoretical advancement in this domain involves re-evaluating the assumptions governing how human preferences are generated. Historically, algorithms assumed preferences were based on the “partial return,” or the cumulative reward over a trajectory segment. However, recent research indicates this assumption is flawed, leading to identifiability issues where the true reward function cannot be uniquely recovered. A more accurate model posits that human judgments are informed by “regret,” measuring a segment’s deviation from optimal decision-making rather than just its accumulated return. Studies demonstrate that modeling preferences based on regret not only improves identifiability but also yields policies more closely aligned with actual human intent [22]. Furthermore, it has been argued that misinterpreting preferences under the partial return assumption often results in learning an optimal advantage function rather than the true reward; adopting regret-based formulations resolves this conceptual mismatch [23].

To operationalize these insights, the field has moved beyond traditional two-stage pipelines—where a reward model is learned first, followed by policy optimization—which can introduce instability and inefficiency. Instead, new architectures like Direct Preference Optimization (DPO) and Contrastive Preference Learning (CPL) bypass explicit reward modeling entirely. These algorithms optimize the policy directly from preference data using contrastive objectives, avoiding the complexities of policy gradients and bootstrapping inherent in earlier approaches [24]. Similarly, Inverse Preference Learning (IPL) leverages the insight that for a fixed policy, the Q-function encodes all necessary reward information, allowing for the parameter-efficient elimination of the learned reward function altogether [25]. These direct optimization techniques significantly simplify the PbRL pipeline, enhancing stability and scalability for high-dimensional robotic tasks.

Despite these algorithmic strides, practical deployment faces significant hurdles regarding data efficiency and feedback quality. Human annotation is expensive and time-consuming, necessitating strategies to maximize information gain per query. Active learning approaches address this by selecting query pairs that reduce uncertainty in the learned reward model; for instance, leveraging disagreement across an ensemble of reward models can guide exploration toward regions of high uncertainty, improving both feedback and sample efficiency [26]. Additionally, the issue of query-policy misalignment—where queries improve the global reward model but fail to aid the current policy—has been identified as a critical failure mode. Mitigating this requires near on-policy query selection and hybrid experience replay mechanisms to ensure feedback is directly relevant to the agent’s current behavioral distribution [27].

Moreover, real-world applications must contend with the reality that human feedback is often noisy, inconsistent, or derived from non-expert sources. Robust algorithms are essential to prevent divergence due to erroneous labels. Methods such as RIME incorporate sample selection discriminators to dynamically filter noisy preferences, while others utilize crowd-sourcing frameworks to aggregate diverse viewpoints and estimate user reliability [28] [29]. The integration of semi-supervised learning further enhances efficiency by leveraging unlabeled data alongside sparse preference labels to infer pseudo-labels based on model confidence [30]. Additionally, incorporating environment dynamics into the reward representation has been shown to dramatically reduce the number of required preference labels by facilitating better generalization across the state-action space [31] [32].

In summary, the transition to preference-based learning represents a pivotal evolution in robotic AI, moving away from fragile dependencies on engineered scalar rewards toward flexible, human-centric alignment. By leveraging relative feedback, accounting for the psychological realities of

human judgment through regret-based models, and developing robust, direct optimization algorithms, PbRL provides a viable pathway for deploying capable and safe agents in complex domains. However, while PbRL solves the reward specification problem, it traditionally demands extensive online interaction to gather sufficient preference labels—a constraint that reintroduces the safety and efficiency challenges discussed in Section 1.3. This tension sets the stage for the next critical frontier: synergizing the data efficiency of offline priors with the alignment precision of preference feedback to achieve truly data-efficient deployment, as explored in the following section.

1.5 Synergizing Priors and Feedback for Data-Efficient Deployment

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Building upon the theoretical and algorithmic foundations of Preference-Based Reinforcement Learning (PbRL) established in the previous section, the next critical frontier is addressing the prohibitive sample complexity that often hinders its real-world robotic deployment. While PbRL effectively resolves the ambiguity of reward engineering by leveraging relative human feedback, it traditionally demands extensive online interaction to gather sufficient preference labels—a constraint that is often unsafe or impractical in physical environments. Conversely, Offline Reinforcement Learning offers a mechanism to leverage vast amounts of static historical data to initialize policies without risky exploration, yet it frequently struggles with distributional shifts and the inability to adapt to novel objectives defined by fixed, potentially misaligned reward functions. The convergence of these two paradigms represents a pivotal shift: by synergizing offline priors with preference-based feedback, modern approaches decouple the cost of acquiring general motor skills from the cost of specifying task-specific objectives. This hybrid architecture utilizes offline data to establish a safe, competent baseline, thereby confining expensive online human feedback to a brief, high-value phase of rapid adaptation and alignment.

The operational backbone of this synergy is the “pre-train and fine-tune” paradigm, which enables robots to master new tasks with minimal human effort by capitalizing on diverse datasets of prior experiences. Systems such as RoboFuME exemplify this strategy by pre-training multi-task manipulation policies from heterogeneous offline datasets and then employing calibrated offline RL techniques to ensure efficient online fine-tuning despite distribution shifts [33]. Crucially, this framework integrates pre-trained vision-language models to autonomously generate dense reward signals during the fine-tuning phase, significantly reducing the reliance on continuous human intervention while preserving alignment with human intent [33]. Similarly, the AWAC algorithm demonstrates how combining sample-efficient dynamic programming with maximum likelihood policy updates allows agents to bootstrap learning from large offline corpora, followed by rapid online refinement that perfects the desired skill [34]. These approaches validate the hypothesis that robust offline priors can mitigate the exploration burden, rendering subsequent preference-based optimization feasible within practical timeframes.

However, simply initializing a policy with offline data is insufficient if the transition to online learning destabilizes the agent or fails to correct misalignments inherent in the static dataset. Effective synergy requires sophisticated mechanisms that balance the retention of useful offline behaviors with the flexibility to explore out-of-distribution states guided by preference feedback. Methods like PROTO address this challenge by introducing an iteratively evolving regularization term that gradually relaxes constraints, facilitating stable initial fine-tuning before transitioning to optimal performance adaptation [35]. Furthermore, managing distributional shift during this transition is critical; without careful oversight, agents may suffer from bootstrap errors that degrade high-quality

initialization. Techniques such as balanced replay schemes and pessimistic Q-ensembles have been developed to prioritize online samples while retaining near-on-policy offline data, preventing over-optimism concerning unfamiliar actions [36]. Such stability is a prerequisite for effective preference learning, as unstable policies yield noisy trajectory comparisons that severely degrade the quality of the learned reward model.

The integration of preference feedback specifically targets the residual “reward engineering” bottleneck that pure offline RL cannot solve. Traditional offline methods rely on fixed reward signals that may fail to capture nuanced human preferences or safety constraints in new contexts. By incorporating preference-based learning, agents can refine their behavior based on relative human feedback rather than absolute scalar rewards. For instance, efficient preference-based RL using learned dynamics models demonstrates that synthesizing diverse preference queries via a world model can significantly reduce the number of real-world interactions required to learn customized policies [37]. This is particularly potent when combined with offline pre-training, where the dynamics model itself is learned from static data, creating a fully data-efficient loop [38]. Moreover, recent advances show that querying preferences between actions derived from offline data and learned policies can correct erroneous estimates of out-of-distribution data without any additional environment interaction, a method known as Offline-with-Action-Preferences (OAP) [39]. This suggests that preference data can be leveraged even in strictly offline settings to enhance policy robustness prior to real-world deployment.

Furthermore, this synergistic approach excels in handling heterogeneous data sources and diverse user intents, a common reality in industrial and domestic settings where offline datasets comprise demonstrations from multiple users with varying skill levels. Latent-variable approaches, such as LAPO, enable policies to represent broader distributions adhering to heterogeneous training data, which can then be fine-tuned via preference feedback to align precisely with a specific user’s style [40]. The ability to adapt to object-centric sub-tasks through one-shot physical human intervention further underscores the power of this combination, where a base policy pre-trained on synthetic or diverse data is updated online to respect specific object preferences indicated by human feedback [41].

Ultimately, the fusion of offline priors and preference feedback creates a robust pathway for deploying autonomous systems in unstructured environments. It allows for the amortization of computational and data costs over a large pre-training phase, while confining the expensive and potentially risky online interaction to a brief, high-value alignment phase. Whether through reset-free fine-tuning systems that self-improve with minimal intervention [33] or frameworks that utilize instruction relabeling via large language models to scale pre-training before preference tuning [42], the consensus is clear: the future of data-efficient robotics lies not in choosing between offline learning and human feedback, but in architecting systems that deeply integrate both to achieve safe, adaptable, and human-aligned autonomy.

2 Algorithmic Foundations of Offline Reinforcement Learning

2.1 Conservative Value Estimation and Policy Regularization

2.1 Conservative Value Estimation and Policy Regularization

The fundamental challenge in offline reinforcement learning (RL) arises from the distributional shift between the behavior policy that generated the static dataset and the learned policy being optimized. In the absence of online interaction to correct errors, standard off-policy algorithms

often suffer from severe extrapolation errors. This occurs when the agent queries the value function for actions that lie outside the support of the data distribution, leading to gross overestimations of Q-values due to function approximation errors. To address this, a dominant paradigm in model-free offline RL has emerged, focusing on conservative value estimation and explicit policy regularization. These methods aim to constrain the learning process such that the agent remains within the data manifold or learns value functions that are provably lower-bounded for out-of-distribution (OOD) actions. Unlike the model-based approaches discussed in the following subsection, which mitigate uncertainty by penalizing rewards or truncating trajectories based on learned dynamics, this section reviews techniques that directly modify the value function optimization or policy constraints within a model-free framework.

One of the most influential approaches in this domain is Conservative Q-Learning (CQL), which directly tackles the overestimation problem by modifying the Bellman error objective. CQL introduces a regularizer that penalizes high Q-values for actions sampled from a wide distribution while maximizing Q-values for actions present in the dataset. Theoretically, this ensures that the learned Q-function serves as a lower bound on the true value of the policy, thereby preventing the agent from exploiting erroneous high-value estimates for unseen actions [43]. Empirical results have demonstrated that CQL substantially outperforms existing offline RL methods, particularly in complex and multi-modal data distributions where standard algorithms fail. However, while CQL effectively curbs overestimation, it can sometimes be overly pessimistic, potentially suppressing the generalization capabilities of the value function and hindering performance improvements over the behavior policy.

To mitigate the excessive conservatism of methods like CQL, researchers have developed variants that seek a more balanced trade-off. For instance, Mildly Conservative Q-learning (MCQ) proposes actively training OOD actions by assigning them appropriate pseudo Q-values rather than uniformly penalizing them. This approach induces a policy that performs at least as well as the behavior policy without inducing erroneous overestimation, showing superior generalization ability when transferring from offline pre-training to online fine-tuning [44]. Similarly, Contextual Conservative Q-Learning (C-CQL) leverages inverse dynamics models to capture contextual information, enabling the policy to generate stable transitions even at perturbed states, which are a common form of OOD state [45]. Furthermore, addressing the issue of imbalanced datasets where state coverage follows a power law distribution, recent work has augmented CQL with retrieval processes to recall past related experiences, effectively alleviating challenges posed by skewed data distributions [46].

Parallel to value-based conservatism, another significant line of research focuses on Implicit Q-Learning (IQL), which adopts a fundamentally different strategy to avoid extrapolation errors. Instead of explicitly regularizing the Q-function against OOD actions, IQL never evaluates actions outside the dataset during training. It achieves policy improvement implicitly by treating the state value function as a random variable and estimating the value of the best actions via an upper expectile of this variable [47]. This allows IQL to leverage the generalization capacity of function approximators to identify optimal actions without directly querying the Q-function with unseen inputs. The policy is subsequently extracted via advantage-weighted behavioral cloning. IQL has demonstrated state-of-the-art performance on standard benchmarks and exhibits remarkable resilience to data corruption, a critical feature for real-world applications where datasets may be noisy or maliciously altered [48].

Recent advancements have further refined the theoretical understanding and practical implementation of these implicit methods. For example, IDQL reinterprets IQL as an actor-critic method,

connecting the critic objective to a behavior-regularized implicit actor. Recognizing that standard Gaussian actors may struggle with multi-modal policies, IDQL utilizes diffusion policies for behavior modeling, combining the ease of implementation of IQL with the expressiveness of generative models [49]. Additionally, the concept of Implicit Value Regularization (IVR) has been proposed to explain the success of in-sample learning paradigms like IQL, leading to new algorithms such as Sparse Q-learning (SQL) and Exponential Q-learning (EQL) that introduce sparsity in value learning for enhanced robustness in noisy regimes [50].

Beyond these foundational methods, other techniques have been developed to handle specific nuances of policy regularization. Quantile Filtered Imitation Learning (QFIL) performs policy improvement by running imitation learning on a dataset filtered by Q-value quantiles, providing a safe policy improvement step with a natural hyperparameter to trade off bias and variance [51]. Moreover, approaches like Robust Offline Reinforcement Learning with Gradient Penalty utilize gradient penalties and constraint relaxation to tame non-optimal trajectories in contaminated datasets, ensuring stability even when the data purity is compromised [52]. Collectively, these conservative and regularization-based methods form the backbone of modern model-free offline RL, providing the necessary theoretical guarantees and empirical stability required for deploying agents in safety-critical robotic systems. While these methods successfully constrain the policy through value estimation or direct regularization, the subsequent subsection explores an alternative paradigm: leveraging learned dynamics models to generate synthetic data while employing pessimistic planning to navigate the uncertainties inherent in those models.

2.2 Dynamics Learning and Pessimistic Planning in Model-Based Methods

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While the preceding subsection explored model-free strategies that constrain policies through conservative value estimation or implicit regularization, this section examines an alternative paradigm: Model-Based Offline Reinforcement Learning (MBRL). MBRL represents a critical shift by leveraging learned dynamics models to generate synthetic data, thereby expanding the effective dataset beyond the limited static buffer. However, this approach introduces a distinct challenge: when a policy explores regions of the state-action space not covered by the training data, the learned dynamics model often produces highly inaccurate predictions. These errors compound over time, leading to catastrophic policy degradation. To address this, the core algorithmic strategy involves incorporating pessimism into the planning process, ensuring that the agent prefers actions within the support of the data or those where model uncertainty is low. This subsection reviews the theoretical underpinnings and practical implementations of dynamics learning coupled with pessimistic planning, focusing on uncertainty quantification, reward penalization, and conservative value estimation.

The foundation of pessimistic model-based RL lies in the accurate quantification of epistemic uncertainty within the learned dynamics model. Traditional approaches often rely on ensemble methods to estimate this uncertainty, where the variance across multiple independently trained models serves as a proxy for model confidence. However, naive ensemble usage can sometimes lead to overconfident predictions or fail to capture the true posterior variance effectively. Recent advancements have sought to refine these estimates to avoid the inefficiencies caused by over-approximation. For instance, research has demonstrated that previous methods upper-bounding the posterior variance via uncertainty Bellman equations may result in suboptimal exploration due to loose bounds [53]. By deriving a new uncertainty Bellman equation that converges to the true posterior variance, algo-

gorithms can achieve sharper uncertainty estimates, which are crucial for distinguishing between genuinely risky out-of-distribution (OOD) states and those that are merely uncertain due to sampling noise [54]. This precise quantification is essential for constructing effective pessimistic penalties that guide the agent away from unreliable model predictions without unnecessarily restricting the policy’s capacity to improve.

A seminal approach to integrating this uncertainty into the learning loop is the modification of the reward function through uncertainty penalization, as exemplified by the Model-Based Offline Policy Optimization (MOPO) framework. In this paradigm, the agent optimizes a policy within a “pessimistic MDP,” where the reward signal is augmented with a penalty term proportional to the estimated model uncertainty. This mechanism effectively lower-bounds the true return, ensuring that the agent only ventures into unknown regions if the potential gain outweighs the risk of model error. The theoretical justification for such approaches often rests on the total variation distance between the learned and true dynamics, although practical implementations frequently rely on heuristic uncertainty estimates. To bridge the gap between theory and practice, rigorous comparisons of various uncertainty heuristics have been conducted, revealing that hyperparameters such as the number of ensemble members and the horizon of imaginary rollouts significantly influence performance [55]. Furthermore, to prevent the generation of unreliable synthetic trajectories, advanced techniques like Trajectory Truncation with Uncertainty (TATU) have been proposed. TATU adaptively truncates synthetic rollouts when the accumulated uncertainty along a trajectory exceeds a specific threshold, thereby preventing the propagation of errors from highly uncertain states into the training data [56]. This method has been shown to significantly boost the performance of both model-based and model-free offline algorithms by ensuring that only high-fidelity synthetic data is utilized.

Beyond reward penalization, another powerful avenue for incorporating pessimism is through the modification of the Bellman operator itself, leading to conservative value estimation. Instead of altering the rewards, these methods directly constrain the value function updates to remain conservative in the face of distributional shift. For example, Conservative State Value Estimation (CSVE) proposes learning a conservative V-function by explicitly imposing penalties on OOD states during the Bellman iteration, rather than focusing solely on OOD actions as in conservative Q-learning variants [57]. This approach allows for more effective state value estimation and facilitates better policy optimization by ensuring that the critic does not overestimate the value of states lying outside the data manifold. Similarly, the concept of Bellman-consistent pessimism offers a general framework for function approximation where pessimism is implemented over the set of functions consistent with the Bellman equations, rather than calculating point-wise lower bounds [58]. This method automatically adapts to the optimal bias-variance tradeoff and provides stronger theoretical guarantees under weaker assumptions compared to bonus-based pessimism.

The interplay between optimism and pessimism also plays a nuanced role in model-based planning, particularly when dealing with risk-sensitive objectives. While pessimism is vital for safety in offline settings, certain scenarios benefit from a tactical balance. Algorithms like Tactical Optimistic and Pessimistic (TOP) estimation demonstrate that the degree of optimism required can vary across tasks and learning stages, suggesting that rigid pessimism might sometimes hinder the discovery of optimal policies in regions with manageable uncertainty [59]. In the context of risk-aware planning, uncertainty-guided strategies such as the uncertainty-guided cross-entropy method have been developed to penalize action sequences that result in high-variance state predictions, effectively guiding the agent toward high-confidence areas of the state space [60]. Moreover, the integration of Bayesian principles allows for dynamic adaptation of the uncertainty set, as seen in approaches

that utilize an Uncertainty Robust Bellman Equation (URBE) to encourage safe exploration while adapting to new observations [61].

Finally, the theoretical landscape of offline RL has been enriched by proofs establishing the minimax optimality of pessimistic approaches. It has been shown that pessimism is not merely a heuristic but a provably efficient principle for eliminating spurious correlations arising from irrelevant trajectories in the dataset [62]. Distributionally Robust Optimization (DRO) based approaches further solidify this by modeling uncertainty in the transition kernel directly, constructing uncertainty sets of statistically plausible dynamics, and optimizing for the worst-case performance within these sets [63]. These theoretical advances confirm that pessimistic planning, when grounded in rigorous uncertainty quantification and conservative Bellman updates, provides a robust foundation for learning safe and effective policies from static datasets. While these model-based methods navigate distributional shift by penalizing rewards or truncating trajectories based on learned dynamics, the subsequent subsection explores a transformative paradigm that bypasses traditional bootstrapped value estimation entirely, reframing offline RL as a sequence modeling task to leverage the scalability of large-scale generative architectures.

2.3 Sequence Modeling and Generative Policy Representations

2.3 Sequence Modeling and Generative Policy Representations

While the preceding subsection detailed how model-based methods mitigate distributional shift through explicit dynamics learning and pessimistic planning, a parallel and transformative paradigm has emerged that bypasses traditional bootstrapped value estimation entirely. This approach reframes offline reinforcement learning as a sequence modeling or generative modeling task, leveraging the scalability of large-scale architectures to address the limitations of standard actor-critic methods. Unlike the uncertainty-driven penalization strategies discussed previously, these methods rely on the expressive power of Transformers and diffusion models to directly capture multi-modal behavioral distributions and long-horizon dependencies within static datasets, thereby avoiding the extrapolation errors associated with estimating Q-values for out-of-distribution actions.

The inception of treating reinforcement learning as a sequence modeling problem marked a pivotal shift in this domain. The seminal work [64] introduced the Decision Transformer (DT), an architecture that abandons conventional Bellman equation updates in favor of a causally masked Transformer. In this framework, the agent generates optimal actions by conditioning on a desired future return, past states, and previous actions, effectively treating the trajectory as a sequence of tokens similar to natural language. This approach simplifies algorithmic design by removing the need for explicit value function estimation, allowing the model to capitalize on the powerful context-handling capabilities of Transformers. However, while DT demonstrated that sequence modeling could match or exceed state-of-the-art offline RL baselines, it was observed that the model primarily relies on memorizing and recalling high-return trajectories from the dataset, often struggling to “stitch” together sub-optimal segments to form novel, optimal paths. This limitation spurred further research into enhancing the stitching capability of sequence models, leading to frameworks that integrate hierarchical structures or advantage conditioning to better decompose tasks and improve robustness against environmental stochasticity [65], [66].

Parallel to the development of autoregressive sequence models, the field has seen a surge in adopting generative models, specifically diffusion models, to represent policies. Traditional policy classes, such as unimodal Gaussian distributions, often fail to capture the complexity of human demonstra-

tions or expert behaviors that are inherently multi-modal. Diffusion models, with their ability to model complex data distributions through iterative denoising, offer a highly expressive alternative. The introduction of Diffusion Q-learning (Diffusion-QL) exemplifies this synergy, where a conditional diffusion model represents the policy while being guided by a learned action-value function. This hybrid approach combines the stability of behavior cloning with the optimality-seeking nature of Q-learning, allowing the agent to generate actions that are both consistent with the data manifold and optimized for high returns [67]. The efficacy of this approach was further validated in robotic manipulation, where [68] demonstrated that representing visuomotor policies as conditional denoising processes significantly outperforms existing methods across diverse benchmarks, particularly in handling high-dimensional action spaces and multimodal distributions.

Beyond single-task policy learning, generative sequence models have proven instrumental in tackling long-horizon planning and multi-task generalization. The flexibility of diffusion models allows for the generation of diverse trajectories, enabling agents to plan milestones in latent spaces for complex, sparse-reward tasks [69]. Furthermore, the integration of language conditioning into diffusion models has facilitated the creation of generalist agents capable of scaling across space, time, and tasks. For instance, [70] proposed a hierarchical planner that leverages natural language instructions to guide diffusion models, achieving state-of-the-art results in multi-task success rates while drastically improving inference speeds. Similarly, [71] introduced a probabilistic framework that composes learned skill distributions to solve unseen long-horizon tasks, demonstrating robust reasoning about action dependencies and geometric constraints.

The versatility of these generative approaches extends to multi-agent systems and world modeling as well. The [72] successfully cast cooperative multi-agent RL into a sequence modeling problem, utilizing an encoder-decoder architecture to achieve monotonic performance improvement and superior data efficiency compared to traditional MARL algorithms. In the realm of world models, [73] proposed a non-autoregressive approach that generates entire on-policy trajectories in a single pass, mitigating the compounding prediction errors common in autoregressive dynamics models. Additionally, the exploration of consistency models as a policy class offers a promising direction for balancing the expressiveness of diffusion models with the computational efficiency required for real-time control, demonstrating significant speedups over standard diffusion policies in both offline and online settings [74].

Collectively, these advancements underscore a fundamental reorientation in offline RL algorithmic design, moving from explicit uncertainty quantification and conservative value updates toward implicit distribution learning via generative processes. By leveraging the representational power of Transformers and the distributional flexibility of diffusion models, researchers are overcoming the bottlenecks of value estimation errors and unimodal policy constraints. Whether through the return-conditioned generation of Decision Transformers, the score-based optimization of Diffusion-QL, or the hierarchical planning of language-controlled diffusion agents, sequence modeling and generative representations have established themselves as indispensable tools for developing robust, scalable, and data-efficient robotic learning systems. While these methods implicitly manage distributional shift through the constraints of the generative process, the following subsection explores an alternative, statistically principled approach that explicitly corrects for stationary distribution mismatches to ensure policy validity without relying on complex generative architectures or pessimistic penalties.

2.4 Distribution Correction and Stationary Distribution Matching

2.4 Distribution Correction and Stationary Distribution Matching

While the preceding subsection highlighted how generative sequence models implicitly manage distributional shift through the constraints of expressive architectures like Transformers and diffusion processes, this section explores an alternative, statistically principled paradigm that explicitly corrects for stationary distribution mismatches. Rather than relying on complex generative sampling or pessimistic value penalization, this approach reformulates offline reinforcement learning (RL) as a distribution correction task. The core objective is to directly estimate and align the occupancy measure of the target policy with the available data, thereby maximizing expected returns while ensuring the learned policy remains strictly within the valid support of the offline buffer. By operating in the space of stationary distributions rather than through direct policy gradients or bootstrapped value updates, these methods offer rigorous guarantees against overestimation and provide a robust mechanism for handling extrapolation errors.

The cornerstone of this framework is the estimation of density ratios between the optimal policy’s stationary distribution and the behavior policy’s distribution. A prominent algorithm exemplifying this strategy is OptiDICE (Offline Policy Optimization via Stationary Distribution Correction Estimation), which bypasses traditional bootstrapping and policy gradient updates to directly estimate these corrections [75]. OptiDICE formulates the optimization problem by deriving a convex objective that minimizes the divergence between the corrected distribution and the optimal distribution, effectively neutralizing the action value overestimation that plagues many model-free offline RL algorithms. This method demonstrates that solving for the stationary distribution correction directly yields competitive performance on standard benchmarks while avoiding the hyperparameter sensitivity often associated with conservative Q-learning variants. The theoretical strength of such approaches lies in their ability to accurately model transition dynamics and stationary behavior within the data’s latent space, ensuring policies do not venture into regions lacking data coverage.

Beyond direct density ratio estimation, the concept of invariant representation learning offers a complementary framework for addressing distributional shifts. Although originally developed for supervised learning and domain adaptation, the principles of learning features that remain stable across different environments are increasingly critical for robust offline RL. The central idea is to identify “invariant” features of the state space that causally determine reward and transition dynamics, while disregarding “spurious” correlations present in the offline dataset that fail to generalize under distribution shifts. Research into heterogeneous risk minimization and invariant learning suggests that models trained to uncover invariant relationships across latent sub-groups can achieve superior generalization when facing unobserved heterogeneity or shifts in the data-generating process. For instance, frameworks like Heterogeneous Risk Minimization (HRM), which jointly learn latent heterogeneity and invariant relationships, provide a theoretical basis for handling datasets where explicit environment labels are absent—a scenario common in aggregated real-world robotic data [76].

Furthermore, the integration of distributionally robust optimization (DRO) techniques significantly enhances the stability of stationary distribution matching. DRO frameworks aim to learn policies that perform well against worst-case perturbations within a specified uncertainty set of the data distribution. This is particularly vital in offline RL, where static datasets may contain biases or represent only a subset of the possible state space. By minimizing worst-case risk over a set of plausible distributions, agents can develop policies robust to covariate shifts and unmodeled temporal effects [77]. Methods leveraging Maximum Mean Discrepancy (MMD) have been proposed to mini-

mize the distance between training and test distributions in feature spaces, effectively regularizing the learning process to prevent dangerous extrapolation [78]. These techniques ensure that learned policies do not rely on fragile correlations that might collapse upon deployment in slightly altered conditions, thereby improving the reliability of stationary distribution estimates.

The intersection of stationary distribution matching and fairness constraints also presents a promising direction for robust offline RL. In scenarios where offline data is collected from diverse operators or environments, ensuring that the learned policy performs equitably across different sub-populations is essential. Approaches enforcing individual fairness or enhancing distributional stability among sub-populations can be adapted to offline RL to prevent policies from exploiting biases inherent in the behavior data [79]. By quantifying the stability of prediction mechanisms across different data segments, algorithms can derive generalization error bounds that hold even under significant distribution shifts [80]. This process is akin to identifying “sufficient invariant features” that persist across domains, ensuring that policy performance does not degrade when encountering out-of-distribution states during deployment [81].

Moreover, recent advances in detecting distribution shifts using latent space representations provide valuable tools for validating the quality of stationary distribution correction. By analyzing the geometry of the embedding space, researchers can establish robustness boundaries and detect when a target policy’s induced distribution diverges significantly from the reference data [82]. This capability is crucial for off-policy evaluation, allowing practitioners to assess the validity of a learned policy without costly online interactions. Techniques utilizing self-organizing clustering or statistical tests in latent spaces can identify violations of the sufficient overlap assumption between behavior and target policies, prompting the adoption of more conservative correction strategies when necessary [83].

In summary, distribution correction and stationary distribution matching represent a sophisticated class of offline RL algorithms that address the core issue of distributional shift through principled statistical estimation. By directly optimizing the stationary distribution via methods like OptiDICE and incorporating insights from invariant learning and distributionally robust optimization, these approaches mitigate the risks of extrapolation error and model misspecification. The synergy between correcting occupancy measures and learning invariant representations ensures that robotic agents can derive safe and effective policies from static datasets, even amidst heterogeneous data sources and complex environmental shifts. These distribution-aware mechanisms lay the necessary groundwork for the theoretical analysis discussed in the following subsection, which establishes the rigorous guarantees—such as single-policy concentrability and realizability—that justify why such correction-based approaches succeed under partial data coverage.

2.5 Theoretical Guarantees Under Partial Data Coverage

2.5 Theoretical Guarantees Under Partial Data Coverage

Building upon the distribution correction mechanisms discussed in the previous section, which aim to align stationary distributions without exhaustive data coverage, it is crucial to establish the theoretical foundations that justify why such approaches succeed under partial observability. The theoretical landscape of offline reinforcement learning (RL) has undergone a profound transformation, shifting from reliance on stringent uniform coverage assumptions to the establishment of rigorous guarantees under significantly weaker conditions. Historically, sample-efficient offline RL was thought to require that the behavior policy covers the entire state-action space uniformly, a

condition known as all-policy concentrability. However, such requirements are practically infeasible in high-dimensional robotic domains where data is typically collected by sub-optimal or specialized policies. Consequently, contemporary research has focused on deriving sample complexity bounds and sub-optimality gaps under the relaxation of uniform overlap requirements, primarily leveraging the concepts of single-policy concentrability and function class realizability.

A pivotal breakthrough in this domain is the realization that valid learning guarantees can be established even when the dataset only covers the optimal policy or a specific comparator policy, rather than all possible policies. This shift is epitomized by the development of algorithms that operate under single-policy concentrability assumptions, providing the mathematical backing for the density-ratio methods highlighted earlier. For instance, recent works have demonstrated that by utilizing a primal-dual formulation of Markov Decision Processes (MDPs), it is possible to achieve polynomial sample complexity requiring only realizability of the value function and single-policy concentrability [84]. This approach models the dual variables, representing discounted occupancy, using density-ratio functions against the offline data, thereby circumventing the need for exhaustive data coverage. Similarly, other studies have introduced algorithms based on marginalized importance sampling (MIS) that provide finite-sample guarantees for general MDPs under partial coverage, provided there is access to weak realizable function classes and potentially additional side information regarding covering distributions [85].

The role of function approximation realizability has also been scrutinized deeply in this context. It was previously conjectured that realizability combined with standard concentrability might be insufficient for sample-efficient learning. This conjecture was resolved positively, proving that without stronger assumptions such as Bellman completeness or restrictive coverage, any algorithm may require sample complexity polynomial in the size of the state space [86]. This fundamental barrier highlights the necessity of either stronger coverage conditions or representation conditions that exceed those required for supervised learning. In response, newer frameworks have emerged that successfully relax both factors simultaneously. For example, optimal conservative offline RL algorithms utilizing augmented Lagrangian methods have been shown to be statistically optimal under general function approximation and single-policy concentrability, effectively bypassing the need for explicit uncertainty quantification which often leads to computational intractability [87].

Furthermore, the literature has expanded to include instance-dependent bounds that offer sharper guarantees based on the specific structure of the problem, such as the presence of sub-optimality gaps. Research indicates that when a positive sub-optimality gap exists in the optimal Q-function, the sample complexity can be improved from the standard $O(1/\epsilon^2)$ rate to $O(1/\epsilon)$ under optimal policy coverage assumptions [88]. This gap-dependent analysis has been extended to linear function approximation settings, where algorithms like Bootstrapped and Constrained Pessimistic Value Iteration (BCP-VI) achieve fast rates of $\tilde{O}(1/K)$ and even absolute zero sub-optimality error beyond a certain data threshold, provided the linear features of optimal actions span those of the behavior policy [89]. These findings suggest that the difficulty of offline RL is not uniform but highly dependent on the intrinsic hardness of the specific MDP instance and the alignment between the behavior and optimal policies.

The robustness of these theoretical guarantees has also been tested against more complex data collection scenarios, including adaptively collected data and the presence of corruption, which are common in real-world robotic deployments. Theoretical frameworks have been developed to handle datasets gathered by evolving behavior policies without assuming independent and identically distributed (i.i.d.) samples. For instance, analysis of the TMIS Offline Policy Evaluation estimator has yielded high-probability, instance-dependent bounds in adaptive settings, recovering minimax-

optimal learning rates [90]. Additionally, the field has addressed the challenge of dataset corruption, proposing algorithms that maintain sub-optimality bounds worsened only by an additive factor proportional to the corruption level, even under single-policy coverage assumptions [91].

Moreover, the interplay between offline and online learning has been explored through the lens of policy finetuning, where theoretical bounds demonstrate that a hybrid approach can outperform purely offline or purely online strategies when a reference policy is available [92]. This line of work underscores that the “best” algorithm depends on the degree of partial coverage; in some regimes, simple offline reduction is optimal, while in others, strategic online exploration yields superior sample efficiency. The introduction of intrinsic offline RL bounds further unifies these perspectives by providing a per-instance lower bound that directly implies existing optimal results under various coverage assumptions, from uniform data coverage to assumption-free regimes [93].

Finally, the extension of these guarantees to multi-agent systems and constrained settings marks another frontier, bridging the gap between theoretical single-agent bounds and complex robotic applications. Recent frameworks for offline learning in Markov games have established sample efficiency under general function approximation by employing Bellman-consistent pessimism and relaxing unilateral concentrability requirements [94]. Similarly, in constrained RL, algorithms like COptiDICE and Primal-Dual-Critic Algorithm (PDCA) have been proven to find near-optimal policies satisfying cost constraints under only concentrability and realizability assumptions, avoiding the need for strong Bellman completeness [95; 96]. Collectively, these theoretical advances provide a robust foundation for deploying offline RL in robotics, assuring that safe and effective policies can be learned from static, partially covered datasets without the prohibitive costs of active exploration, thereby setting the stage for preference-based refinements where human feedback further guides policy optimization within these theoretically sound boundaries.

3 Preference-Based Learning and Reward Modeling from Human Feedback

3.1 Foundations of Preference Modeling and Reward Inference

3.1 Foundations of Preference Modeling and Reward Inference

The cornerstone of Preference-Based Reinforcement Learning (PbRL) and Reinforcement Learning from Human Feedback (RLHF) lies in the accurate mathematical formulation of how humans express preferences and how these expressions can be inverted to recover an underlying reward function. This foundational step is critical, as the fidelity of the subsequent policy optimization—discussed in the following section on two-stage pipelines—depends entirely on the validity of the reward model derived here. Traditionally, the dominant paradigm has relied on the assumption that human preferences between two trajectory segments are determined solely by the difference in their cumulative rewards, often referred to as the “partial return” assumption. Under this framework, typically modeled via the Bradley-Terry-Luce (BTL) model, a human is assumed to prefer the trajectory with the higher sum of rewards. This formulation has powered numerous successes in aligning large language models and robotic policies, providing a tractable likelihood function for training reward models [97]. However, recent theoretical and empirical investigations have increasingly challenged the validity of this assumption, suggesting that it fails to capture the nuanced cognitive processes humans employ when evaluating behavior, particularly in complex, sequential decision-making tasks.

A critical limitation of the partial return model is its failure to account for the counterfactual nature of human judgment. Humans often evaluate a trajectory not just by what happened, but

by how much better it could have been had optimal decisions been made. This insight has led to the emergence of regret-based preference models, which posit that human preferences are informed by a segment’s deviation from optimal decision-making rather than its absolute return. Research indicates that the partial return model suffers from fundamental identifiability issues; specifically, multiple distinct reward functions can generate the exact same preference data under this assumption, making it impossible to uniquely recover the true intent of the human stakeholder [22]. In contrast, modeling preferences based on regret offers a path to identifiability. Theoretical proofs demonstrate that given infinitely many preferences generated according to a regret model, one can uniquely identify the reward function that generated those preferences, a property that the partial return model lacks in multiple contexts [22]. Empirical studies further support this shift, showing that regret-based models not only better predict real human preferences but also lead to policies that are more faithfully aligned with human intent when trained on finite datasets [22].

The implications of mis-specifying the preference model are profound and directly impact the robustness challenges addressed in modern RLHF pipelines. When algorithms assume preferences are based on partial return while the underlying human behavior is actually driven by regret, the learned function often converges to an approximation of the optimal advantage function rather than the true reward function [23]. While this misspecification can sometimes yield a highly shaped reward function that performs adequately if specific pitfalls are addressed, it represents a theoretically unsound foundation that complicates the optimization landscape. Furthermore, the reliance on the BTL model and the assumption of independence of irrelevant alternatives (IIA) can introduce perverse incentives, where the learning algorithm is driven to optimize for response formats that game the preference model rather than genuinely satisfying human values [98]. This highlights the necessity of moving beyond static, simplistic preference oracles toward more general formulations that can accommodate diverse human cognitive strategies, a prerequisite for developing the robust reward models discussed in Section 3.2.

Parallel to the debate between partial return and regret models is the enduring challenge of identifiability in Inverse Reinforcement Learning (IRL), which serves as a foundational precursor to modern PbRL. The core difficulty in IRL is that an observed policy can be explained by an infinite number of reward functions, rendering the problem ill-posed without additional constraints or assumptions about the demonstrator’s rationality [99]. Standard behavioral models, such as assuming the demonstrator is noisily optimal or follows Boltzmann rationality, are often misspecified when applied to real human data, as humans exhibit systematic biases like risk aversion, myopia, and non-stationary learning behaviors [100]. Rigorous mathematical analysis reveals that IRL is highly sensitive to such misspecification; even mild deviations between the assumed behavioral model and the true human decision-making process can lead to significant errors in the inferred reward function [101]. These sensitivities underscore why simple reward models are prone to the overoptimization and fragility issues that necessitate the advanced ensemble and meta-learning techniques explored later.

To mitigate these identifiability challenges, researchers have explored various avenues, including leveraging data from multiple experts operating under different discount factors or in different environments, which can theoretically render the reward function identifiable up to a constant [102]. Additionally, incorporating constraints into the IRL framework, such as safety constraints in Constrained Markov Decision Processes, alters the conditions required for identifiability and generalizability, often necessitating stronger assumptions to ensure the recovered reward transfers correctly to new environments [103]. Another promising direction involves relaxing the assumption of static human demonstrators. By modeling the human as a learning agent whose policy evolves

over time, it is possible to extract more information from the demonstration data than if the human were assumed to be stationary, thereby improving the accuracy of preference inference [104].

Furthermore, the integration of prior knowledge and structural assumptions into the reward learning process has proven essential for reducing the sample complexity and improving the robustness of inference. Approaches that incorporate priors over the structure of the reward function, such as imposing soft constraints or utilizing abstract state spaces, can significantly reduce the amount of human feedback required to recover the target reward [105]. Similarly, methods that explicitly learn features from human input, rather than relying on hand-crafted features or implicit deep learning representations, offer a divide-and-conquer strategy that enhances both sample efficiency and the generalization capability of the learned reward [106]. These advancements collectively underscore that the foundation of preference modeling is not merely a choice of loss function, but a careful consideration of the cognitive mechanisms driving human feedback, the identifiability properties of the resulting mathematical models, and the robustness of these models to the inevitable noise and bias present in real-world human data. Addressing these foundational theoretical gaps is a prerequisite for building the reliable, high-fidelity reward models required to prevent the collapse of downstream policy optimization in complex robotic environments.

3.2 Two-Stage RLHF Pipelines and Reward Model Robustness

3.2 Two-Stage RLHF Pipelines and Reward Model Robustness

Building upon the foundational theories of preference modeling and the critical identifiability challenges outlined in Section 3.1, the practical implementation of aligning artificial intelligence systems predominantly relies on the two-stage Reinforcement Learning from Human Feedback (RLHF) pipeline. This standard paradigm, widely adopted in large language models and increasingly in robotics, decouples the learning process into two distinct phases: first, training a reward model (RM) to approximate human preferences from pairwise comparisons or rankings; and second, optimizing a policy to maximize the scores predicted by this learned model, typically via algorithms like Proximal Policy Optimization (PPO). While this framework has driven remarkable successes, its efficacy is fundamentally bottlenecked by the fidelity and robustness of the intermediate reward model. As highlighted in the previous section, if the underlying preference assumptions are misspecified or the data is noisy, the resulting reward model acts as a flawed proxy for true human values. This imperfection precipitates phenomena such as reward overoptimization, where policies exploit inaccuracies in the reward function to achieve high scores while generating outputs that remain misaligned with actual human intent. Consequently, a significant body of recent research has shifted focus from theoretical formulation to engineering robustness, developing advanced training objectives, ensemble strategies, and meta-learning techniques to stabilize this two-stage architecture.

A primary manifestation of reward model fragility is overoptimization, where the policy drives the reward signal to extremes that no longer correlate with genuine quality. To mitigate this vulnerability, ensemble-based methods have emerged as a powerful strategy for enhancing reliability. By aggregating predictions from multiple reward models, researchers can quantify uncertainty and penalize actions where models disagree, effectively preventing the policy from exploiting the idiosyncratic errors of a single model. Systematic studies demonstrate that employing ensemble-based conservative optimization objectives, such as worst-case and uncertainty-weighted optimization, can practically eliminate overoptimization in both best-of-n sampling and PPO settings [107]. These approaches leverage variance within the ensemble to identify regions of the state space where the

reward signal is unreliable, acting as a crucial regularizer during policy optimization. Furthermore, to address the computational costs associated with ensembling, efficient methods utilizing Low-Rank Adaptation (LoRA) have been proposed. These lightweight ensembles maintain the benefits of diverse reward perspectives while significantly reducing the computational burden, proving that even resource-constrained setups can achieve substantial improvements in alignment performance [108].

Beyond architectural ensembling, the structural design of the reward learning objective itself is pivotal for robustness. Traditional ranking losses often fail to capture the nuanced margins between preferred and non-preferred responses, leading to models that struggle to generalize to out-of-distribution samples. In response, contrastive learning frameworks have been introduced to enhance the discriminative power of reward models. By incorporating penalty terms that explicitly contrast chosen responses against baseline or rejected responses, these methods encourage the model to focus on meaningful semantic differences rather than superficial artifacts [109]. This approach not only improves the distinction of subtle quality variations but also calibrates the reward signal according to task difficulty, thereby reducing the variance inherent in downstream PPO training. Additionally, the concept of reward consistency—the ability of a model to adapt its assignments based on semantic changes in prompts—has gained traction. Techniques such as ConvexDA and RewardFusion have been developed to improve this consistency through extrapolation during training and inference, demonstrating that more consistent reward models yield downstream policies with superior utility [110].

To address the challenge of diverse and shifting preference distributions, meta-learning represents another critical frontier for enhancing reward model robustness. In scenarios where human feedback originates from multiple sources with conflicting viewpoints, a single static reward model is often insufficient. Meta-learning approaches enable reward models to rapidly adapt to new preference distributions or specific user groups by learning a shared initialization that captures commonalities across tasks. This adaptability is particularly crucial for iterative RLHF, where the policy distribution shifts continuously, potentially moving data out of the support of the original reward training set. By employing meta-learning, reward models can maintain their discriminative abilities on out-of-distribution samples, ensuring stable alignment throughout the fine-tuning process [111]. Moreover, theoretical work has extended these concepts to multi-party settings, establishing sample complexity bounds for learning diverse social welfare functions. This underscores the necessity of modeling heterogeneous preferences explicitly rather than aggregating them into a single scalar, a step essential for handling the complex social dynamics often present in robotic deployment [112].

The integrity of the data used to train these models is equally paramount, as noisy, ambiguous, or adversarially corrupted preference labels can severely degrade the entire pipeline. Recent investigations into “reward hacking” and preference poisoning reveal that injecting even a small fraction of malicious preference pairs can manipulate the resulting policy to generate targeted, undesirable outputs [113]. To counteract such vulnerabilities, methods like Iterative Data Smoothing have been proposed, which dynamically update training labels based on the model’s current beliefs. By replacing hard binary labels with soft probabilities, these techniques mitigate overfitting to noisy annotations [114]. Similarly, information-theoretic approaches introduce variational information bottlenecks to filter out irrelevant information during reward modeling, effectively detecting and mitigating overoptimization by monitoring deviations in the latent space [115].

Finally, the issues of reward sparsity and the correlation between reward scores and superficial features, such as response length, pose significant hurdles to faithful alignment. Empirical analyses have shown that optimizing for helpfulness often inadvertently optimizes for verbosity, as many

reward models exhibit a strong positive correlation between sequence length and reward score [116]. Addressing this requires not only better data curation but also architectural innovations that decouple desirable traits from confounding variables. Dense reward shaping techniques, which redistribute sparse episode-level rewards to individual tokens using attention mechanisms, offer a promising direction for stabilizing training. These methods ensure that the policy learns from granular feedback rather than global heuristics, providing a more precise optimization signal [117]. Collectively, these advancements in ensemble methods, contrastive objectives, meta-learning, and data robustness strategies are essential for evolving the two-stage RLHF pipeline from a fragile experimental setup into a reliable foundation. However, despite these improvements, the inherent complexity and instability of maintaining a separate reward model motivate the exploration of alternative architectures that bypass this intermediate step entirely, a direction we explore in the following section on direct and reward-free preference optimization.

3.3 Direct and Reward-Free Preference Optimization

3.3 Direct and Reward-Free Preference Optimization

While the previous section detailed strategies to strengthen the intermediate reward model within the standard two-stage RLHF pipeline, a parallel and increasingly influential research direction seeks to eliminate this bottleneck entirely. The traditional paradigm—training a scalar reward model followed by policy optimization via algorithms like Proximal Policy Optimization (PPO)—introduces significant complexity, instability, and computational overhead. These issues stem not only from the difficulty of training a robust reward model but also from the high variance inherent in policy gradient updates and the sensitivity to hyperparameters during the reinforcement learning phase. In response, a new class of algorithms has emerged that bypasses explicit reward modeling, opting instead to directly optimize the policy using preference data. This shift towards direct and reward-free preference optimization represents a fundamental architectural change, offering improved stability, reduced memory usage, and greater sample efficiency, which are critical for scaling alignment in robotic systems.

The seminal work in this domain is Direct Preference Optimization (DPO), which reparameterizes the reward function in terms of the optimal policy and a reference policy. By deriving a closed-form solution for the optimal policy under a Bradley-Terry preference model, DPO transforms the complex RLHF objective into a simple classification loss that can be minimized using standard supervised learning techniques [118]. This approach effectively decouples alignment from the need for sampling from the policy during training and removes the requirement for a separate reward model, thereby simplifying the training loop and reducing the hyperparameter tuning burden. Theoretical analyses have further clarified the mechanics of DPO, showing that it implicitly performs inverse Q-learning at the token level, allowing for some degree of credit assignment despite its bandit-derived formulation [119]. However, subsequent research has also highlighted limitations, noting that DPO may decrease the probability of generating preferred data more slowly than it suppresses dispreferred data, potentially hindering the full realization of human-preferred behaviors [120].

Building upon the foundation of direct optimization, several variants have been proposed to address specific shortcomings related to regularization, data quality, and objective formulation. For instance, Identity Preference Optimization (IPO) was introduced to incorporate KL-regularization more effectively by using a root-finding mean squared error loss, addressing issues where DPO might overfit to deterministic preferences or ignore the reference distribution [121]. Furthermore,

Maximum Preference Optimization (MPO) offers an intuitive off-policy perspective via importance sampling, combining the objectives of RLHF and IPO to ensure that regularization remains effective even when the support of the preference distribution differs from the reference distribution [121]. To enhance robustness against noisy or ambiguous human labels—a challenge also discussed in the context of reward model poisoning in the previous section—Robust DPO (rDPO) introduces a de-biased loss function that theoretically guarantees sub-optimality bounds even in the presence of random preference flips, making it suitable for real-world datasets where label noise is prevalent [122].

Beyond the DPO family, other frameworks have sought to ground direct preference optimization in alternative theoretical assumptions about human behavior, further diversifying the toolkit for reward-free learning. Contrastive Preference Learning (CPL) challenges the standard assumption that human preferences follow a reward-maximizing distribution, proposing instead that they are better modeled by regret minimization under an optimal policy [24]. By leveraging the principle of maximum entropy, CPL derives a fully off-policy contrastive objective that optimizes policies directly from preferences without ever constructing a reward function. This method is particularly advantageous for high-dimensional and sequential tasks, such as robotics, where the optimization challenges of traditional RL are most acute. Similarly, Inverse Preference Learning (IPL) operates on the insight that for a fixed policy, the Q-function encodes all necessary information about the reward function, rendering the explicit learning of a reward model redundant [25]. IPL is specifically designed for offline preference data, achieving competitive performance on continuous control and robotics benchmarks with significantly fewer parameters and hyperparameters than transformer-based reward models.

The landscape of direct optimization also includes methods that generalize the underlying divergence constraints or incorporate curriculum learning strategies to refine convergence. For example, f -DPO generalizes DPO by incorporating diverse f -divergence constraints, such as Jensen-Shannon or forward KL divergences, allowing for a more flexible balance between alignment performance and generation diversity [123]. Additionally, Curry-DPO enhances alignment by utilizing curriculum learning with ranked preferences, systematically ordering multiple preference pairs from easy to hard to improve training stability and final performance [124]. Other approaches like Statistical Rejection Sampling Optimization (RSO) aim to source preference data closer to the target optimal policy using rejection sampling, thereby providing a more accurate estimation of the optimal policy compared to methods restricted to static datasets [125].

Collectively, these direct and reward-free methods represent a paradigm shift towards more stable and efficient alignment algorithms, serving as a crucial alternative to the fragile two-stage pipelines discussed previously. By removing the intermediate reward modeling step, they mitigate the risks of reward hacking and model misspecification while simplifying the deployment pipeline. Whether through the closed-form derivations of DPO, the regret-based foundations of CPL, or the Q-function equivalence of IPL, these approaches demonstrate that high-quality alignment can be achieved through direct policy optimization. As we transition to the next section, it becomes evident that while these methods reduce algorithmic complexity, the demand for high-quality preference data remains a critical constraint, necessitating advanced strategies for data efficiency, active learning, and synthetic feedback to make these approaches viable for real-world robotic deployment.

3.4 Data Efficiency, Active Learning, and Synthetic Feedback

3.4 Data Efficiency, Active Learning, and Synthetic Feedback

As established in the preceding discussion on direct and reward-free preference optimization, bypassing explicit reward modeling significantly enhances algorithmic stability and simplifies the training pipeline. However, while these methods mitigate the computational fragility of the RLHF loop, they do not resolve the fundamental bottleneck of data acquisition: the prohibitive cost and latency associated with collecting high-quality human preference annotations. Whether employing a two-stage pipeline or a direct optimization framework like DPO or CPL, the performance of the resulting policy remains strictly contingent on the volume and quality of the underlying preference dataset. In real-world robotic applications, where physical interactions are slow, expensive, and potentially hazardous, relying on vast quantities of human-labeled comparisons is often impractical. Consequently, research has pivoted toward strategies that decouple performance gains from the linear scaling of human effort, focusing on maximizing data efficiency through active learning, synthetic feedback generation, and the robust exploitation of noisy, crowd-sourced annotations.

Active learning serves as a primary mechanism for enhancing data efficiency by shifting from passive random sampling to intelligent query selection. The core objective is to identify trajectory pairs or state-action sequences that yield the highest information gain regarding the latent preference distribution. While traditional approaches rely heavily on uncertainty estimation, selecting instances solely based on model uncertainty can be detrimental in complex domains where high uncertainty often correlates with noise rather than informativeness. To counter this, diversity-based query strategies have been introduced to ensure selected batches cover a broad spectrum of the behavior space, thereby improving generalization with fewer annotations [126]. This concept has been formalized within the RLHF paradigm as a contextual dueling bandit problem, giving rise to algorithms like Active Preference Optimization (APO). These methods provide theoretical guarantees on sample efficiency by actively selecting prompts that minimize the sub-optimality gap of the learned policy, achieving parity with standard methods using significantly fewer human queries [127]. In the specific context of robotics, where serial query generation creates a wall-clock time bottleneck, batch active preference-based learning algorithms employ techniques such as Determinantal Point Processes (DPP) to generate diverse, informative query sets simultaneously. This parallelization retains the benefits of active querying while drastically accelerating the data collection cycle [128].

Complementing the strategic selection of human queries, synthetic preference generation has emerged as a powerful paradigm for augmenting limited datasets. By leveraging the policy itself or auxiliary models to simulate human judgment, these techniques reduce reliance on explicit external labeling. A prominent example is the “Best-of-N” sampling strategy, which generates multiple candidate responses for a given prompt and constructs synthetic preference pairs from the best and worst candidates. This self-training mechanism amplifies the signal from a small seed of human data, enabling iterative reward model improvement without additional human intervention [129]. Pushing this further, adversarial preference optimization frameworks establish a min-max game between the policy and the reward model, where the system autonomously generates challenging preference pairs to adapt to shifting output distributions. This creates a self-correcting loop that bridges the distribution gap without continuous external annotation [130]. Additionally, automated methods now synthesize fine-grained attribution preference data from existing corpora, demonstrating that large-scale synthetic pairs can drive state-of-the-art performance in high-fidelity tasks such as attributed text generation [131].

Beyond algorithmic innovations in query selection and data synthesis, scaling data collection also

requires robust strategies for handling diverse and imperfect crowd-sourced feedback. In practical deployments, annotators exhibit varying levels of expertise, fatigue, and subjective interpretation, leading to heterogeneous and sometimes contradictory labels. Rather than filtering out this variability, modern approaches explicitly model annotator performance to incorporate multi-perspective feedback. Annotator-centric active learning strategies, for instance, optimize not only which samples to label but also which specific annotators to query, ensuring the dataset captures a wide diversity of judgments including minority perspectives often lost in majority-vote aggregation [132]. This is critical for subjective tasks where a single ground truth is non-existent. Furthermore, robust optimization formulations have been developed to handle preference elicitation under noise, utilizing de-biased loss functions that maintain policy stability even in the presence of random label flips [122]. Empirical studies reinforce that training on datasets reflecting a range of annotator agreements, rather than enforcing strict consensus, yields more nuanced reward landscapes that better mirror the complexity of human values [133].

Finally, the integration of domain knowledge with active learning offers a pragmatic pathway for specialized robotic applications. By infusing domain-specific constraints or heuristics into the reward modeling process, the dependency on extensive preference annotations can be dramatically reduced. For example, leveraging domain priors in e-commerce opinion summarization has been shown to reduce the required preference dataset size by over twenty-fold while simultaneously boosting performance, illustrating how intelligent priors can substitute for massive data volumes [134]. Similarly, interactive frameworks combining active learning with preference learning enable dynamic, few-shot adaptation, allowing systems to converge to user preferences through minimal targeted interactions [135]. Collectively, these advancements in active query selection, synthetic augmentation, and robust crowd-sourcing form a critical toolkit for overcoming data inefficiency. Yet, as we maximize the utility of available data, a new challenge arises: how to aggregate these diverse, multi-faceted, and potentially conflicting preferences into a coherent objective. Addressing this necessitates a shift from single-objective optimization to multi-objective alignment frameworks, a topic explored in the following section.

3.5 Multi-Objective Alignment and Handling Diverse Preferences

3.5 Multi-Objective Alignment and Handling Diverse Preferences

While previous strategies have focused on maximizing data efficiency and synthesizing feedback to overcome annotation bottlenecks, a fundamental challenge remains in how these diverse and potentially conflicting preferences are aggregated into a usable objective. The prevailing paradigm in Reinforcement Learning from Human Feedback (RLHF) has historically relied on collapsing heterogeneous human opinions into a single, scalar reward model. Although this approach has driven significant advances in aligning policies with general intent, it inherently struggles to capture the pluralistic nature of human values. In real-world robotic and interactive applications, users possess distinct cultural backgrounds, personal tastes, and situational contexts. A monolithic reward function, trained to maximize average utility, inevitably fails to satisfy minority groups or address specific user needs, resulting in a “one-size-fits-all” solution that may be suboptimal or even harmful for significant portions of the population. Consequently, recent research has pivoted from merely acquiring more data to developing multi-objective alignment frameworks that explicitly model, learn, and optimize for diverse preference distributions rather than erasing their variance.

A critical theoretical limitation of standard single-reward RLHF is its inability to equitably represent these diverse preferences. Research has demonstrated an impossibility result wherein aligning

a model to a single aggregated reward function cannot simultaneously satisfy conflicting preference distributions inherent in data collected from multiple users. To address this, equitable alignment strategies have been proposed, such as MaxMin-RLHF, which draws inspiration from the Egalitarian principle in social choice theory. This approach learns a mixture of preference distributions via an expectation-maximization algorithm and optimizes a policy using a MaxMin objective. By focusing on maximizing the minimum utility across different user groups, this method ensures that the performance for minority groups is significantly improved without compromising the majority, thereby achieving a more robust and fair alignment [136]. Similarly, theoretical studies in multi-party RLHF have established that traditional approaches fail when learning a single reward function, necessitating the incorporation of meta-learning to handle multiple preferences and the adoption of various social welfare functions, such as Nash, Utilitarian, and Leximin, to aggregate conflicting viewpoints [112].

Beyond equitable aggregation, another powerful direction involves enabling fine-grained, directional control over model behavior to suit individual user needs. The Directional Preference Alignment (DPA) framework introduces a novel mechanism where user preferences are modeled as directions (unit vectors) within a multi-objective reward space. Unlike scalar RLHF, DPA trains a multi-objective reward model and utilizes a preference-conditioned variant of Rejection Sampling Fine-tuning. This allows users to arithmetically specify their desired trade-offs between competing objectives, such as prioritizing helpfulness while minimizing verbosity, offering intuitive and precise control over the agent’s output [137]. This concept of decomposing preferences into multiple dimensions is further explored in Personalized Soups, which treats alignment as a Multi-Objective Reinforcement Learning (MORL) problem. By training models independently on different personalization dimensions and merging their parameters post-hoc, this approach efficiently achieves personalized alignment that respects individual declared desires [138].

The challenge of handling diverse objectives also extends to the optimization algorithms themselves, aiming to reduce the computational burden often associated with multi-objective settings. Traditional MORL methods often require unstable and computationally expensive reinforcement learning fine-tuning. To mitigate this, Multi-Objective Direct Preference Optimization (MODPO) was developed as an RL-free algorithm that extends Direct Preference Optimization (DPO) to multiple objectives. MODPO folds language modeling directly into reward modeling, training models as implicit collective reward models that can combine objectives with specific weightings. This method not only matches the theoretical guarantees of MORLHF but also offers superior stability and computational efficiency, consistently producing a Pareto front of models that cater to diverse preferences [139]. Furthermore, the “Rewarded Soups” approach embraces reward heterogeneity by fine-tuning multiple networks on diverse proxy rewards and linearly interpolating their weights. This strategy leverages the linear connectivity of fine-tuned weights to achieve Pareto-optimal generalization across the entire space of preferences, proving effective across text, vision, and control tasks [140].

Addressing the complexity of multi-faceted human feedback also requires sophisticated reward modeling architectures capable of disentangling conflicting goals. For instance, hierarchical reward modeling frameworks like ALaRM integrate holistic rewards with aspect-specific rewards to provide more precise guidance, filtering and combining multiple rewards based on consistency to handle sparse and inconsistent supervision signals [141]. Additionally, safe alignment necessitates decoupling conflicting objectives such as helpfulness and harmlessness. Safe RLHF formalizes this by training separate reward and cost models and using Lagrangian methods to dynamically adjust the balance during fine-tuning, ensuring that safety constraints are strictly satisfied while maximizing

performance [142]. The importance of distinguishing between these objectives is underscored by findings that diversified preference data can negatively affect the calibration of reward models on shared preferences, necessitating specialized multi-objective reward learning methods to enhance calibration and overall alignment performance [143].

Finally, the intersection of multi-objective learning and active strategies has led to frameworks like Multi-Objective Reinforced Active Learning (MORAL), which maintains a distribution over scalarization weights to interactively tune agents toward a variety of preferences without computing multiple distinct policies. This is particularly vital in scenarios involving normative conflicts, such as delivery or emergency tasks, where agents must navigate complex ethical landscapes [144]. Collectively, these advancements signify a paradigm shift from seeking a universal optimal policy to developing flexible, multi-objective systems capable of adapting to the rich tapestry of human values. However, the efficacy of these multi-objective frameworks is fundamentally contingent upon the quality, robustness, and transparency of the underlying reward models themselves, a topic that necessitates a deeper examination of reward shaping and interpretability in the following section.

3.6 Advanced Reward Shaping and Interpretability Mechanisms

3.6 Reward Shaping, Interpretability, and Robustness

As established in the preceding discussion on multi-objective alignment, the efficacy of Preference-based Reinforcement Learning (PbRL) and Reinforcement Learning from Human Feedback (RLHF) is fundamentally contingent upon the quality, robustness, and transparency of the underlying reward models. While the previous section highlighted the necessity of modeling diverse preference distributions to avoid “one-size-fits-all” solutions, this section addresses the structural and methodological challenges inherent in constructing these reward signals themselves. The prevailing reliance on deep neural networks as standard architectures for inferring rewards from human preferences introduces a critical bottleneck: their opaque nature often obscures the rationale behind agent behaviors, making it difficult to verify alignment or diagnose failure modes, particularly when aggregating conflicting objectives. To address these challenges, recent research has pivoted towards advanced reward shaping techniques and interpretable reward structures that enhance both the sample efficiency of learning and the trustworthiness of deployed systems.

A primary avenue for improving transparency involves transitioning from black-box neural approximators toward intrinsically interpretable representations. In this context, tree-structured reward functions have emerged as a powerful alternative capable of satisfying the need for explicit logic in multi-objective settings. By learning reward models that possess a compositional, hierarchical structure, researchers can provide clear, traceable explanations for why specific actions or trajectories are preferred over others. The work titled [145] demonstrates that such tree models can achieve performance competitive with deep neural networks on high-dimensional tasks while offering superior robustness to limited or corrupted data. Furthermore, the interpretability inherent in these structures facilitates rigorous verification, allowing designers to inspect the logical conditions governing the reward signal and ensuring that the agent’s objectives align with human intent. This approach is further corroborated by [146], which proposes an online, active preference learning algorithm specifically designed to construct these tree-based rewards, thereby enabling developers to debug alignment issues effectively.

Parallel to structural interpretability, the integration of Large Language Models (LLMs) has revolutionized the automated design and refinement of reward functions, offering a scalable solution for the complex reward engineering required in diverse robotic domains. LLMs, endowed with

extensive world knowledge and reasoning capabilities, serve as automated critics or designers that generate dense, informative reward signals from sparse human feedback or natural language descriptions. For instance, the framework presented in [147] utilizes the critique capability of LLMs to produce intermediate, step-level rewards during training. By coupling a policy model with a critic language model, this method translates comprehensive feedback into token or span-level rewards, significantly mitigating the sparsity issues that plague traditional RLHF pipelines. Similarly, in physical skill learning, [148] proposes a two-component approach where an LLM first proposes features and parameterizations for a reward function, which are then refined through an iterative self-alignment process that minimizes ranking inconsistencies with new environmental observations. This synergy between language models and environment feedback allows for the creation of efficient reward functions without the need for costly expert demonstrations. The automation of reward design is further advanced by systems like [149] and [150], which leverage LLMs to write executable code for dense reward functions. These systems employ iterative loops involving reward designers, critics, and trajectory analyzers to refine reward codes, achieving success rates comparable to or exceeding those of manually engineered rewards in complex manipulation and locomotion tasks.

Beyond automation and structure, ensuring the reliability of learned rewards requires adaptive mechanisms that can balance domain knowledge with data-driven signals while guarding against misspecification and overoptimization—risks that are amplified in multi-objective scenarios. Reward shaping remains a critical technique for incorporating prior knowledge, yet naive application can lead to unintended behaviors if the shaping rewards are misaligned. To address this, adaptive frameworks have been developed to learn how to utilize shaping rewards dynamically. The study [151] formulates this as a bi-level optimization problem, where the system learns a weighting function to adaptively blend shaping rewards with the true objective, effectively ignoring unbeneficial heuristics. This concept is extended in [12], which introduces a framework to learn behavior alignment reward functions that integrate auxiliary heuristics with primary rewards, demonstrating robustness even when provided with poorly specified auxiliary signals. Furthermore, the challenge of distinguishing between accidental success and purposeful behavior in sparse reward settings is addressed by [152], which constructs auxiliary reward functions optimized via meta-learning to provide refined feedback.

The interpretability of these learned rewards is further enhanced by methods that explain reward assignments through counterfactual analysis, providing a diagnostic lens for the complex reward landscapes discussed earlier. [153] proposes generating Counterfactual Trajectory Explanations (CTEs) that contrast original trajectories with counterfactual variants to elucidate the reward landscape, thereby helping users identify flaws in learned reward functions. Additionally, the issue of reward model divergence and the gap between the explicit reward model and the implicit rewards learned by the policy during RLHF is explored in [154]. This work defines the “Implicit Reward Model” (IRM) to characterize the actual objectives pursued by the fine-tuned model, offering a diagnostic tool to minimize discrepancies between training objectives and emergent behaviors. Finally, to ensure that reward models do not overfit to specific biases or fail under distribution shifts, ensemble methods and information-theoretic approaches are employed. [155] and [107] highlight the utility of ensembling to reduce overoptimization, while [115] introduces an information bottleneck objective to filter irrelevant information and detect overoptimization via latent space deviations. Collectively, these advanced mechanisms represent a paradigm shift towards reward systems that are not only high-performing but also transparent, adaptable, and robust against the complexities of real-world deployment, laying the necessary groundwork for the subsequent exploration of offline data utilization and scalability challenges.

4 Synergizing Offline Data with Preference Feedback

4.1 Direct Optimization from Offline Preference Datasets

4.1 Direct Optimization from Offline Preference Datasets

The integration of offline reinforcement learning with preference-based feedback represents a pivotal advancement in robotic learning, particularly for scenarios where active environment interaction is prohibited due to safety constraints, high operational costs, or physical limitations. Traditional Reinforcement Learning from Human Feedback (RLHF) pipelines typically operate in a two-stage fashion: first, a reward model is trained on human preference data, and second, a policy is optimized against this learned reward using online reinforcement learning algorithms such as Proximal Policy Optimization (PPO). While effective in simulation or low-risk domains, this paradigm faces significant hurdles in robotics. The reliance on online rollouts for policy optimization introduces the risk of catastrophic failures during exploration, and the separation between reward learning and policy optimization can lead to reward hacking, where the agent exploits imperfections in the learned reward model rather than aligning with true human intent. To address these challenges, recent research has shifted towards direct optimization methods that bypass the explicit reward modeling stage and eliminate the need for online interaction, enabling robust policy learning directly from static, offline preference datasets. This transition establishes a foundational framework for leveraging historical data, setting the stage for addressing the critical issue of data integrity discussed in the subsequent section.

Direct Preference Optimization (DPO) and its variants have emerged as leading frameworks in this domain, originally gaining traction in large language model alignment before being adapted for continuous control and robotic tasks. The core theoretical insight driving these methods is that the optimal policy can be expressed in closed form as a function of the reward function and a reference policy. By substituting this optimal policy parameterization directly into the Bradley-Terry model of human preferences, the learning objective can be re-formulated as a simple classification loss over preference pairs, effectively treating the policy model itself as an implicit reward model [118]. This reformulation allows for stable, off-policy training using only static datasets, removing the instability and hyperparameter sensitivity associated with actor-critic algorithms. In the context of robotics, where data collection is expensive and dangerous, the ability to learn solely from a fixed buffer of historical trajectories and human comparisons is transformative.

However, the direct application of standard DPO to robotic settings requires careful adaptation to handle the complexities of continuous state-action spaces and the specific nature of robotic preference data. Unlike textual generation, where responses are discrete tokens, robotic behaviors are continuous trajectories that must satisfy dynamic constraints. Contrastive Preference Learning (CPL) addresses some of these limitations by deriving algorithms based on a regret-based model of human preferences rather than the traditional partial return assumption. CPL demonstrates that by leveraging the principle of maximum entropy, one can derive a fully off-policy, contrastive objective that learns optimal policies without ever constructing an explicit reward function or engaging in online reinforcement learning [24]. This approach is particularly advantageous for robotics as it scales elegantly to high-dimensional sequential decision-making problems and avoids the unwieldy optimization challenges inherent in policy gradient methods. Furthermore, CPL’s ability to operate entirely off-policy makes it ideally suited for leveraging diverse, sub-optimal, and expert data often found in static robotic datasets.

The efficacy of direct optimization methods compared to traditional two-stage RLHF approaches

has been a subject of rigorous theoretical and empirical scrutiny. Theoretical analyses suggest that while two-stage methods incur errors from both reward model estimation and subsequent policy optimization, direct methods like DPO can achieve comparable or superior sub-optimality gaps under certain conditions, particularly when the ground-truth reward is not perfectly realizable [156]. In offline settings, the absence of online exploration means that the policy cannot correct for reward model errors through new data collection, making the robustness of the optimization procedure paramount. Direct methods mitigate the risk of distributional shift by constraining the learned policy to remain close to the reference policy (typically the behavior policy that generated the offline data) via implicit regularization. This is critical in robotics, where deviating too far from the data manifold can result in physically impossible or unsafe actions.

Despite their advantages, direct optimization from offline preferences is not without challenges, particularly regarding data quality. One significant issue is the sensitivity to the quality and coverage of the offline dataset; if the preference data does not adequately cover the state space relevant to the task, the learned policy may fail to generalize. To mitigate this, researchers have proposed enhancements such as Inverse Preference Learning (IPL), which exploits the interchangeability between the Q-function and the reward function for a fixed policy. IPL eliminates the need for a learned reward function entirely, offering a parameter-efficient alternative that attains competitive performance on continuous control benchmarks while reducing algorithmic complexity [25]. Additionally, the problem of noisy or corrupted labels in static datasets remains a pressing concern that limits the reliability of these direct methods. Algorithms like RIME incorporate sample selection mechanisms to filter denoised preferences dynamically, ensuring robust training even when human feedback is inconsistent or adversarial [28]. While RIME offers a specific mitigation strategy, the broader challenge of learning from fundamentally corrupted or ambiguous labels necessitates a more comprehensive theoretical treatment, which is the focus of the following section.

Furthermore, the integration of dynamics models into preference-based learning offers a promising hybrid direction for offline settings. While pure direct optimization methods avoid model learning, incorporating a learned dynamics model can enable the synthesis of diverse preference queries and facilitate reward pre-training without any environmental interaction, thereby enhancing sample efficiency and safety [37]. This suggests that while direct optimization provides a powerful baseline, combining it with model-based insights or advanced regularization techniques like those found in Constrained DPO can further improve safety and alignment guarantees [157]. Ultimately, the shift towards direct optimization from offline preference datasets marks a paradigm change in robotic learning, moving away from risky online exploration and fragile reward engineering toward stable, data-efficient, and safe alignment strategies that leverage the full potential of static historical data, even as the field grapples with the inherent imperfections of such data.

4.2 Robustness Against Noisy and Corrupted Human Labels

4.2 Robustness Against Noisy and Corrupted Human Labels

While Section 4.1 established direct optimization as a powerful paradigm for learning from static preference datasets, it also highlighted a critical vulnerability: the sensitivity of these methods to the quality and integrity of the offline data. Specifically, the reliance on historical buffers introduces the risk that policies may overfit to spurious correlations or adversarial manipulations inherent in real-world human feedback. In practical robotic applications, preference data is rarely pristine; it is often aggregated from diverse annotators with varying expertise, susceptible to fatigue, inconsistency, or even malicious corruption. Consequently, ensuring policy stability and

safety when learning from such imperfect buffers is a paramount challenge that bridges the gap between theoretical direct optimization and practical deployment. This section addresses this challenge by exploring theoretical frameworks and algorithmic innovations—ranging from pessimistic confidence sets to advanced uncertainty quantification—designed to fortify learning against corrupted preference data, thereby laying the groundwork for the query-efficient strategies discussed in the subsequent section.

A foundational approach to mitigating the impact of corrupted labels involves the rigorous application of pessimism within the learning framework. The principle of pessimism, long recognized as essential in standard offline RL for handling distributional shift, becomes even more critical when the reward signal itself is derived from potentially faulty human comparisons. When an ε -fraction of trajectory pairs in an offline dataset is corrupted—whether through flipped feedback or manipulated trajectory features—standard RLHF algorithms can fail catastrophically, converging to policies that optimize for the noise rather than the true human intent. To counteract this, novel corruption-robust offline RLHF methods have been proposed that explicitly model the uncertainty introduced by corruption. These methods typically operate by first learning a reward model accompanied by rigorous confidence sets that account for the potential extent of data corruption. Subsequently, a pessimistic optimal policy is learned over these confidence sets, effectively treating the worst-case scenario within the plausible range of reward functions as the target for optimization [158]. This two-stage process ensures that the resulting policy remains near-optimal even in the presence of adversarial attacks, provided that the corruption level is bounded and the data coverage assumptions are met.

The theoretical underpinnings of these robust methods draw heavily upon the broader literature of distributionally robust optimization and corruption-tolerant learning. The challenge of learning from contaminated datasets parallels issues found in standard offline RL where reward signals or transitions may be impure; in such contexts, algorithms relying on data purity exhibit severe performance degradation. To address this, researchers have introduced techniques such as gradient penalties over learned value functions to prevent the explosion of Q-values caused by outliers, alongside constraint relaxation mechanisms that allow the policy to deviate slightly from non-optimal actions identified in the noisy data [52]. When adapted to the preference-based setting, similar principles apply: the learning objective must be regularized to prevent the policy from overfitting to spurious correlations introduced by noisy human labels. Furthermore, the concept of “double pessimism” has emerged as a powerful framework for robust offline RL, addressing both the mismatch between the behavior policy and the target policy, and the perturbation of the nominal model or reward function [159]. In the context of RLHF, this implies being pessimistic not only about the value of out-of-distribution actions but also about the validity of the inferred reward model in regions where human feedback is sparse or contradictory.

Beyond pessimistic optimization, the robustness of RLHF systems can be significantly enhanced through sophisticated uncertainty quantification. Traditional methods often struggle to distinguish between epistemic uncertainty (due to lack of data) and aleatoric uncertainty (due to inherent noise or corruption in the labels). Advanced algorithms leverage ensemble methods to estimate the posterior distribution of value functions or reward models, providing lower confidence bounds that serve as a robust proxy for the true utility of a policy. For example, diverse randomized value functions can be employed to estimate the posterior distribution of Q-values, offering a provably pessimistic approach that fosters reliable value estimation even when the data contains significant noise [18]. Similarly, the use of bootstrapped ensembles allows for the characterization of uncertainty without explicit policy constraints, enabling the algorithm to penalize value estimates based

on the disagreement among ensemble members [160]. When applied to preference data, these techniques allow the agent to identify and down-weight trajectory pairs that yield inconsistent reward predictions, thereby isolating the signal from the noise.

Moreover, the nature of human decision-making itself introduces a layer of complexity that robust algorithms must accommodate. Humans are not optimal decision-makers; they often act under uncertainty and exhibit bounded rationality. Ignoring these characteristics can lead to the misinterpretation of rational responses to uncertain environments as mere noise. Theoretical analyses in inverse decision theory suggest that preference learning can paradoxically become easier when the decision problem involves higher uncertainty, as the variation in choices provides more information about the underlying trade-offs [161]. Robust RLHF frameworks must therefore incorporate models of human behavior, such as the Dynamic Discrete Choice (DDC) model, which accounts for forward-looking behavior and bounded rationality. By integrating such models with pessimistic policy optimization, algorithms like DCPPO can recover human reward functions and learn near-optimal policies even from datasets generated by suboptimal, noisy human agents [162].

Finally, the robustness of these systems is contingent upon the ability to handle heavy-tailed reward distributions and extreme outliers that often arise from erroneous human annotations. Standard mean-based estimators are highly sensitive to such outliers, necessitating the adoption of robust statistical estimators like the median-of-means. Frameworks such as ROAM and ROOM strategically incorporate the median-of-means method to enable straightforward uncertainty estimation and manage heavy-tailed rewards effectively, adhering to the principle of pessimism while ensuring stability against extreme label corruption [163]. Additionally, in scenarios where the dataset may suffer from unobserved confounders or missing observations alongside noise, causal-adjusted pessimistic policies offer a pathway to learn optimal behaviors by forming confidence sets that account for these structural deficiencies [164]. Collectively, these advancements signify a maturing field where the synergy of offline data and preference feedback is no longer limited by the assumption of label purity. While these methods provide robust mechanisms to tolerate existing noise, they do not reduce the cost of acquiring large volumes of data; addressing this economic bottleneck through intelligent sample selection is the focus of the following section.

4.3 Active Elicitation and Query-Efficient Labeling Strategies in Static Buffers

4.3 Active Elicitation and Query-Efficient Labeling Strategies in Static Buffers

While the preceding section established methods for ensuring robustness against noisy and corrupted labels within static datasets, a complementary challenge persists: the prohibitive cost of acquiring high-quality human feedback at scale. Even with robust algorithms capable of tolerating imperfection, the sheer volume of data in offline buffers often far exceeds the limited budget available for human annotation. Consequently, the paradigm must shift from the passive consumption of pre-labeled datasets to active elicitation strategies that intelligently select the most informative trajectory pairs for annotation. The core objective here is to maximize the utility of every human query by identifying samples that offer the highest information gain regarding the underlying reward function or policy optimization, all while operating strictly within the constraints of a fixed, static dataset. This focus on query efficiency serves as a critical bridge between the need for robust data handling discussed previously and the complex reality of heterogeneous, multi-user preferences addressed in the subsequent section.

Traditional active learning frameworks have long sought to reduce annotation costs by selecting uncertain or diverse samples from a pool of unlabeled data. In the realm of classification and regression, strategies such as uncertainty sampling, diversity-based selection, and hybrid approaches have been extensively studied to optimize the trade-off between model performance and labeling effort, including methods that adjust annotation precision to explore larger input spaces [165]. These methods typically rely on acquiring labels for individual instances where the model’s confidence is lowest. However, applying these concepts to offline preference-based reinforcement learning requires a fundamental adaptation. Instead of labeling single states or actions, the agent must select pairs of trajectories to present to a human oracle for relative comparison. This shift necessitates acquisition functions that can estimate the potential reduction in policy regret or reward model uncertainty derived specifically from pairwise feedback, rather than absolute ground truth labels.

A significant challenge in static buffer settings is the “cold-start” problem, where the initial policy or reward model lacks sufficient calibration to identify truly informative queries. In standard active learning, the model iteratively updates based on new labels, but in an offline setting, the selection must be performed in batches or through a simulated loop without real-time environment feedback. Research into cold-start scenarios suggests that leveraging pre-trained representations or self-supervised signals can provide a robust initial estimate of sample importance, such as using language modeling loss as a proxy for uncertainty when labeled data is scarce [166]. For robotic trajectories, this might involve using latent space embeddings derived from unsupervised pre-training on the static buffer to identify regions of the state-action space that are under-represented or exhibit high variance in predicted outcomes. By treating the selection of trajectory pairs as a covering problem, algorithms can aim to maximize the probability coverage of the latent space, ensuring that the annotated preferences span the diverse behaviors present in the offline data [167].

Furthermore, the reliability of human annotations in static buffers cannot be taken for granted, echoing the concerns regarding label corruption discussed in Section 4.2. Human preferences are often noisy, inconsistent, or subjective, particularly when evaluating complex, long-horizon robotic tasks. Effective active elicitation strategies must therefore account for annotator reliability and the potential for label corruption. Methods that jointly estimate worker performance and annotation reliability have shown promise in crowdsourced settings, suggesting that similar mechanisms could be adapted for offline preference learning [168]. By incorporating uncertainty about the annotator’s consistency into the acquisition function, the system can avoid querying pairs that are likely to yield ambiguous or contradictory feedback, thereby preserving the annotation budget for high-confidence, high-impact comparisons. This proactive filtering is essential not only for efficiency but also for mitigating the impact of the very noise sources that robust algorithms are designed to handle.

The concept of “query efficiency” also extends to the computational cost of selecting these pairs. In large-scale robotic datasets, evaluating an acquisition function over all possible trajectory pairs is computationally intractable. Recent advances in active learning for deep networks highlight the importance of efficient approximation methods, such as using adversarial examples to approximate the distance to decision boundaries [169]. In the context of static buffers, techniques like batch active learning formulated through sparse approximation can be employed to select a diverse subset of trajectory pairs that collectively approximate the information content of the entire dataset [170]. This allows for parallel annotation efforts, significantly speeding up the data preparation phase for offline preference optimization.

Moreover, the nature of robotic data often involves high redundancy, where many trajectories represent similar behaviors or successful executions of the same task. Naive sampling strategies may waste annotation budgets on redundant pairs that provide little new information. Studies in se-

mantic segmentation and object detection have demonstrated that accounting for data redundancy and class imbalance is crucial for effective active learning [171]. Translating this to robotics, active elicitation strategies must prioritize trajectory pairs that highlight subtle differences in execution quality, safety violations, or alignment with nuanced human preferences that are not captured by simple success metrics. Techniques that focus on “hard” examples—those where the current policy or reward model struggles to distinguish between trajectories—can drive faster convergence in preference learning, as seen in strategies guided by local sensitivity and hardness [172].

Finally, the emergence of Large Language Models (LLMs) offers a novel avenue for enhancing query efficiency in static buffers. LLMs can serve as synthetic annotators or “critics” to pre-filter trajectory pairs, identifying those that are likely to be confusing or informative for human reviewers [173]. By leveraging the reasoning capabilities of foundation models to simulate human preferences, systems can generate a preliminary ranking of trajectory pairs, allowing human annotators to focus only on the most contentious or critical comparisons. This “human-in-the-loop” approach, augmented by AI-driven pre-screening, represents a promising direction for scaling offline preference-based learning to complex, real-world robotic applications where human time is the ultimate bottleneck [174]. Through the synergistic combination of robust uncertainty estimation, redundancy awareness, and AI-assisted filtering, active elicitation strategies can transform static offline buffers into highly efficient resources. This efficiency is particularly vital when addressing the heterogeneity of multi-user preferences, as discussed in the following section, where distinguishing between genuine preference diversity and mere noise requires careful, targeted querying.

4.4 Handling Heterogeneous and Multi-User Preferences in Offline Settings

4.4 Handling Heterogeneous and Multi-User Preferences in Offline Settings

While Section 4.3 established strategies for intelligently eliciting high-quality labels within a static budget, those methods often presuppose a coherent underlying preference signal. In reality, the “noise” filtered by active learning is not always random error; it frequently stems from the fundamental heterogeneity of human values. Real-world robotic datasets aggregate feedback from diverse operators with conflicting objectives, varying cultural backgrounds, and distinct task-specific priorities. Traditional Reinforcement Learning from Human Feedback (RLHF) paradigms, which typically assume a singular reward function derived from aggregated preferences, fail to capture this plurality. When applied to such heterogeneous offline datasets, these monolithic approaches risk optimizing for a “majority vote” that systematically marginalizes minority viewpoints or produces policies that satisfy no one fully. Addressing this requires a paradigm shift from noise reduction to diversity preservation, employing sophisticated algorithms capable of inferring latent user intent, clustering diverse preference distributions, and optimizing for fairness without the benefit of the active online querying discussed previously.

A primary difficulty in offline multi-user settings is the identifiability of distinct preference sources within a static buffer. Unlike online settings where an agent can actively query specific users to disambiguate intent, offline learning must rely entirely on the structural patterns within the data to separate conflicting signals. Recent theoretical work highlights that learning a single reward function is insufficient for representing diverse human preferences, often leading to an impossibility result where no single policy can satisfy all stakeholders equitably [136]. To overcome this, researchers have begun treating the problem as one of learning a mixture of preference distributions.

By employing expectation-maximization algorithms, it becomes possible to disentangle the latent preferences of different user groups directly from the dataset. This approach allows the system to model the underlying diversity rather than collapsing it into a noisy average, thereby preserving the nuance required for equitable alignment [136].

Clustering preference distributions serves as a critical mechanism for managing this heterogeneity in static datasets. When feedback originates from a crowd of unknown expertise and reliability, simply averaging preferences can be detrimental, especially when error rates vary significantly across the population. The [29] framework demonstrates that robustly aggregating crowd feedback requires estimating the reliability of each user solely from noisy preference comparisons. By identifying clusters of consistent raters and weighing their contributions accordingly, agents can outperform those trained on majority-vote preferences. This is particularly vital in robotics, where a minority of expert operators might provide high-quality feedback that would otherwise be drowned out by a larger pool of non-expert or contradictory annotations. Furthermore, the ability to identify minority viewpoints within the crowd ensures that specialized operational requirements are not ignored during the policy learning phase [29].

Beyond clustering, the aggregation of these diverse preferences into a final policy necessitates a shift from utilitarian maximization to social choice theories that prioritize fairness. In offline settings, where the cost of correcting unfair behaviors post-deployment is high, the selection of the social welfare function is paramount. Provable multi-party RLHF approaches have established sample complexity bounds for optimizing various social welfare functions, such as Nash, Utilitarian, and Leximin welfare, directly from offline preference data [112]. These methods reveal a statistical separation between multi-party and single-party RLHF, indicating that achieving fairness among diverse groups demands more data but yields policies that are robust to conflicting objectives. Specifically, the MaxMin alignment objective, inspired by the Egalitarian principle, seeks to maximize the minimum utility across all user groups, ensuring that the performance for minority groups is not compromised for the sake of the majority [136].

The challenge of heterogeneity is further compounded by the subjective nature of human feedback, which often varies based on socio-demographic factors and individual values. The [175] underscores the importance of mapping sociodemographics to contextual preferences, revealing that alignment norms are heavily dependent on which humans provide the feedback. In an offline robotic context, ignoring these demographic axes can lead to policies that are culturally insensitive or unsafe for specific user groups. Consequently, advanced offline learning frameworks must incorporate mechanisms to infer latent user attributes or intent clusters to apply appropriate fairness constraints. For instance, in kidney exchange markets, algorithms have been developed to learn a distribution over preference functions based on heterogeneous moral preferences, dynamically determining weights to align with community values rather than a static average [176]. This concept translates directly to robotics, where a robot might need to adapt its behavior based on the inferred preferences of the specific human it is collaborating with, even when trained on a heterogeneous dataset.

Moreover, the issue of annotator disagreement is not merely noise to be filtered but often contains meaningful signal regarding the complexity of the task or the subjectivity of the domain. Conventional methods that resolve disagreements via majority voting before training discard valuable information about the spectrum of acceptable behaviors. Approaches like [177] argue for leveraging label distributions rather than single labels, using unsupervised learning to pool similar items and preserve the diversity of opinion. In offline RL, this suggests that policy learning should aim to reproduce the distribution of preferred behaviors observed in the data, rather than converging to a single deterministic mode. This is essential for generating diverse and adaptable robotic behaviors

that can cater to different users’ needs without requiring retraining.

Finally, ensuring fairness in these offline multi-user settings often requires optimizing for specific metrics that balance utility and equity. Methods such as [178] illustrate how adaptive sampling strategies can correct for group biases inherent in the data distribution, a technique that can be adapted for offline preference learning to ensure that underrepresented preference groups are adequately modeled. Similarly, the integration of fairness constraints directly into the optimization objective, as seen in consequentialist approaches to equitable decision-making, allows for the explicit trade-off between overall performance and parity across different operator groups [179]. By synthesizing these techniques—latent intent inference, robust clustering of unreliable crowds, and social-welfare-based optimization—offline preference-based RL can move beyond the limitations of homogeneous assumptions. This progression sets the stage for the critical challenge discussed in the following section: rigorously evaluating these complex, multi-objective policies in static environments where traditional validation metrics may fail to capture the nuances of diverse stakeholder satisfaction.

4.5 Off-Policy Evaluation and Validation Protocols for Preference Policies

4.5 Off-Policy Evaluation and Validation Protocols for Preference Policies

Building upon the complexities of handling heterogeneous and multi-user preferences discussed in the previous section, the deployment of offline preference-based reinforcement learning faces a subsequent, equally formidable challenge: the rigorous evaluation of learned policies without the luxury of online rollouts. In high-stakes robotic domains, such as healthcare automation or autonomous driving, deploying a candidate policy to gather fresh trajectories for validation is often prohibitively expensive, unsafe, or ethically restricted. This constraint is particularly acute when dealing with the diverse stakeholder groups identified earlier; a policy that appears optimal under a single metric may fail specific minority groups, yet verifying these failures requires granular assessment capabilities. Consequently, the community must rely on Off-Policy Evaluation (OPE) techniques capable of estimating the performance of a target policy solely from static datasets generated by different behavior policies. However, when the feedback signal consists of human preferences rather than explicit scalar rewards, and when these preferences reflect a mixture of latent user intents, traditional OPE methodologies face significant adaptations. The core difficulty lies in reconstructing or approximating the value function from pairwise comparisons while accounting for the distributional shift between the behavior policy that collected the data and the target policy being evaluated, all while preserving the nuance of the underlying preference distributions.

A primary avenue for addressing this challenge involves adapting importance sampling (IS) and doubly robust (DR) estimators to the preference setting. Traditional IS methods estimate the target policy’s value by reweighting logged rewards based on the likelihood ratio of actions under the target versus behavior policies. However, in preference-based learning, the reward signal is latent and inferred from relative rankings of trajectory segments, complicating the direct application of these estimators. Recent theoretical advances have begun to bridge this gap by establishing sample complexity guarantees for OPE with preference data. Specifically, researchers have demonstrated that by leveraging fitted-Q-evaluation with deep neural networks, it is possible to learn value functions that generalize efficiently even in high-dimensional state spaces, provided the underlying Markov Decision Process exhibits low-dimensional manifold structures [180]. This work is pivotal as

it provides the first provably efficient guarantees for OPE in Reinforcement Learning from Human Feedback (RLHF), showing that estimation errors can be controlled without suffering from the curse of dimensionality, assuming sufficient smoothness in the reward function.

Beyond direct value estimation, validating improvements against behavior baselines requires robust statistical tools that can handle the noise, sparsity, and subjectivity inherent in human preference labels. One critical issue is that human feedback is often conditioned on multiple underlying factors—such as the diverse socio-demographic backgrounds highlighted in Section 4.4—and is available only sporadically, making the extrapolation of accurate OPE estimations challenging. To resolve this, frameworks such as OPE for Human Feedback (OPEHF) have been introduced. These approaches reconstruct immediate human rewards by regularizing the estimation with environmental knowledge distilled into a latent space, which captures the dynamics of state transitions alongside the issuance of preference signals [181]. By aligning the latent representation of the environment with the sparse preference data, these methods significantly improve the accuracy of performance estimation compared to directly applying standard OPE variants that assume dense, agent-defined rewards.

Furthermore, the reliability of any OPE protocol depends heavily on the quality of the behavior policy model used for weighting, a factor that becomes increasingly critical in heterogeneous datasets. In scenarios where the true behavior policy is unknown—a common occurrence in datasets collected from diverse human demonstrators, crowd-sourced annotators, or legacy systems—the calibration of the estimated behavior policy becomes paramount. Empirical studies indicate that powerful parametric models, such as deep neural networks, may produce highly uncalibrated probability estimates, leading to biased importance weights and erroneous policy evaluations. Conversely, simpler non-parametric approaches, such as k-nearest neighbors, can sometimes yield better-calibrated behavior policy estimates, thereby producing superior importance sampling-based evaluations [182]. This insight suggests that for preference-based offline RL, where data heterogeneity is high and user clusters may exhibit distinct behavioral modes, careful consideration of behavior policy estimation techniques is essential to avoid invalidating the evaluation process.

To further enhance the robustness of validation protocols, researchers have explored the use of confidence intervals and pessimistic bounds to quantify uncertainty in policy value estimates. Given the potential for dataset shifts, model misspecification, and the inherent variance in inferred reward landscapes, providing a point estimate of performance is often insufficient for safety-critical deployment. Methods leveraging distributionally robust optimization have been proposed to generate robust and optimistic cumulative reward estimates that serve as confidence bounds with non-asymptotic guarantees [183]. Additionally, primal-dual optimization approaches utilizing kernel Bellman losses have been developed to construct non-asymptotic confidence intervals for infinite-horizon evaluation, making minimal assumptions about the data generating process [184]. These statistical safeguards are particularly relevant for preference policies, where the reward landscape is inferred from subjective comparisons and thus subject to higher variance than environments with ground-truth rewards.

Another sophisticated direction involves the decomposition of returns to isolate the agent’s contribution from stochastic environmental factors, a concept known as “skill vs. luck.” By understanding the advantage function as the causal effect of an action on the return, it is possible to decompose trajectory returns into parts caused by the agent’s skill and parts due to luck. This decomposition enables the extension of Direct Advantage Estimation to off-policy settings without relying on high-variance importance sampling, offering a more stable pathway for evaluating policy improvements in noisy preference datasets [185]. Moreover, when dealing with large action spaces common in

robotics, where positivity violations (actions taken by the target policy but not the behavior policy) are frequent, techniques such as policy convolution or learning action embeddings can mitigate variance by leveraging latent structures within the action space [186; 187].

Finally, the validity of offline evaluation protocols must be tested against comprehensive benchmarks that reflect the complexity of real-world robotic tasks and the diversity of human preferences. The lack of unified benchmarks has historically hindered the comparison of OPE methods, but recent efforts have established collections of policies and datasets specifically designed to challenge the limits of existing evaluation techniques [188]. These benchmarks facilitate the assessment of how well different estimators perform under varying degrees of distribution shift and data coverage, which is crucial for validating preference-based policies before any real-world deployment. By synthesizing advanced estimators, robust confidence bounding, and calibrated behavior modeling, the field is moving toward validation protocols that can confidently certify the safety and efficacy of robots learned from static preference data, effectively closing the loop between offline training and safe deployment.

5 Hybrid Strategies: Offline Pre-training, Online Fine-Tuning, and Sim-to-Real Transfer

5.1 Stabilizing the Transition: Adaptive Regularization and Policy Expansion

5.1 Stabilizing the Transition: Adaptive Regularization and Policy Expansion

While Section 5.2 delves into strategic exploration driven by uncertainty quantification, the foundational challenge in offline-to-online reinforcement learning lies first in stabilizing the transition itself to prevent catastrophic performance degradation. This phase represents a critical juncture in robotic deployment: offline pre-training provides a robust initialization from static datasets, yet policies confined strictly to the behavior policy’s support often plateau at sub-optimal levels. Conversely, naive fine-tuning via standard online RL algorithms frequently triggers “policy collapse,” where performance deteriorates rapidly due to the sudden distributional shift between high-quality, constrained offline data and the noisy, out-of-distribution samples generated by early exploration. To mitigate these risks before engaging in active strategic exploration, recent research has prioritized stabilizing mechanisms through adaptive regularization and novel policy expansion architectures that preserve beneficial pre-trained behaviors while enabling safe capability growth.

A primary strategy for preventing performance collapse involves the dynamic adjustment of behavior cloning (BC) constraints during the fine-tuning phase. Traditional methods often employ a fixed BC loss to regularize the policy, ensuring it remains close to the data manifold. However, a static constraint can be overly restrictive, hindering the discovery of superior actions outside the original dataset, or insufficient if relaxed too prematurely. To address this, adaptive approaches modulate the weight of the BC loss based on real-time metrics of agent performance and training stability. For instance, the method proposed in [189] introduces a mechanism to adaptively weigh the behavior cloning loss during online fine-tuning. By monitoring the agent’s return and the stability of value estimates, this approach automatically reduces the regularization strength as the agent demonstrates competence, thereby allowing for greater exploration while preventing the initial divergence that leads to collapse. Similarly, [190] explores the concept of tipping the balance between RL and BC objectives over time. The authors demonstrate that continuing to train a policy offline while gradually reducing the influence of the BC component yields refined policies

that are robust enough to withstand the transition to online interaction, effectively matching or exceeding the performance of more complex alternatives.

Beyond simple weight adjustment, iterative refinement of the regularization target offers a more sophisticated means of stabilization. Rather than anchoring the learning policy to the static behavior policy throughout training, iterative methods update the reference policy to reflect the agent’s improving capabilities. The framework presented in [191] enhances behavior regularization by iteratively updating the reference policy used for constraints. This approach, grounded in conservative policy iteration, ensures that the policy improves gradually while implicitly avoiding queries of out-of-sample actions that could cause learning failures. By evolving the regularization target, the algorithm maintains a “safe corridor” for improvement that expands as the agent learns, theoretically guaranteeing convergence to the optimal policy within the dataset’s coverage. Complementing this, [35] proposes a trust-region-style update where the regularization term evolves iteratively. PROTO bridges offline pre-training and standard off-policy RL by gradually relaxing the constraint strength, ensuring stable initial fine-tuning and optimal final performance with minimal computational overhead.

An alternative paradigm to modifying loss functions is the architectural innovation of policy expansion. Instead of forcing a single policy network to adapt from an offline distribution to an online one—a process prone to interference—policy expansion schemes introduce additional policy components dedicated to online learning. The method described in [192] exemplifies this by maintaining the pre-trained offline policy as a fixed candidate within a policy set and introducing a separate, expandable policy responsible for further learning. These policies are composed adaptively during environment interaction. This architecture ensures that the useful behaviors encoded in the offline policy are fully retained and never destroyed by the stochastic gradients of early online updates, while the new policy component is free to capture novel, high-reward behaviors discovered through exploration. This decoupling effectively mitigates the risk of catastrophic forgetting and allows the offline policy to participate in exploration naturally.

Furthermore, the stability of the transition is heavily dependent on how uncertainty is managed during the shift in data distribution, serving as a conceptual bridge to the uncertainty-aware strategies discussed in the following section. Techniques that leverage ensemble methods to quantify uncertainty can guide the relaxation of constraints. As noted in [193], incorporating behavioral cloning into policy updates allows for sufficient penalization of uncertain actions even with smaller ensemble sizes. This balance facilitates stable online fine-tuning by preventing severe performance drops when the agent encounters states where the value estimates have high variance. Additionally, [194] proposes the RO2O algorithm, which incorporates Q-ensembles for uncertainty penalties and adversarial samples for policy smoothness. This combination enables a consistent learning procedure during online adaptation, theoretically providing tighter optimality bounds against distribution shifts.

Finally, the decoupling of exploration and exploitation policies offers another robust solution to stabilize the learning trajectory. The Offline-to-Online-to-Offline (OOO) framework introduced in [195] utilizes an optimistic policy for data collection and a separate pessimistic policy trained on all observed data for evaluation. This decoupling reduces bias from online interactions in the final evaluation policy and allows for more aggressive exploration behaviors during the fine-tuning phase, which in turn generates higher quality data for the exploitation policy. By synthesizing these adaptive regularization techniques, iterative refinement strategies, and architectural expansions, the robotics community establishes a stable foundation upon which more advanced, uncertainty-driven exploration strategies can be built.

5.2 Strategic Exploration and Uncertainty-Aware Fine-Tuning

5.2 Strategic Exploration and Uncertainty-Aware Fine-Tuning

Building upon the stabilization mechanisms and adaptive regularization strategies established in Section 5.1, which secure the initial transition from offline pre-training to online interaction, this section addresses the subsequent challenge of strategic exploration. While Section 5.1 focuses on preventing catastrophic collapse by maintaining a “safe corridor” through behavior cloning constraints and policy expansion, Section 5.2 explores how agents can actively and safely venture beyond the support of the static dataset. The core objective here shifts from mere survival to efficient capability growth: guiding the agent toward high-value, out-of-distribution (OOD) regions that promise policy improvement without destabilizing the learning process through unchecked extrapolation errors. Strategic exploration and uncertainty-aware fine-tuning have emerged as the dominant paradigms to address this delicate balance, utilizing sophisticated mechanisms to quantify epistemic uncertainty and adaptively regulate the exploration-exploitation trade-off.

A fundamental approach to strategic exploration in this hybrid setting involves the use of ensemble-based uncertainty quantification to distinguish between known and unknown states. By maintaining an ensemble of Q-functions or dynamics models, agents can estimate the variance of their predictions, which serves as a proxy for epistemic uncertainty. This uncertainty signal is pivotal for directing exploration; actions leading to states with high prediction variance are prioritized to reduce model ignorance, provided they also exhibit high potential value. For instance, the concept of using ensembles to drive exploration is well-established, where algorithms leverage the disagreement among ensemble members to construct upper confidence bounds or intrinsic rewards [196]. However, in the context of offline-to-online transfer, naive application of such strategies can be detrimental if the uncertainty estimates are not calibrated against the specific constraints of the pre-trained policy. Recent advancements have shown that simply increasing the number of Q-networks within an ensemble, when combined with clipped Q-learning, can effectively penalize OOD data points and foster a more robust policy update mechanism [197]. Furthermore, the independence of ensemble members is crucial; shared pessimistic targets can paradoxically lead to optimistic value estimates, whereas independently trained networks provide more reliable uncertainty quantification necessary for safe fine-tuning [198].

To operationalize these uncertainty estimates during fine-tuning, several frameworks propose adaptive mechanisms that smoothly bridge the conservative nature of offline learning with the aggressive requirements of online exploration. A notable strategy involves state-adaptive balancing, where the learning objective dynamically shifts based on the estimated uncertainty of the current state-action pair. The Simple Unified uNcertainty-Guided (SUNG) framework exemplifies this by applying conservative offline RL objectives to high-uncertainty samples to prevent extrapolation errors, while simultaneously employing standard online RL objectives for low-uncertainty samples to facilitate efficient exploitation [199]. This dual-objective approach ensures that the agent does not blindly trust its value estimates in uncharted territories, thereby mitigating the risk of performance collapse during the initial stages of online interaction. Similarly, the Robust Offline-to-Online (RO2O) algorithm enhances this stability by incorporating adversarial samples to enforce policy and value smoothness, theoretically demonstrating that such regularization leads to tighter optimality bounds under distribution shift [194].

Beyond static uncertainty penalties and adaptive loss weighting, planning-based exploration offers a proactive method for navigating the state space. Instead of relying solely on myopic uncertainty bonuses, agents can perform look-ahead simulations using learned dynamics models to identify tra-

jectories that maximize information gain or expected return in high-reward regions. The “Planning to Go Out-of-Distribution” (PTGOOD) algorithm illustrates this by framing the fine-tuning process as a non-myopic planning problem. PTGOOD targets exploration in regions that are unlikely to be visited by the behavior policy but possess high reward potential, utilizing concepts from the Conditional Entropy Bottleneck to ensure that collected online data provides novel information relevant to the final deployment policy [200]. This approach avoids the instability often introduced by modifying reward functions with intrinsic bonuses, offering a more principled path toward policy expansion. Additionally, model-based approaches like MUSBO (Model-based Uncertainty regularized and Sample Efficient Batch Optimization) further refine this by quantifying uncertainty to bootstrap policy updates, aggressively updating policies in low-uncertainty regions while downweighting updates in high-uncertainty areas to maintain safety during limited deployment windows [201].

The efficacy of these strategies also depends on the efficient management of computational resources and the diversity of the uncertainty estimates. Traditional deep ensembles can be computationally prohibitive, prompting research into diversified randomized value functions that achieve provable pessimism with fewer networks [18]. Moreover, the integration of uncertainty quantification into the sampling strategy itself, such as in the MEET framework, allows agents to adaptively sample transitions that balance exploration and exploitation based on Q-value uncertainty, leading to faster convergence [202]. In continuous control domains, methods like Wasserstein Actor-Critic (WAC) employ approximate Q-posteriors and Wasserstein barycenters to propagate epistemic uncertainty across the state-action space, enforcing exploration in a principled manner that accounts for the continuous nature of robotic action spaces [203].

Ultimately, successful offline-to-online fine-tuning in robotics requires a symbiotic relationship between uncertainty estimation and strategic planning. By leveraging ensemble discrepancies to identify epistemic gaps and employing adaptive regularization to constrain policy updates, modern algorithms enable robots to safely expand their capabilities beyond the confines of static datasets. These uncertainty-aware fine-tuning strategies not only prevent the destabilization common in pure online RL but also ensure that every online interaction yields maximal informational value. While this section has detailed how to navigate the uncertainty of online data distribution, a distinct yet parallel challenge arises when the source of prior knowledge is a simulation rather than real-world data. The following section, 5.3, addresses this “sim-to-real” gap, exploring how dynamics-aware frameworks bridge the discrepancy between simulated training environments and physical deployment, extending the principles of robust transfer to scenarios where the domain itself shifts.

5.3 Bridging the Reality Gap: Dynamics-Aware Sim-to-Real Transfer

5.3 Bridging the Reality Gap: Dynamics-Aware Sim-to-Real Transfer

While Section 5.2 established how uncertainty-aware fine-tuning guides agents from static datasets into unknown online regions, a parallel and equally critical challenge arises when the source of prior knowledge is a simulation rather than real-world data. The transition of reinforcement learning policies from high-fidelity simulations to physical robotic systems—known as the “sim-to-real” gap—remains a persistent bottleneck in deploying data-efficient algorithms. This discrepancy stems from fundamental mismatches in dynamics, sensor noise, and actuator latency between the simulated source domain and the real-world target. Whereas traditional domain randomization served as a foundational technique for robustifying policies by blindly varying parameters, recent advancements have shifted toward sophisticated, dynamics-aware frameworks. These approaches explicitly model, penalize, or adapt to discrepancies, leveraging hybrid offline-online strategies, kinematic parameter

adaptation, and value-guided filtering to ensure safe and effective transfer.

A critical evolution in this domain involves refining the scope and mechanism of domain randomization itself. Contrary to the conventional wisdom of carefully measuring kinematic parameters while randomizing only dynamic ones, empirical evidence suggests that virtually randomizing kinematic parameters, such as link lengths, during simulation training often yields superior transferability [204]. This finding underscores the importance of kinematic uncertainty in bridging the reality gap, prompting the development of algorithms like Multi-Policy Bayesian Optimization, which trains ensembles of universal policies conditioned on virtual kinematic parameters to efficiently adapt to target environments with limited rollouts [204]. Furthermore, theoretical frameworks have been established to explain the efficacy of domain randomization, modeling the simulator as a set of Markov Decision Processes (MDPs) with tunable parameters and proving that memory-dependent policies can succeed under mild conditions without real-world training samples [205]. However, static randomization distributions are often suboptimal; adaptive methods like Bayesian Domain Randomization (BayRn) utilize Bayesian optimization to search the space of source domain distribution parameters, dynamically adjusting them based on sparse real-world data to maximize target domain performance [206].

To address the limitations of purely simulation-based training and better align with the hybrid paradigms discussed previously, researchers have developed frameworks that combine the safety and abundance of offline simulation data with the fidelity of limited online real-world interactions. The Dynamics-Aware Hybrid Offline-and-Online Reinforcement Learning (H2O) framework exemplifies this approach by introducing a dynamics-aware policy evaluation scheme [207]. H2O adaptively penalizes Q-function learning on simulated state-action pairs that exhibit large dynamics gaps relative to real-world data, thereby preventing the agent from overfitting to imperfect simulator dynamics while still leveraging the vast offline dataset [207]. An improved iteration, H2O+, offers greater flexibility in bridging various offline and online learning choices, explicitly accounting for dynamics gaps to achieve superior performance in complex real-world robotics tasks [208]. Complementary to this, the Value-Guided Data Filtering (VGDF) algorithm addresses cross-domain policy adaptation by selectively sharing transitions from the source domain based on the proximity of paired value targets across domains, effectively filtering out source data that would mislead the policy due to dynamics mismatches [209].

Beyond hybrid learning, specific mechanisms for reward augmentation and model correction have been developed to enhance robustness against local misspecification. Augmented World Models (AugWM) tackle the issue of changing dynamics during transfer by augmenting a learned dynamics model with simple transformations that capture potential changes in physical properties [210]. By providing the sampled augmentation as context, the policy can adapt to environmental changes in a zero-shot manner, significantly improving generalization across hundreds of altered dynamics settings [210]. Similarly, the concept of “pessimism”—a theme resonant with the uncertainty quantification in Section 5.2—has been integrated into model-based offline RL to handle local misspecification, where safe policy improvement theorems are established by jointly selecting dynamics models and policies that accurately represent the visited state-action pairs [211]. In scenarios where the source domain lacks full support for the target dynamics, novel approaches focus on skewing and extending the source support towards the target, moving away from the stringent requirement that the simulator must encompass all possible target transitions [212].

Task-driven adaptation represents another frontier, where the goal shifts from matching global dynamics to optimizing specific task performance. The AdaptSim framework meta-learns an adaptation policy to adjust simulation parameter distributions iteratively based on real-world task

performance, rather than attempting to globally align simulator dynamics with reality [213]. This task-centric approach has demonstrated significant improvements in data efficiency and asymptotic performance compared to traditional system identification methods [213]. Additionally, causal discovery methods like COMPASS automate the parameter-tuning process by learning a differentiable mapping between environment parameters and the sim-to-real gap, pruned by a learned causal graph to improve interpretability and generalization [214].

Finally, ensuring safety during this transfer is paramount, mirroring the safety concerns addressed in uncertainty-aware fine-tuning. Frameworks such as SafeAPT leverage diverse repertoires of policies evolved in simulation, using probabilistic reward and safety models to select the most promising safe policy for real-world episodic interaction [215]. Moreover, multiplicative controller fusion approaches allow learned policies to fall back on algorithmic priors in uncertain states, providing a reliable strategy for safe sim-to-real transfer without fine-tuning [216]. Collectively, these dynamics-aware strategies represent a paradigm shift from treating the simulator as a perfect proxy to viewing it as an imperfect, adaptable prior that must be rigorously corrected and constrained by real-world dynamics. Having established how to bridge the gap between simulation and reality, the subsequent challenge lies in executing these adaptations efficiently across distributed robotic fleets, which is the focus of the following section on federated and parameter-efficient strategies.

5.4 Efficiency in Adaptation: Federated, Compression-Aware, and Parameter-Efficient Strategies

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Having established how to bridge the dynamics gap between simulation and reality through adaptive priors and hybrid learning, the subsequent challenge lies in executing these adaptations efficiently across distributed robotic fleets. While Section 5.3 addressed the *fidelity* of transfer by correcting model mismatches, this section focuses on the *feasibility* of deployment, tackling the severe computational, communication, and energy constraints inherent to edge robotics. The transition from offline pre-training to online fine-tuning is frequently hindered by the necessity for low-latency responsiveness and data privacy, which often precludes reliance on massive cloud resources during the adaptation phase. Consequently, efficient adaptation strategies have become paramount, evolving into a triad of cloud-edge architectures, compression-aware mechanisms, and parameter-efficient fine-tuning (PEFT) to mitigate the burdens of continuous learning in real-world settings.

A primary architectural solution to these constraints involves the strategic distribution of compute, storage, and networking resources between the cloud and the edge, a paradigm often referred to as “Fog Robotics.” In this setup, deep models are initially trained on public synthetic data in the cloud and subsequently adapted to private real-world environments at the edge. This hierarchical approach significantly reduces inference cycle times and improves performance in time-critical tasks such as object recognition and grasp planning, leveraging powerful server-side resources for heavy lifting while maintaining the agility required for real-time robotic control [217].

To further alleviate the communication bottlenecks associated with transmitting large model updates between edge robots and central servers—a critical hurdle in scaling the fleet-level adaptation discussed above—compression-aware strategies have gained significant traction. These methods aim to reduce the size of transmitted parameters without compromising the fidelity of the learned policy. For instance, error-bounded lossy compression techniques, such as those utilized in the FedSZ

framework, can compress model states by over an order of magnitude while maintaining inference accuracy within a negligible margin of error, effectively turning communication constraints into a manageable variable [218]. Similarly, adaptive compression methods that dynamically adjust compression rates based on the error induced during training have been proposed to optimize the trade-off between communication efficiency and model utility without requiring manual tuning [219]. In the context of split learning, where models are partitioned between devices and servers, adaptive compression-aware methods like ‘deprune’ have demonstrated the ability to reduce network usage by up to four times compared to standard split learning approaches, making them ideal for bandwidth-constrained robotic networks [220]. Furthermore, joint optimization of compression ratios and transmission deadlines has been shown to accelerate federated edge learning by nearly thirty times compared to vanilla algorithms, addressing both the straggler effect and bandwidth limitations [221].

Beyond compression, parameter-efficient fine-tuning (PEFT) strategies are essential for adapting large foundation models on resource-constrained robotic hardware, complementing the dynamics-aware adaptations of the previous section with computational frugality. Traditional full-model retraining is often infeasible on edge devices; thus, methods that freeze the majority of pre-trained weights and only update a small subset of parameters are critical. The FedCLIP method exemplifies this by designing attention-based adapters for large vision-language models, allowing for fast generalization and personalization in federated settings while reducing computational and communication costs by hundreds of times compared to full fine-tuning [222]. Similarly, the REFT framework employs variable pruning and knowledge distillation to accommodate heterogeneous client devices, enabling efficient training in environments where devices possess varying computational capabilities [223]. Another effective approach involves training only a random subset of model layers while freezing the rest, a strategy that has been empirically shown to reduce data transmission by over 75% without harming global model accuracy, thereby efficiently utilizing on-device resources [224].

Federated transfer learning represents another cornerstone for privacy-preserving robot networks, enabling fleets of robots to share knowledge without exposing sensitive sensory data, a requirement that becomes increasingly stringent as robots operate in diverse user environments. This is particularly relevant for social robots and mobile manipulators that must adapt to diverse user preferences and environmental dynamics. The concept of “Elastic Transfer” combines federated learning with continual learning to preserve relevant parameters across interactions, facilitating decentralized adaptation while mitigating catastrophic forgetting—a precursor to the stability challenges explored in the following section [225]. To address the heterogeneity of data distributions across different robots, semi-decentralized frameworks like FedSR utilize incremental subgradient optimization within ring clusters to improve generalization and alleviate cloud communication bottlenecks [226]. Moreover, sophisticated client selection mechanisms, such as those based on Deep-Q-Reinforcement Learning and spectral clustering, can identify the most informative subsets of devices for participation in each communication round, thereby accelerating convergence and reducing the total number of required rounds [227].

Finally, the integration of reinforcement learning for orchestrating these efficiency mechanisms offers a dynamic solution to the complex trade-offs between latency, energy, and accuracy. Online learning frameworks can learn optimal offloading policies and model selection strategies in end-edge-cloud systems, achieving significant speedups in response time while adhering to strict accuracy requirements [228]. Additionally, sample-driven control mechanisms built on soft actor-critic algorithms can dynamically adapt sampling processes in real-time sensing applications to minimize overfitting and maximize energy efficiency [229]. By synergizing cloud-edge architectures, advanced

compression techniques, parameter-efficient adapters, and federated protocols, the robotics community can achieve rapid, safe, and scalable online adaptation. These efficiency-focused strategies provide the necessary infrastructure to support the next evolutionary step: maintaining performance in non-stationary environments where dynamics shift unpredictably and labels remain scarce, as detailed in the following section.

5.5 Continual Adaptation in Non-Stationary and Label-Scarce Environments

5.5 Continual Adaptation in Non-Stationary and Label-Scarce Environments

Building upon the efficient adaptation architectures and parameter-efficient strategies discussed in the previous section, the deployment of robotic systems in real-world scenarios introduces the compounded challenge of non-stationary environments where dynamics shift unpredictably and human feedback remains sparse or costly. While Section 5.4 addressed *how* to adapt efficiently under resource constraints, this section focuses on *what* to adapt to when ground truth is scarce and distributions drift over time. Maintaining high performance under these conditions requires a paradigm shift from static offline training to dynamic, lifelong adaptation strategies that balance plasticity for new tasks with stability for previously acquired skills. A primary obstacle in this domain is catastrophic forgetting, where the acquisition of new knowledge leads to the rapid degradation of performance on prior tasks. To address this, recent research has increasingly focused on online active continual adaptation, self-training mechanisms augmented by active querying, and sophisticated test-time adaptation (TTA) protocols designed to mitigate forgetting without requiring extensive labeled datasets.

In label-scarce environments, the inability to rely on dense reward signals or ground-truth annotations necessitates the development of online learning frameworks that can leverage unlabeled data streams while selectively querying human experts only when necessary. The framework of Online Active Continual Adaptation addresses this by combining online self-training with active label queries to track changing distributions efficiently. For instance, methods utilizing an online teacher-student structure have been proposed to enable self-training from unlabeled data, employing margin-based criteria to decide when to query limited labels, thereby achieving favorable regret bounds even in separable cases with shifting domains [230]. This approach is particularly relevant for robotics, where continuous interaction with a changing physical world generates vast amounts of unlabeled sensory data, but human intervention for correction or preference labeling must be minimized. Furthermore, in scenarios involving sequential tasks without explicit boundaries, active continual learning strategies have revealed critical trade-offs between retaining old knowledge and acquiring new capabilities, suggesting that query strategies conditioned on previous annotations can significantly improve performance in domain-incremental settings [231].

Beyond active querying, self-training mechanisms face the dual threats of error accumulation and catastrophic forgetting when pseudo-labels become unreliable due to distribution shifts. To counteract these issues, advanced TTA methods have been developed to ensure long-term stability. For example, approaches that incorporate weight-averaged and augmentation-averaged predictions have shown efficacy in reducing error accumulation caused by noisy pseudo-labels in continually changing target domains [232]. Additionally, the integration of small memory buffers into self-training frameworks has proven essential for preserving source model knowledge while adapting to dynamically changing, temporally correlated data streams, a capability crucial for robots operat-

ing in unstructured environments [233]. The challenge is further compounded in universal TTA settings, where models must handle varying sequence lengths and multiple domain shifts without succumbing to model bias or trivial solutions; here, techniques such as diversity weighting and adaptive prior correction have been introduced to maintain generalization and prevent the severe performance degradation often seen in in-distribution data after adaptation [234].

To specifically mitigate catastrophic forgetting in resource-constrained robotic platforms, researchers have explored parameter-efficient adaptation strategies that avoid full model updates, echoing the efficiency themes of Section 5.4 but applied to temporal shifts. Distribution-aware tuning methods have demonstrated that adaptively selecting and updating small groups of parameters—distinguishing between domain-specific and task-relevant parameters—can effectively mitigate error accumulation while avoiding the forgetting associated with updating the entire network [235]. Similarly, memory-efficient approaches utilizing lightweight meta-networks and self-distilled regularization allow frozen original networks to adapt to target domains with significantly reduced memory footprints, preventing the deviation of outputs from well-trained source knowledge [236]. These methods are complemented by layer-wise auto-weighting algorithms that leverage Fisher Information Matrices to identify which layers require preservation versus concentrated adaptation, thereby minimizing forgetting in non-stationary settings while reducing computational loads [237].

The intersection of continual learning and reinforcement learning presents unique challenges, particularly regarding the stability of optimization trajectories when replaying past experiences. While experience replay is a standard technique, it can lead to unstable updates in online settings; consequently, modifications to the optimization geometry, such as Layerwise Proximal Replay, have been proposed to balance learning from new and replay data by allowing only gradual changes in hidden activations [238]. Moreover, in multi-modal large language models used for robot instruction tuning, catastrophic forgetting remains a significant hurdle. Strategies such as projecting prompt gradients to residual spaces have been developed to minimize interference between tasks, enabling positive forward transfer and ensuring that new instructions do not degrade performance on pre-trained capabilities [239].

Finally, the robustness of these adaptation mechanisms against adversarial perturbations and the susceptibility of continually learned tasks to security threats remain open concerns. Studies have indicated that continually learned tasks, regardless of the specific incremental setting, can be highly susceptible to adversarial attacks, raising profound questions about the safety of deploying such systems in critical real-world applications without additional verification layers [240]. Therefore, future robotic systems must integrate these adaptive learning mechanisms with rigorous safety constraints and verification protocols. Addressing these stability and security challenges is a prerequisite for the next evolutionary step: achieving fully autonomous reward discovery and reset-free learning, where robots operate with minimal human supervision not only in adapting to shifts but in defining their own objectives and recovering from failures, as explored in the following section.

5.6 Autonomous Reward Discovery and Reset-Free Real-World Learning

5.6 Autonomous Reward Discovery and Reset-Free Real-World Learning

Building on the stability and security mechanisms discussed in Section 5.5, which enable robots to adapt to non-stationary environments while mitigating catastrophic forgetting, the field now confronts the final frontier of true autonomy: eliminating the dependency on hand-crafted reward functions and human-mediated environment resets. While the previous section addressed *how*

to maintain performance amidst distribution shifts and sparse labels, this section focuses on *what* drives learning when explicit reward signals are absent and episodic resets are infeasible. Traditional reinforcement learning (RL) paradigms rely on the luxury of instant returns to initial states and precisely engineered scalar rewards, assumptions that crumble in physical deployments where resets are time-consuming, dangerous, or impossible without human aid. To achieve fully autonomous deployment, recent research has pivoted toward synergistic frameworks that integrate foundation models for autonomous reward discovery and develop robust protocols for reset-free, continual learning. These approaches aim to minimize human supervision not only during data collection but throughout the entire lifecycle of continuous adaptation and skill acquisition.

A primary avenue for reducing human intervention is the replacement of manual reward engineering with autonomous reward generation leveraging large-scale foundation models. In complex manipulation tasks, defining a scalar reward function that captures nuanced success criteria is often intractable. Recent systems address this by utilizing pre-trained Vision-Language Models (VLMs) to act as dense reward classifiers during the online fine-tuning process. For instance, the RoboFuME system demonstrates a reset-free fine-tuning architecture that leverages pre-trained VLMs to autonomously provide reward signals, allowing a robot to self-improve on a target task with minimal human input [33]. By grounding reward signals in the semantic understanding of VLMs, agents can evaluate their own performance against natural language descriptions of goals, effectively bypassing the need for domain experts to engineer reward shapes. Similarly, the SALMON framework introduces the concept of instructable reward models trained on synthetic preference data derived from human-defined principles, enabling alignment with minimal human supervision [241]. This paradigm shifts the burden from providing thousands of labeled trajectories to defining high-level principles, which foundation models can then interpret to generate dense feedback signals for policy optimization.

Parallel to autonomous reward discovery is the critical challenge of reset-free learning, where agents must operate in non-episodic, continual environments. The inability to reset the environment necessitates algorithms that can recover from failures and sequence tasks such that the completion of one task naturally sets up the initial conditions for another. Multi-task reinforcement learning has emerged as a powerful strategy to solve this “reset problem.” The key insight is that in a sufficiently diverse repertoire of tasks, the terminal state of one task often serves as a valid starting state for another, effectively allowing the agent to perform its own resets. This approach is exemplified by systems that leverage multi-task RL bootstrapped with prior data to enable continuous autonomous practicing, minimizing the number of external resets needed while learning temporally extended behaviors [242]. Furthermore, research has shown that appropriately chosen multi-task settings can directly alleviate the reset-free learning challenge, as solving multiple tasks together allows different combinations of skills to serve as implicit resets for one another [243]. This capability is crucial for scaling RL to real-world dexterous manipulation, where human intervention for resetting objects is a major bottleneck.

To further enhance autonomy and address the label scarcity issues highlighted in Section 5.5, these systems increasingly incorporate offline preference-based apprenticeship learning. This allows agents to acquire new skills from static datasets without requiring active human feedback loops during deployment, bridging the gap between safe offline pre-training and risky online exploration. By combining offline RL with learned human preferences, agents can generalize to novel tasks not explicitly demonstrated in the training data, effectively performing apprenticeship learning without physical rollouts [244]. This is particularly valuable when online interaction is costly or unsafe. Moreover, techniques such as Human Guided Exploration (HuGE) demonstrate how spo-

radic, asynchronous, and low-quality human feedback can be used to steer exploration in reset-free settings, bifurcating the feedback process from policy learning to allow for unbiased self-supervised improvement [245]. Such methods ensure that even when human input is available, it is utilized efficiently to guide exploration rather than micromanage every step, complementing the active querying strategies discussed previously.

The integration of these components—foundation model-based reward generation, multi-task reset mechanisms, and offline preference learning—creates a synergistic framework for autonomous real-world learning. Systems like RoboFuME illustrate that by utilizing calibrated offline RL techniques alongside VLM-derived rewards, robots can incorporate data from disparate sources and improve on target tasks within hours of autonomous experience [33]. Additionally, the development of benchmarks and formalisms for Autonomous Reinforcement Learning (ARL) highlights the necessity of moving away from episodic assumptions toward continual, non-episodic world models where agents contend with a lack of human supervision [246]. Future directions in this domain will likely focus on refining the reliability of foundation model rewards to prevent hallucination-induced failures and developing more sophisticated multi-task curricula that maximize the probability of self-resetting in unstructured environments. By addressing both the reward specification and the environmental reset bottlenecks, these hybrid strategies pave the way for robots that can learn and adapt continuously in the real world with unprecedented levels of independence, setting the stage for the next generation of embodied AI systems capable of open-ended exploration.

6 Advanced Architectures for Complexity, Generalization, and Multi-Agent Systems

6.1 Hierarchical Skill Discovery and Latent Abstraction

6.1 Hierarchical Skill Discovery and Latent Abstraction

Bridging the gap between reactive, short-horizon control and the deliberate, long-horizon planning discussed in the subsequent section requires a fundamental shift in how agents represent behavior. Traditional reinforcement learning algorithms often falter when faced with the sparse reward signals and compounding errors inherent in tasks spanning hundreds or thousands of timesteps. To overcome these limitations, the field has increasingly adopted hierarchical architectures that decompose complex behaviors into reusable, temporally extended primitives known as skills. These skills serve as latent abstractions, enabling high-level planners to reason over compressed action spaces while low-level controllers execute robust motor patterns. A primary challenge in this domain is extracting such skills from unsegmented, offline data without explicit task boundaries or dense reward annotations. Recent advancements demonstrate that unsupervised and self-supervised methods can effectively uncover latent structures within raw trajectory data, allowing agents to synthesize novel behaviors by chaining discovered primitives.

A foundational approach involves identifying recurring patterns in demonstration data to construct a library of goal-conditioned policies. For instance, bottom-up skill discovery methods utilize agglomerative clustering on unsegmented demonstrations to identify hierarchical task structures, allowing agents to learn sensorimotor policies that can be composed by a meta-controller [247]. This approach bypasses the need for manual segmentation, making it highly scalable for real-world deployment where human annotations are costly. Similarly, other frameworks propose learning continuous spaces of recurring primitive behaviors directly from undirected offline experience. By delineating behaviors supported by the data from those that are not, these methods mitigate

distributional shift while providing the temporal abstraction necessary for efficient long-horizon reasoning [248]. The integration of such latent primitives allows agents to “stitch” together suboptimal trajectory segments to solve tasks that were never explicitly demonstrated in their entirety [249].

The role of latent variable models has further evolved to capture both discrete behavioral modes and continuous execution variances. Hierarchical mixture latent variable models have been proposed to cluster offline data into distinct, executable behaviors while retaining the flexibility to adapt to new tasks and unseen objects [250]. These models encourage directed exploration, proving particularly effective in sparse-reward settings where standard exploration strategies fail. Furthermore, the concept of “play” has emerged as a rich source for unsupervised skill acquisition. Methods like Play-LMP organize teleoperated play data into a latent space, enabling agents to reuse these self-supervised behaviors to achieve specific user-defined goals at test time, significantly outperforming policies trained only on narrow task demonstrations [251]. This shift from task-specific training to continuum-based behavior learning enhances robustness and enables the retrying of behaviors critical for physical interaction.

To further enhance the composability and generalization of these skills, researchers have integrated language and symbolic reasoning into the abstraction process. Language provides a unique compositional structure that facilitates fast learning and combinatorial generalization. Approaches utilizing language as an abstraction allow high-level policies to reason using structured instructions, while low-level policies follow these commands to execute diverse sub-skills [252]. This synergy is evident in methods that use Large Language Models (LLMs) to propose initial trajectory segmentations, which are then refined through variational inference to discover reusable skills [253]. Additionally, frameworks like CompILE utilize unsupervised sequence segmentation to learn latent encodings of behavior that can be re-composed to perform new tasks, even in environments not seen during training [254]. These latent codes act as actionable representations that bridge the gap between high-level intent and low-level control.

Advanced techniques also leverage visual representations and world models to facilitate skill discovery and planning. The Universal Visual Decomposer (UVD) operates purely on visual demonstrations to detect phase shifts in embedding spaces, effectively extracting visual subgoals without auxiliary information [255]. These visually discovered subgoals can then guide goal-conditioned policies or provide reward shaping for reinforcement learning. In parallel, model-based approaches like Choreographer exploit learned world models to discover and adapt skills in imagination, decoupling the exploration process from physical deployment [256]. This allows for the efficient evaluation of skill combinations in latent space before real-world execution. Moreover, the integration of skills with symbolic planners enables agents to solve complex tasks requiring both continuous control and logical reasoning. Systems like SEADS learn diverse skills alongside a symbolic forward model, allowing the agent to plan over abstract state transitions and execute the resulting plan with continuous controllers [257].

Finally, the efficiency of skill discovery has been improved through novel tokenization and compression strategies. Inspired by natural language processing, some methods discretize continuous action spaces and apply tokenization techniques to generate temporally extended actions, significantly reducing the computational burden of skill generation in sparse-reward domains [258]. Other approaches, such as SKID RAW, employ Bayesian and variational inference to simultaneously segment raw trajectories and learn the skills required to reconstruct them, facilitating natural human-robot interaction through adaptive behavior [259]. Collectively, these advancements in hierarchical skill discovery and latent abstraction establish the foundational primitives required for the more

sophisticated long-horizon planning strategies explored in the following section, moving robotics closer to the flexibility and adaptability observed in biological systems.

6.2 Long-Horizon Planning via World Models and Graph Search

6.2 Long-Horizon Planning via World Models and Graph Search

Building upon the library of reusable, temporally extended primitives established in the previous section, the next critical challenge lies in orchestrating these skills to solve complex, long-horizon tasks. While Section 6.1 focused on discovering latent abstractions to compress action spaces, this section addresses how agents leverage these abstractions within structured world models and graph-based representations to perform deliberate, multi-step planning. Traditional model-based reinforcement learning often falters in high-dimensional visual domains due to compounding errors during step-by-step rollouts, causing a rapid divergence between the learned model and reality. To overcome this, recent advancements have shifted towards architectures that decouple low-level control from high-level strategic reasoning. By operating over temporally abstract dynamics and graph-structured state spaces, agents can execute efficient tree-search planning and robust multi-step outcome prediction, effectively navigating sparse reward landscapes with unprecedented sample efficiency.

A primary strategy for extending planning horizons involves the construction of hierarchical world models that operate at multiple temporal scales, naturally aligning with the skill hierarchies discussed previously. By learning abstract representations of the environment, agents can plan over high-level goals rather than individual motor commands. For instance, the Forecaster framework introduces a deep hierarchical approach that learns a temporally abstract world model to facilitate tree-search planning over high-level goals [260]. This method trains a low-level policy to execute these abstract goals while the high-level planner utilizes the world model to envision long-term repercussions, significantly enhancing performance in domains like AntMaze where goals are distant and rewards are sparse. Similarly, the Director architecture demonstrates the efficacy of planning inside the latent space of a learned world model, where a high-level policy selects latent goals to maximize task and exploration rewards [261]. This approach allows for interpretable decisions, as the world model can decode latent goals back into images, bridging the gap between abstract planning and visual verification without requiring global position information.

Beyond hierarchical abstraction, the incorporation of graph structures into world models has emerged as a powerful paradigm for managing long-horizon reasoning, transforming continuous control problems into discrete search problems amenable to classical algorithms. Graph-based approaches treat significant states or landmarks as nodes and feasible transitions—often realized as the skills discovered in Section 6.1—as edges. The L3P method exemplifies this by learning graph-structured world models composed of sparse, multi-step transitions between latent landmarks [262]. By distilling reachability estimates from Q-functions into graph edges, L3P enables agents to leverage the robustness of model-free RL alongside the generalization capabilities of graph-search algorithms, proving particularly effective in high-dimensional continuous control tasks. Complementing this, the Latent Space Roadmap (LSR) framework constructs a graph-based structure in a low-dimensional latent space to globally capture system dynamics for complex manipulation tasks [263]. This roadmap connects clusters of similar states, enabling the agent to find latent plans between start and goal configurations before proposing specific actions, thus facilitating planning in high-dimensional spaces such as those involving deformable objects.

The integration of tree-search mechanisms with these learned models further refines planning capabilities, allowing agents to evaluate potential skill sequences before execution. While Monte Carlo Tree Search (MCTS) has long been a staple in discrete domains, its application to continuous control via latent dynamics is a recent breakthrough. Research has demonstrated that tree-based planning over learned latent dynamics can yield significant bootstrapping benefits and improved sample efficiency in continuous control benchmarks [264]. Furthermore, the Broadly-Exploring, Local-policy Trees (BELT) approach merges classical motion planning concepts with learned policies, using an RRT-inspired tree search guided by task-conditioned policies to explore high-dimensional state spaces efficiently [265]. This hybrid method allows for temporally extended dynamics propagation, enabling robust sequential reasoning over long horizons that pure learning-based methods often fail to achieve.

To ensure that these world models remain accurate over long horizons, researchers have also focused on learning representations that prioritize task-relevant information and causal structures. Temporal predictive coding has been employed to encode elements of the environment that are predictable across time, implicitly filtering out nuisance information and improving planning in latent spaces plagued by visual distractions [266]. Additionally, methods like PlaNet utilize latent dynamics models with both deterministic and stochastic components, optimized via a multi-step variational inference objective known as latent overshooting, to achieve high-performance planning from pixels [267]. The concept of “jumpy models” further advances this by learning multi-step dynamics prediction offline, allowing for zero-shot generalization and faster inference of high-level plans in downstream tasks [268].

Moreover, the intersection of generative modeling and planning has opened new avenues for visual planning, complementing the symbolic and skill-based approaches. The Diffused Task-Agnostic Milestone Planner leverages diffusion models to generate sequences of milestones in a latent space, exploiting the flexibility of generative models to plan diverse trajectories for multi-task decision-making [69]. In parallel, approaches like Hallucinative Topological Memory build connectivity graphs directly in image space using energy-based functions and hallucinated samples from conditional VAEs, enabling zero-shot planning with high interpretability [269]. These methods collectively illustrate a paradigm shift where world models are not merely predictors of the next frame but structured, abstract, and searchable representations. While this section establishes how hierarchical world models and graph search enable long-horizon planning, the execution of these plans in unstructured, real-world environments requires a critical missing ingredient: the ability to ground high-level strategic goals in low-level sensory perception, a topic addressed in the subsequent section through multimodal fusion and vision-language grounding.

6.3 Multimodal Fusion and Vision-Language Grounding for Generalist Agents

6.3 Multimodal Fusion and Vision-Language Grounding for Generalist Agents

While Section 6.2 established how hierarchical world models and graph search enable long-horizon planning, the execution of these plans in unstructured, real-world environments requires a critical missing ingredient: the ability to grounding high-level strategic goals in low-level sensory perception. The realization of embodied generalist agents hinges on the successful fusion of linguistic reasoning with 3D visual understanding. Although Large Language Models (LLMs) possess vast repositories of commonsense knowledge and planning capabilities, they historically lack the spatial grounding

necessary to interact with the physical world. To bridge this gap, recent research has pivoted toward integrating LLMs with Vision-Language Models (VLMs) and 3D perception modules, creating architectures that can perceive, reason, and act within complex spatial contexts. This integration is not merely a concatenation of modalities but requires sophisticated alignment mechanisms that allow natural language instructions to be grounded in dynamic visual observations and 3D geometric structures, effectively translating the abstract plans discussed previously into actionable physical behaviors.

A primary challenge in this domain is enabling agents to understand and interact with the 3D world, a capability that standard 2D-centric VLMs often lack. The work [270] addresses this by introducing LEO, an agent trained through a two-stage process involving 3D vision-language alignment and instruction tuning. This approach demonstrates that shared model architectures can effectively excel in perceiving, grounding, and planning within 3D environments, thereby overcoming the limitations of models restricted to textual or 2D visual domains. Similarly, [271] proposes a family of 3D-LLMs that directly ingest 3D point clouds and features. By utilizing specialized prompting mechanisms and a 3D localization mechanism, these models capture rich spatial relationships and affordances, significantly outperforming 2D VLMs in tasks such as 3D question answering and task decomposition. These advancements highlight that true generalization requires the injection of 3D structural information directly into the language model’s reasoning stream, providing the geometric fidelity necessary for the precise control required in robotic manipulation.

Beyond static understanding, embodied agents must possess the ability to plan and execute actions based on a dynamic understanding of world dynamics, reinforcing the world model concepts introduced in the preceding section. [272] extends this by linking 3D perception, reasoning, and action through a generative framework. By introducing interaction tokens and aligning embodied diffusion models with a 3D-based LLM, this architecture predicts future goal images and point clouds, effectively simulating the consequences of actions before execution. This mimics human-like imagination, allowing for more robust planning in embodied environments. Furthermore, [273] argues that LLMs can be enhanced by fine-tuning them with experiences acquired from world model simulators. This paradigm allows smaller language models to acquire diverse embodied knowledge, such as object permanence and physical planning, matching or even outperforming much larger models on downstream physical reasoning tasks.

The fusion of modalities also dictates how agents interpret and execute complex instructions in open-world settings, serving as the interface between abstract intent and concrete motion. [274] illustrates a powerful synthesis where LLMs infer affordances and constraints from language, which are then composed with vision-language model outputs to generate 3D value maps. These maps guide model-based planners to synthesize zero-shot, closed-loop robot trajectories, demonstrating how high-level semantic reasoning can be grounded into precise geometric control signals. In navigation tasks, [275] proposes VLMaps, a representation that fuses pretrained visual-language features with 3D reconstructions. This enables natural language indexing of the map without additional labeled data, allowing robots to navigate to spatially complex goals described in free-form language, such as “three meters to the right of the chair.”

However, the effectiveness of these generalist agents is often constrained by the quality of semantic grounding and the ability to handle diverse linguistic inputs. [276] highlights that prevalent misgrounding in current LVLMs poses significant safety and reliability risks. The study advocates for data-centric enhancement through fine-grained multimodal instruction tuning to improve the connection between language and physical entities. Additionally, [277] reveals that existing 3D-VL models are fragile to variations in language style, often failing to understand semantically identical

sentences phrased differently. This underscores the need for robust fusion modules that can handle the unpredictability of human language in real-world interactions, a prerequisite for the reliable deployment of neuro-symbolic systems discussed next.

To address the scarcity of 3D data and the cost of training, novel training strategies have emerged that leverage cross-modal distillation. [278] introduces a cross-modality imitation learning approach where a VLM agent learns from an LLM expert operating in a parallel text world. By distilling reflection outcomes and improved actions from the text domain, the embodied agent adapts quickly to visual world dynamics without extensive real-world trial and error. Similarly, [279] leverages pre-trained VLMs to provide retrospective supervision for embodied agents, using hindsight experience replay to teach task-relevant groundings for novel objects and abstract categories.

The scope of these generalist agents is further expanded by frameworks that unify diverse tasks under a single multimodal interface. [280] explores the potential of models like GPT-4V to handle perception, cognition, and action in an end-to-end manner, showing strong decision-making capabilities compared to collaborative multi-agent frameworks. Moreover, [281] utilizes schema-based instructions to cast various navigation tasks into a unified generation problem, achieving state-of-the-art performance and strong generalizability across unseen tasks. Finally, [282] demonstrates that incorporating continuous sensor modalities directly into language models enables positive transfer across diverse embodied tasks, from manipulation to visual question answering, establishing a link between words and percepts that is essential for general intelligence.

In conclusion, the trajectory toward embodied generalist agents is defined by the deepening integration of LLMs and VLMs with 3D perception and world modeling. By addressing challenges in 3D grounding, semantic robustness, and data-efficient learning through architectures like 3D-LLMs, generative world models, and composable value maps, the field is moving closer to agents that can flexibly interpret natural language and execute complex tasks in the physical world. While this section focuses on the perceptual and representational foundations of grounding, the following section will explore how these capabilities are combined with explicit symbolic reasoning and compositional logic to achieve rigorous, verifiable, and long-horizon task execution.

6.4 Compositional Reasoning and Neuro-Symbolic Integration

6.4 Compositional Reasoning and Neuro-Symbolic Integration

While Section 6.3 established the perceptual foundations for grounding high-level goals in 3D sensory data, translating these grounded intentions into rigorous, long-horizon execution requires a mechanism to enforce logical consistency and causal structure. Purely neural approaches, despite their semantic flexibility, often lack the interpretability and reliability necessary for safe deployment, while traditional symbolic planners struggle with the ambiguity of raw sensory inputs. Neuro-symbolic integration addresses this dichotomy by combining the commonsense reasoning of Large Language Models (LLMs) with the verifiable constraints of symbolic logic. This synergy enables agents to decompose complex objectives into causally linked sub-tasks, correct hallucinated plans through formal verification, and reason effectively over the spatio-semantic memories constructed in the previous section. By bridging the gap between the continuous, probabilistic nature of perception and the discrete, deterministic requirements of action, this paradigm serves as the critical intermediate layer before scaling to the decentralized multi-agent coordination discussed subsequently.

A core component of this paradigm is the utilization of structured scene representations, such as

scene graphs, to anchor the abstract knowledge of LLMs in the physical environment. Purely language-based planners often generate infeasible action sequences due to hallucinations or a lack of awareness regarding specific objectology and object states. To mitigate this, recent frameworks leverage scene graphs as an intermediate representation that unifies symbolic, topological, and metric information. For instance, the DELTA framework utilizes scene graphs to provide environmental topology to LLMs, enabling the decomposition of long-term goals into autoregressive sequences of sub-goals that are physically actionable, thereby significantly reducing planning time and increasing success rates compared to naive LLM prompting [283]. Similarly, the SayNav approach incrementally builds 3D scene graphs from exploration data to serve as inputs for LLMs, allowing for dynamic, step-by-step navigation planning in unknown large-scale environments that outperforms oracle baselines [284]. These methods demonstrate that grounding language models in structured spatial representations is essential for translating natural language instructions into executable robot plans.

Beyond static scene understanding, robust task execution requires the ability to perform compositional reasoning over time and space, effectively breaking down complex procedures into causal chains of events. Neuro-symbolic frameworks address this by embedding symbolic logic directly into the reasoning loops of neural models. The PLAN framework, for example, employs commonsense-infused prompting combined with symbolic program executors to formalize procedural knowledge, effectively acting as a causal intervention to mitigate spurious correlations between goals and steps that often plague zero-shot LLM planning [285]. This approach ensures that generated plans adhere to logical constraints derived from commonsense knowledge bases. Furthermore, the JARVIS framework illustrates a modular architecture where LLMs are prompted to generate symbolic representations for sub-goal planning, which are then reasoned over by a symbolic module based on task-level commonsense, achieving state-of-the-art results in conversational embodied agents [286]. By separating the generative capability of the LLM from the verification and execution logic of the symbolic module, these systems achieve a level of reliability unattainable by monolithic neural networks.

The challenge of hallucination in foundation models remains a critical barrier to deployment, necessitating robust mechanisms for plan correction and verification. Neuro-symbolic integration provides a natural solution by employing symbolic solvers to validate the feasibility of neural-generated plans. The Teriyaki framework trains LLMs to act as neuro-symbolic planners compatible with the Planning Domain Definition Language (PDDL), leveraging the linear scaling of LLM response times to synthesize plans action-by-action while maintaining compatibility with traditional symbolic verification tools [287]. Similarly, the LLM-DP framework employs a hybrid approach where an LLM works in tandem with a traditional planner; the LLM handles noisy observations and uncertainty, while the symbolic planner ensures the logical consistency and optimality of the action sequence [288]. This division of labor allows the system to benefit from the flexibility of neural models in interpreting ambiguous inputs while retaining the guarantee of goal achievement provided by symbolic search.

Furthermore, effective compositional reasoning often requires the agent to maintain and query spatio-semantic memories that evolve over time, extending the grounding capabilities discussed in Section 6.3. The CoPa framework demonstrates how foundation models can be used to identify spatial geometry constraints of object parts, which are then used to derive precise post-grasp poses for manipulation, effectively grounding high-level task planning in fine-grained physical understanding [289]. In parallel, research into differentiable scene graphs has shown that it is possible to learn these structured representations in an end-to-end manner, allowing visual reasoning systems to

optimize directly on downstream tasks without discrete, non-differentiable bottlenecks [290]. Additionally, the integration of causal reasoning into knowledge graphs, as seen in CausalKG, enables agents to perform interventional and counterfactual reasoning, asking “what-if” questions to better understand the consequences of potential actions before execution [291].

Finally, the scalability of these neuro-symbolic approaches to complex, large-scale domains is being addressed through advanced graph-based planning, laying the groundwork for the multi-agent systems explored in the next section. The TASKOGRAPHY benchmark highlights the necessity of sparsifying 3D scene graphs to enable real-time planning, proposing methods like SCRUB and SEEK to make classical and learning-based planners viable over large-scale hierarchical maps [292]. Moreover, hierarchical planning algorithms that utilize both geometric and symbolic scene graphs have demonstrated the ability to scale to long-horizon manipulation tasks, achieving high success rates in real-world kitchen storage scenarios by jointly optimizing high-level task plans and low-level motions [293]. These developments underscore that the future of robotic intelligence lies in the seamless fusion of neural perception and generation with symbolic reasoning and verification. Having established how single agents can achieve rigorous, verifiable planning through neuro-symbolic integration, the following section extends these principles to the challenges of decentralized coordination and communication in multi-robot fleets.

6.5 Decentralized Multi-Robot Coordination and Communication

6.5 Decentralized Multi-Robot Coordination and Communication

Building upon the neuro-symbolic frameworks that enable robust single-agent reasoning and plan verification, the scaling of robotic systems from individual units to large-scale fleets necessitates a fundamental shift from centralized control architectures to decentralized coordination strategies. While the compositional reasoning discussed previously allows an agent to decompose complex goals, coordinating a swarm of such agents introduces the curse of dimensionality, rendering centralized planners computationally intractable for real-time deployment in dynamic environments. Consequently, recent research has increasingly focused on decentralized multi-robot coordination, where agents make local decisions based on partial observations and limited communication. A pivotal development in this domain is the integration of Large Language Models (LLMs) as reasoning engines for both global orchestration and local decision-making, offering unprecedented capabilities in handling complex, long-horizon tasks through natural language interfaces. However, deploying LLMs in multi-agent settings introduces unique challenges regarding token efficiency, context window limitations, and the reliability of emergent communication protocols, distinguishing this challenge from the single-agent hallucination issues addressed in neuro-symbolic integration.

A primary consideration in decentralized systems is the architectural choice between centralized, decentralized, and hybrid communication frameworks. Recent comparative studies have evaluated these paradigms in the context of LLM-based task planning, revealing that purely centralized approaches often struggle with the exponential growth of token requirements as agent counts rise. In contrast, fully decentralized systems may lack the global perspective necessary for complex coordination, potentially leading to sub-optimal collective behaviors. Research indicates that hybrid frameworks, which combine local autonomy with selective global synchronization, often achieve superior task success rates and scale more effectively to larger teams [294]. These hybrid models leverage the reasoning power of LLMs to manage high-level goal decomposition—akin to the symbolic planners in the previous section—while allowing low-level execution to remain distributed, thereby balancing computational load and coordination fidelity. Furthermore, the organization

of these agents plays a critical role; inspired by human organizational structures, frameworks that impose designated leadership roles and specific communication hierarchies have been shown to mitigate information redundancy and enhance team efficiency [295]. Such structured approaches enable LLM agents to exhibit spontaneous cooperative behaviors and reduce the cognitive load associated with unstructured peer-to-peer negotiation.

Token efficiency remains a critical bottleneck when utilizing LLMs for multi-agent communication, particularly in bandwidth-constrained or real-time scenarios where the transmission of verbose natural language is impractical. To address this, novel algorithms have been developed to optimize the information exchange process. For instance, methods that employ deterministic messaging policies, where senders selectively transmit only the most informative observations required for receiver inference, have demonstrated convergence to optimal policies in partially observable environments where traditional decentralized methods fail [296]. Additionally, learning-based approaches that utilize Graph Neural Networks (GNNs) to facilitate efficient local broadcasting allow agents to maintain situational awareness in crowded environments without overwhelming the communication channel [297]. These techniques are essential for scaling multi-agent path finding (MAPF) to dense environments, where implicit coordination mechanisms often degrade. By integrating GNNs with reinforcement learning, systems like CRAMP have achieved significant improvements in makespan and collision avoidance compared to state-of-the-art decentralized baselines [297].

In partially observable environments, the ability of agents to model teammate behavior and predict future actions is paramount for successful collaboration, extending the concept of causal reasoning to the multi-agent domain. Decentralized planning algorithms augmented with learned teammate models have shown promise in this regard. By combining Monte Carlo Tree Search (MCTS) with deep learning-based policy approximators, agents can iteratively refine their understanding of teammates' strategies, leading to convergence toward Nash equilibria in cooperative tasks [298]. This capability is further enhanced by frameworks that enable agents to learn communication protocols dynamically. For example, actor-critic schemes have been employed to learn optimal policies where the emergent communication consists not of raw sensor data encoding, but of directive actions tailored to the specific task, thereby improving interpretability and coordination efficiency [299]. Moreover, in scenarios characterized by intermittent connectivity or line-of-sight constraints, specialized planning methods ensure safety by explicitly avoiding configurations where communication loss could lead to collisions, demonstrating the necessity of robustness in decentralized deployment [300].

The application of these principles extends to complex navigation and formation control tasks, requiring the synthesis of high-level strategic planning and low-level geometric constraints. Hierarchical learning approaches have been proposed to decompose multi-objective problems, such as simultaneous pathfinding and formation maintenance, into manageable sub-tasks. These methods calculate theoretical weights to balance reward influences and utilize communication channels to foster cooperation, successfully scaling to large world sizes where centralized planners falter [301]. Similarly, scalable GNN-based planners like MASP have demonstrated the ability to handle navigation tasks with substantial numbers of agents by dividing the search space and modeling interactions through graph structures, achieving near-perfect success rates even with unseen team sizes [302].

Despite these advances, challenges persist in ensuring reliable collaboration among heterogeneous agents and adapting to dynamic team compositions, a prerequisite for the distributed mapping systems discussed in the following section. Ad hoc teamwork, where an agent must cooperate with unfamiliar teammates without pre-established protocols, remains a difficult frontier. Recent work

utilizing code-driven reasoning and enhanced memory mechanisms, such as CodeAct, has shown potential in repurposing partial information for rapid adaptation to new team configurations [303]. Furthermore, the development of self-organizing agent systems that can dynamically adjust their workflow and information priorities without human intervention represents a significant step toward truly autonomous multi-robot swarms capable of operating in open-ended environments [304]. As the field progresses, the synergy between learned communication protocols, efficient token usage, and robust decentralized planning algorithms will be instrumental in realizing scalable, resilient, and intelligent multi-robot systems. These coordinated fleets provide the foundational infrastructure necessary for the federated learning and distributed map building paradigms explored next, where the focus shifts from coordinating actions to collaboratively constructing shared environmental representations.

6.6 Federated Learning and Distributed Map Building for Exploration

6.6 Federated Learning and Distributed Map Building for Exploration

Building upon the decentralized coordination and communication protocols established in the previous section, the next critical challenge in scaling multi-robot systems lies in collaboratively constructing shared environmental representations. While Section 6.5 detailed how agents can coordinate actions and negotiate tasks through efficient messaging, these capabilities must be underpinned by a unified understanding of the physical world, particularly in expansive, unstructured, or hazardous environments. Traditional approaches to distributed mapping often rely on centralized data aggregation, where raw sensory inputs from various agents are transmitted to a central server for processing. However, this paradigm faces significant bottlenecks in real-world robotics, including prohibitive bandwidth requirements, latency issues, and critical privacy concerns regarding the exposure of raw environmental data. Federated Learning (FL) has emerged as a transformative paradigm to address these challenges, extending the principles of decentralized decision-making to the domain of perception. By enabling collaborative model training across distributed agents without the need to centralize raw data, FL allows robot teams to jointly learn compact, adaptive representations of their environment while keeping sensitive sensory data localized on edge devices.

A primary application of FL in robotics is distributed mapping, where the goal is to construct a coherent global map from heterogeneous data streams generated by dynamic environments. The sheer volume and variety of data collected by multiple agents pose significant management challenges, echoing the token efficiency constraints discussed in multi-agent communication. To overcome this, recent approaches leverage implicit neural representations, which model maps as continuous functions learned by neural networks rather than discrete grid-based structures. This method offers a highly compact and adaptable representation suitable for transmission over bandwidth-constrained networks. For instance, research has demonstrated that combining implicit neural mapping with meta-initialization techniques allows networks to be pre-trained on Earth datasets, enabling them to quickly learn and generalize to novel map structures in diverse domains such as Martian terrain or glacial environments [305]. This synergy between federated learning and implicit representations ensures that robot teams can share knowledge effectively, achieving strong generalization capabilities even when operating in vastly different physical contexts.

The efficacy of federated learning extends beyond planetary exploration to terrestrial mobile robotics, particularly in vision-based navigation and obstacle avoidance. In the Internet of Robotic Things (IoRT), wheeled robots operating in distinct indoor environments can collaborate to enhance their situational awareness without sharing proprietary visual data. Studies have shown

that federated learning approaches can outperform traditional centralized training processes that rely on a priori aggregated data, offering significant performance benefits for vision-based navigation tasks [306]. Furthermore, this collaborative framework facilitates sim-to-real knowledge transfer, allowing models trained in simulation to be refined through real-world decentralized data collection, all while maintaining the inherent privacy-preserving nature of FL by keeping computation at the edge.

Addressing the heterogeneity of data distributions across different robotic clients is a critical challenge in federated map building, mirroring the difficulties of coordinating heterogeneous agents discussed previously. Robots operating in different locations inevitably encounter non-independent and identically distributed (non-IID) data, which can lead to convergence issues and catastrophic forgetting in standard federated averaging algorithms. To mitigate these effects, architectural innovations have been proposed. Research indicates that self-attention-based architectures, such as Transformers, exhibit greater robustness to distribution shifts compared to traditional convolutional networks, thereby improving federated learning outcomes over heterogeneous data splits [307]. By replacing standard convolutional layers with Transformer-based components, systems can reduce catastrophic forgetting of previously learned map features and accelerate convergence toward a superior global model. Additionally, specific modifications to convolutional neural networks, such as strategic adjustments to activation functions and normalization layers, have been shown to enable pure CNNs to match or exceed the robustness of Vision Transformers in handling heterogeneous federated data [308].

To further enhance the efficiency and scalability of distributed mapping, the field is shifting from server-dependent FL toward fully decentralized and server-less architectures, aligning closely with the hybrid and decentralized frameworks analyzed in Section 6.5. Traditional FL relies on a central server for aggregation, creating a single point of failure and potential communication bottlenecks that contradict the resilience goals of swarm robotics. Decentralized paradigms, such as those utilizing gossip-based shared data structures or mobile agent platforms, eliminate the need for a central coordinator. For example, methodologies have been developed to decentralize federated reinforcement learning in multi-robot scenarios using mobile agents that aggregate Q-tables or model parameters directly among connected robots, ensuring viability even if central controlling entities fail [309]. Similarly, server-less variants like Flow-FL utilize gossip protocols to synchronize learning processes based on data-driven triggers, allowing robots to contribute model updates only when sufficient local data has been collected, thus optimizing communication in staggered online data collection settings [310].

Communication efficiency remains a paramount concern, especially for resource-constrained robotic platforms operating under the bandwidth limitations highlighted in prior sections. Innovative frameworks have been introduced to reduce the communication footprint of federated learning without sacrificing model accuracy. Techniques such as Deep Equilibrium Learning have been employed to create compact shared representations, significantly reducing the communication size and memory footprint during training while allowing for personalized model adaptation at each node [311]. Moreover, dynamic sampling and selective masking strategies can control the fraction of client models selected for aggregation and transmit only the most significant parameter updates, effectively alleviating the burden on communication networks [312].

Finally, the integration of federated learning with continual learning algorithms addresses the need for long-term autonomy in social and collaborative robotics. As robots interact with humans and environments over extended periods, they must adapt to new users and changing contexts without forgetting previous knowledge, a necessity for the self-organizing systems envisioned earlier.

Frameworks combining FL with Continual Learning, such as those employing importance-based regularization, enable decentralized robots to capture interaction dynamics distributed both physically across the swarm and temporally across repeated encounters [225]. This capability is essential for building resilient multi-agent systems that can collectively build and refine maps of complex, evolving environments while respecting data privacy and operational constraints. Through these advanced architectures and strategies, federated learning is redefining the landscape of distributed map building, enabling scalable, private, and efficient exploration for the next generation of autonomous robotic teams, and laying the groundwork for the preference-based alignment techniques explored in subsequent sections.

7 Safety, Robustness, and Formal Verification in Real-World Deployment

7.1 Control Barrier Functions for Constraint Satisfaction and Input Limits

7.1 Control Barrier Functions for Constraint Satisfaction and Input Limits

As robotic systems increasingly rely on data-driven paradigms like Offline and Preference-Based Reinforcement Learning (RL), ensuring that learned policies adhere to strict physical safety constraints is paramount. Control Barrier Functions (CBFs) have emerged as a foundational mathematical framework to bridge this gap, enforcing the forward invariance of a predefined safe set by transforming state constraints into conditions on the control input. These conditions are typically satisfied through a real-time optimization problem, often formulated as a Quadratic Program (QP). However, the practical deployment of CBFs in robotics—particularly when wrapping learning-based controllers—is frequently hindered by the complexity of constructing valid barrier functions that simultaneously address high-relative-degree dynamics, strict actuation limits, and multiple concurrent constraints. Recent advancements have focused on bridging the gap between these theoretical safety guarantees and the physical realities of robotic hardware, laying the groundwork for the resilient systems discussed in subsequent sections.

A primary challenge in CBF synthesis is handling systems where the safety constraint does not directly depend on the control input, known as high-relative-degree systems. In such scenarios, standard zeroing CBFs are insufficient because the control input does not appear in the first time derivative of the constraint function. To address this, researchers have extended the theory to High Order Control Barrier Functions (HOCBFs), which recursively define a series of barrier conditions to propagate the safety requirement down to the control input level [313]. These formulations allow for the management of complex kinematic and dynamic constraints, such as those found in adaptive cruise control or aerial robotics, by ensuring the intersection of a series of sets remains forward invariant. Furthermore, specific methodologies have been developed to construct robust CBFs for high-relative-degree constraints in the presence of input constraints and bounded disturbances, ensuring that the resulting safety conditions remain feasible even when actuators are saturated [314].

The issue of feasibility becomes particularly acute when input constraints are present, a scenario common in real-world robotics where actuators have finite limits. A critical failure mode in CBF-based control is the “constraint compatibility problem,” where the safety requirement demands a control effort that exceeds the robot’s physical actuation limits, inevitably leading to a safety violation. To resolve this, recent work has proposed the construction of optimal CBFs designed specifically to maximize the feasible action space subject to control bounds [315]. By encoding

dynamical motion primitives within the CBF structure, these optimal formulations ensure that the safety filter remains compatible with the system’s actuation capabilities. Additionally, the concept of Input Constrained Control Barrier Functions (ICCBF) has been introduced to explicitly identify subsets of the safe state space where a feasible controller exists, thereby generalizing higher-order approaches to systems with non-uniform relative degrees [316].

When multiple safety constraints must be satisfied simultaneously, ensuring the joint feasibility of the resulting QP becomes non-trivial. The composition of multiple CBFs can lead to conflicting requirements that render the optimization problem infeasible, potentially causing the controller to fail precisely when safety is most critical. To mitigate this, algorithmic solutions have been proposed to verify the robust compatibility of multiple CBFs a priori, leveraging Lipschitz properties to guarantee that a solution exists within the defined safe set [317]. Complementary approaches involve decoupling the design of multiple CBFs so they can be tuned independently while ensuring their composition forms a viability domain [318]. Another innovative strategy involves the synthesis of Consolidated Control Barrier Functions (C-CBFs), where multiple time-varying candidate functions are adapted into a single consolidated candidate using a predictor-corrector optimization law, ensuring sufficient control authority is maintained despite input limits [319].

Beyond manual derivation, data-driven and learning-based methods are increasingly employed to construct CBFs for complex, nonlinear systems where analytical solutions are intractable, aligning well with the data-centric nature of offline RL. One approach involves learning linear CBFs through an iterative, optimization-based framework that samples from a user-specified safe set, explicitly addressing control input constraints during the construction phase [320]. For more expressive representations, neural networks have been utilized to learn CBFs; however, verifying these neural CBFs (NCBFs) remains a challenge. Novel frameworks now exist to simultaneously synthesize and verify neural CBFs using branch-and-bound techniques, ensuring formal safety guarantees over continuous state spaces [321]. Further enhancing data efficiency, prioritized sampling strategies have been developed to refine CBFs from collected data, allowing the recovery of larger portions of the true safe set with fewer samples [322].

To address the conservatism often inherent in point-wise myopic CBF formulations, researchers have explored integrating optimal control principles to balance immediate safety with long-term performance. Approximate optimal control approaches embed the cost of violating safety constraints directly into the value function, balancing long-horizon performance with immediate safety requirements [323]. Similarly, the unification of Hamilton-Jacobi reachability and CBFs has led to the development of Control Barrier-Value Functions (CBVFs), which provide robust safety guarantees against bounded disturbances while preserving the computational efficiency of QP-based synthesis [324]. Additionally, adaptive learning-based approaches parameterize the class- $\mathcal{K}$ functions within CBFs using neural networks, trained via reinforcement learning to mitigate myopic behavior and improve feasibility and stability [325].

Finally, specialized constructions such as Backup Control Barrier Functions offer a tractable method to guarantee feasibility by relying on a fixed backup policy to implicitly define a control invariant set, proving effective in computing sets close to the maximum control invariant set [326]. Collectively, these advancements in constructing zeroing, high-relative-degree, and optimal CBFs represent a significant stride toward deploying safe, reliable, and high-performance controllers in real-world robotic systems operating under strict physical limitations. While these methods ensure safety under nominal conditions and bounded disturbances, the following section expands the scope to address robustness against intentional adversarial perturbations and dataset corruption, which pose distinct threats to the integrity of these safety mechanisms.

7.2 Robustness Against Adversarial Perturbations and Dataset Corruption

7.2 Robustness Against Adversarial Perturbations and Dataset Corruption

While the preceding section established methods to ensure safety under nominal physical constraints and bounded disturbances, the deployment of data-driven paradigms like Offline and Preference-Based Reinforcement Learning (RL) in open-world environments introduces a distinct class of threats: intentional adversarial perturbations and dataset corruption. Unlike the natural uncertainties addressed later, these threats stem from adaptive adversaries capable of manipulating sensory inputs, actuation channels, and training data to induce catastrophic failures. Ensuring robustness in this context requires a multi-layered defense strategy that integrates viability domain verification, resilient control synthesis, and certified defenses against adversarial artifacts. The intersection of machine learning security and cyber-physical system (CPS) resilience has thus become critical, as traditional perimeter defenses are often insufficient against agents who can directly manipulate data distributions or physical signals [327].

A foundational approach to ensuring safety under such attacks involves the concept of viability domains, which define the set of states from which a system can be controlled to remain within safe operational boundaries despite adversarial interference. Recent advances have introduced sampling-based methods to compute these viability domains for nonlinear systems, providing sufficient conditions to guarantee safety even when actuators are compromised [328]. By verifying modified barrier function conditions on a finite set of points, researchers can iteratively compute safe initial conditions and input constraints that limit an adversary’s ability to drive the system into unsafe regions. This is particularly relevant for offline RL policies, which may lack explicit safety guarantees when encountering out-of-distribution states induced by attacks. Furthermore, the notion of structural resilience has been formalized using graph-theoretic frameworks, where the resilience of a linear structured system to Denial-of-Service (DoS) and integrity attacks is characterized by the existence of unmatched vertices in maximum matchings [329]. These theoretical tools allow designers to verify whether a system topology inherently possesses the redundancy required to withstand specific classes of actuator saturation or sensor erasure attacks.

In the realm of perception, which serves as the primary input for many robotic learning agents, adversarial patches pose a significant threat. These physical attacks involve placing contiguous, maliciously crafted patterns in the environment to mislead vision-based controllers. To counter this, the community has moved towards certified defenses that provide mathematical guarantees of robustness against patches of bounded size and location. Novel architectures and certification procedures, such as those combining model design with end-to-end optimization for certified robustness, have demonstrated the ability to maintain high accuracy on clean inputs while certifying defense against physical patches [330]. Furthermore, architecture-agnostic iterative black-box certified defense methods have been proposed to handle scenarios where the position and size of the adversarial patch are unknown, estimating patch parameters through search-based masking before applying certification [331]. Complementary to certification, detection mechanisms based on class-conditional reconstruction networks have shown promise in identifying physical adversarial attacks by leveraging the discrepancy between clean and reconstructed images, although their effectiveness can vary with dataset complexity [332].

Beyond perception, the robustness of the decision-making and control layers is paramount. Learning-enabled controllers are susceptible to state and action space perturbations, where an

attacker manipulates the feedback loop to induce catastrophic failures. Shield-based defenses have emerged as an effective strategy, treating the controller and environment as black boxes to filter unsafe actions in real-time [333]. Additionally, adversarial training techniques specifically tailored for the action space have been developed to robustify Deep RL agents. By exposing agents to optimal perturbations during training, these methods significantly reduce the success rate of targeted actuation attacks, inducing behaviors that are naturally more robust to boundary conditions [334]. The vulnerability of RL agents is further compounded by the potential for attackers to manipulate reward signals or state observations, tricking the agent into learning nefarious policies with minimal effort [335].

The integrity of the dataset itself represents another critical vector for corruption, especially in offline RL where policies are distilled from static buffers. If the historical data contains poisoned transitions or corrupted preference labels, the resulting policy may exhibit degraded or unsafe behavior. Resilience against such dataset corruption often parallels strategies used in secure state estimation, where sliding mode observers and adaptive-gain algorithms are employed to reconstruct and clean corrupted measurements online [336]. In multi-agent settings, resilient coordination protocols have been designed to tolerate unbounded sensor and actuator attacks, ensuring that the collective system maintains containment performance even when individual agents are compromised [337]. Moreover, the integration of digital twin architectures allows for the decoupling of defense protocols, where a secure twin layer can reconstruct leader dynamics and guide followers despite composite attacks on the physical layer [338].

Finally, the broader context of CPS security emphasizes the need for system-centric approaches that consider mission goals alongside technical constraints. The “Mission Aware” perspective argues that resilience evaluation must be driven by traceable engineering evidence that balances trade-offs between competing objectives in the presence of threats [339]. This is complemented by chaos engineering principles, which advocate for runtime analyses to predict environmental changes and apply mitigation measures that minimize the blast radius of adverse events [340]. As robotic systems evolve to incorporate complex foundation models, the challenge of ensuring robustness against both digital and physical adversarial perturbations remains a central pillar of safe deployment. While these strategies defend against intentional malice, the following section shifts focus to addressing the equally critical challenge of managing inherent model uncertainty and natural distributional shifts arising from unstructured environments.

7.3 Handling Model Uncertainty and Distributional Shift via Data-Driven Safety

7.3 Handling Model Uncertainty and Distributional Shift via Data-Driven Safety

While the previous section addressed robustness against malicious adversarial perturbations and dataset corruption, a distinct yet equally critical challenge lies in managing inherent model uncertainty and natural distributional shifts. The deployment of learning-based control policies in real-world robotic systems is frequently hampered by model misspecification and unknown external disturbances that are not adversarial in nature but arise from unstructured environments. Traditional control strategies often rely on precise analytical models; however, in practice, these models are prone to significant errors due to unmodeled dynamics, varying friction coefficients, or interactions with novel objects. When a policy trained in simulation or under idealized conditions encounters such distributional shifts, the risk of catastrophic failure increases dramatically. To address these challenges, the field has increasingly data-driven safety mechanisms that do not assume perfect knowledge but actively quantify and reason about uncertainty. A dominant paradigm

in this space involves the use of probabilistic models, particularly Gaussian Processes (GPs), to construct rigorous safety certificates that adapt as new data is collected, serving as a probabilistic bridge between the defensive strategies of the previous section and the formal guarantees discussed subsequently.

Gaussian Processes have emerged as a cornerstone for safe learning because they provide not only a prediction of the system dynamics but also a calibrated measure of uncertainty through their posterior variance. This property is critical for constructing safety barriers that expand or contract based on the confidence of the model. For instance, research has demonstrated that by modeling the projection of residual dynamics onto a control barrier function using a GP, controllers can generate safe inputs that guarantee safety with high probability, even when the nominal model is inaccurate [341]. This approach allows the safety filter to reason explicitly about the uncertainty of its predictions, significantly reducing safety violations compared to deterministic learning methods that might fail when predictions are imperfect. Furthermore, the application of GPs extends to learning the safety measure itself, where active sampling is used to implicitly capture the relationship between system dynamics and failure states, thereby greatly reducing failures during the learning phase even before the safety measure is fully converged [342].

However, the computational complexity of standard GP inference, which scales cubically with the number of data points, poses a significant bottleneck for high-dimensional robotic systems and embedded platforms. To mitigate this, recent advancements have introduced computationally lightweight alternatives that maintain probabilistic guarantees. One such approach leverages the Nadaraya-Watson estimator to achieve logarithmic scaling with respect to the dataset size, enabling real-time safe learning on complex manipulators without sacrificing the theoretical guarantees provided by more expensive non-parametric methods [343]. Additionally, efficient online data selection algorithms have been developed to strategically identify and utilize only the most informative data points, reducing the time complexity of data-driven safety filters from quadratic to linear while maintaining robust performance in uncertain conditions [344].

Beyond point-wise uncertainty estimation, ensuring safety requires a holistic understanding of the system’s future reachable states under uncertainty. Online data-driven reachability analysis has become a vital tool for verifying that a system will avoid dangerous regions despite unknown disturbances, conceptually linking probabilistic estimation with the rigorous reachability analysis detailed in the following section. Techniques employing optimization-based moving horizon estimators can characterize disturbances in real-time and incorporate them into reachability calculations, providing theoretical bounds on future states that ensure constraint satisfaction [345]. This is particularly important for systems subject to non-stationary disturbances, where context-aware safe reinforcement learning frameworks utilize probabilistic latent variable models to infer environmental changes and adjust safety constraints dynamically [346]. Moreover, by integrating contraction theory with learned disturbance bounds, planners can provide statistical guarantees on both runtime safety and goal reachability for underactuated stochastic systems, ensuring that tracking errors remain within safe limits even when the underlying dynamics are only partially known [347].

The challenge of distributional shift is further addressed through methods that learn robust controllers capable of adapting to sudden changes in system dynamics, such as those caused by physical damage or component failure. Frameworks combining model predictive control with active learning allow robots to infer altered dynamics probabilistically safely at the moment of failure, regaining control within seconds by finding information-rich trajectories that respect chance constraints [348]. Similarly, risk-aware control strategies utilize concepts like “Surface-at-Risk” to learn disturbance bounds online with minimal data, enabling drones to adapt to wind gusts or payload changes after

less than a minute of operation [349]. These methods often rely on deriving uniform error bounds for GP regression under weaker assumptions, allowing for the synthesis of safety conditions that are valid even when hyperparameters are unknown or the data is noisy [350].

Furthermore, the integration of Bayesian learning with control barrier functions allows for the synthesis of safe policies for systems with arbitrary relative degrees. By formulating the safety problem as a second-order cone program that incorporates chance constraints on Control Lyapunov Functions and Control Barrier Functions, systems can autonomously estimate and adapt their dynamics models during operation while maintaining high-probability safety guarantees [351]. This adaptability is crucial for handling the epistemic uncertainty inherent in data-driven models, where the lack of knowledge about the true system dynamics must be distinguished from aleatory uncertainty arising from stochastic noise. Advanced frameworks address both by designing data-based probabilistic safe controllers that leverage distributionally robust optimization techniques, ensuring safety even when the noise distribution deviates from standard Gaussian assumptions [352]. Ultimately, these data-driven approaches represent a paradigm shift from static, worst-case safety analysis to dynamic, probabilistic safety certification. While they offer flexible adaptation to the unknown, they naturally lead to the question of how to provide absolute, mathematically rigorous guarantees for such adaptive behaviors, a topic explored in the subsequent section through formal verification and temporal logic.

7.4 Formal Verification, Reachability Analysis, and Temporal Logic Guarantees

7.4 Formal Verification, Reachability Analysis, and Temporal Logic Guarantees

While the data-driven safety mechanisms discussed in the previous section offer flexible adaptation to model uncertainty through probabilistic bounds, they inherently rely on statistical confidence rather than absolute mathematical certainty. For safety-critical robotic deployments where even a low-probability failure is unacceptable, it becomes necessary to transition from high-probability assurances to rigorous, worst-case guarantees. This section explores formal verification methods, particularly those grounded in reachability analysis and temporal logic, which provide mathematical proofs that a system will satisfy specific specifications under all admissible disturbances and initial conditions. These techniques serve as the definitive bridge between the adaptive, uncertainty-aware controllers of Section 7.3 and the risk-aware planning frameworks that integrate these hard constraints into stochastic learning processes in Section 7.5.

Hamilton-Jacobi (HJ) reachability analysis stands as a cornerstone technique in this domain, offering a powerful framework for computing the set of states from which a system can reach a target or avoid a failure region despite bounded disturbances. The core of this approach involves solving a Hamilton-Jacobi-Isaacs partial differential equation to derive a value function, the zero sublevel set of which defines the reachable tube. While traditionally limited by the curse of dimensionality, recent advancements have significantly expanded its applicability to complex mission profiles. For instance, the integration of HJ reachability with temporal logic trees allows for the verification of multi-stage tasks. By constructing temporal logic trees via reachability analysis, researchers can check for the existence of control policies that satisfy linear temporal logic (LTL) specifications and synthesize the full set of feasible control inputs in real-time [353]. This methodology addresses critical issues such as the “leaking corner issue,” ensuring that the intersection of reachable sets correctly approximates the viable state space for task completion. Furthermore, the construction of controlled temporal logic trees enables online control synthesis for uncertain discrete-time systems, providing recursive feasibility without requiring discrete abstractions of the infinite system

dynamics [354].

To address the computational bottlenecks associated with high-dimensional systems—a challenge also noted in the GP-based methods of the prior section—novel numerical and learning-based approaches have been developed to scale formal verification. The DeepReach algorithm leverages sinusoidal neural networks to solve the underlying PDEs, scaling computational requirements with the complexity of the reachable tube rather than the state dimension, thereby enabling safety verification for systems with up to ten or more dimensions [355]. Enhancements to this framework include optimizing activation functions to improve convergence on high-dimensional collision avoidance problems [356] and deriving rigorous error bounds to provide probabilistic safety assurances for neural approximations [357]. Verification of these neural reachable tubes is further strengthened through scenario optimization and conformal prediction, which offer a trade-off between resilience to outlier errors and the strength of probabilistic guarantees [358]. Additionally, hardware acceleration using Field Programmable Gate Arrays (FPGAs) has demonstrated the potential for real-time formal verification, achieving significant speedups over CPU-based implementations for lower-dimensional models [359].

Beyond spatial reachability, the concept of time-bounded resilience has gained traction as a metric for system robustness against adversarial disruptions, complementing the disturbance rejection strategies discussed previously. This formalization considers the ability of a system to adapt and recover within a specified time horizon. Theoretical analysis reveals that time-bounded resilience problems for certain classes of planning scenarios reside within specific levels of the polynomial hierarchy, highlighting the computational complexity involved in verifying such properties [360]. Complementary work on time-sensitive distributed systems utilizes multiset rewriting languages with explicit time to verify properties like realizability and survivability, proving that bounded-time restrictions can render otherwise undecidable problems tractable within NP-complete or PSPACE-complete classes [361][362]. These theoretical foundations are crucial for designing systems that must maintain operational integrity under strict temporal constraints, a requirement that directly informs the dynamic risk budgets explored in the following section.

Verifying exponential stability and other dynamic properties often requires combining reachability analysis with barrier certificates, creating a synergy with the Control Barrier Functions (CBFs) utilized in data-driven safety. Barrier certificates serve as Lyapunov-like functions that prove a system trajectory remains within a safe set without explicitly computing reachable sets, thus avoiding discretization errors. This approach has been successfully extended to verify stochastic systems against fragments of LTL, providing lower bounds on the probability of satisfying complex temporal properties [363]. For nonlinear systems, the integration of automata-theoretic methods with barrier certificates allows for the decomposition of verification tasks into simpler constraints over the continuous state space, facilitating the verification of LTL properties without explicit system abstraction [364]. Moreover, converse barrier certificates have been established for finite-time safety, proving the existence of time-dependent certificates if and only if the system is safe over a given horizon [365].

The synergy between temporal logic and reachability is further exemplified in the design of behavior trees and non-stationary policies, which provide structured ways to implement verified logic. Goals specified in finite-trace LTL can be systematically compiled into behavior trees that guarantee goal satisfaction for any successful trace, relaxing the synthesis problem while maintaining formal guarantees [366]. Similarly, constraint satisfaction propagation techniques enable the synthesis of non-stationary policies that maximize the probability of satisfying time-sensitive goals by linking high-level logical specifications with low-level Markov decision processes [367]. Finally, the robust-

ness of these guarantees is increasingly being scrutinized through the lens of robust LTL, which introduces many-valued semantics to ensure that small violations of environmental assumptions lead only to small violations of system guarantees, thereby enhancing the reliability of verified systems in unpredictable real-world settings [368]. Together, these advancements in HJ reachability, temporal logic trees, and barrier-based verification form a comprehensive toolkit for ensuring the safety, stability, and mission success of next-generation robotic systems. While these methods provide absolute guarantees for individual agents under known bounds, integrating them into the stochastic, reward-maximizing framework of Reinforcement Learning remains a key challenge; the subsequent section addresses this by exploring how such formal tools can be adapted to define risk-aware constraints and probabilistic safety shields within RL algorithms.

7.5 Risk-Aware Planning and Probabilistic Safety Constraints in Reinforcement Learning

7.5 Risk-Aware Planning and Probabilistic Safety Constraints in Reinforcement Learning

While the formal verification techniques discussed in the previous section provide rigorous, worst-case guarantees for known dynamics, their application to high-dimensional, learning-based robotic systems often faces scalability hurdles or requires precise models that are unavailable in practice. Consequently, the deployment of reinforcement learning (RL) agents in real-world environments necessitates a complementary paradigm: shifting from optimizing expected cumulative rewards to managing risk and ensuring probabilistic safety guarantees. Traditional Constrained Markov Decision Processes (CMDPs) typically formulate safety as a constraint on the expected cumulative cost; however, this expectation-based formulation can be insufficient for safety-critical applications where even a single transient violation may lead to catastrophic failure. To address this, recent research has focused on frameworks that integrate safety directly into the learning process by adapting concepts from formal verification—such as reachability analysis and barrier functions—to stochastic settings, thereby creating dynamic risk budgets that account for model uncertainty and environmental noise.

A pivotal advancement in this domain is the integration of reachability analysis with constrained RL to define and maintain feasible sets of states, effectively bridging the gap between the deterministic guarantees of Section 7.4 and the stochastic nature of RL. Unlike standard CMDP approaches that may allow transient violations provided the long-term average cost remains low, reachability-constrained methods enforce persistent, state-wise safety. The “Reachability Constrained Reinforcement Learning” framework addresses the limitations of prior conservative estimations by using reachability analysis to establish a self-consistency condition, characterizing the feasible set via a safety value function [369]. This approach ensures that the learned policy operates within the largest possible feasible set while guaranteeing convergence to a local optimum where safety constraints are strictly satisfied. Similarly, the “Iterative Reachability Estimation for Safe Reinforcement Learning” introduces a framework known as RESPO, which optimizes for rewards within a feasible set while guaranteeing entry into this set from outside regions with minimal cumulative discounted violations [370]. These methods translate the theoretical rigor of Hamilton-Jacobi reachability into practical policy optimization tools for stochastic settings.

Complementing reachability-based approaches, Lyapunov-based methods offer a powerful mechanism for enforcing stability and safety through local linear constraints that guarantee global properties, serving as a probabilistic analogue to the barrier certificates discussed previously. The

“Lyapunov Barrier Policy Optimization” (LBPO) method exemplifies this by utilizing a Lyapunov-based barrier function to restrict policy updates to a safe set at every training iteration [371]. This technique not only significantly reduces constraint violations during training compared to baselines like CPO and SDDPG but also allows users to explicitly control the conservativeness of the agent. Furthermore, the integration of Lyapunov functions with uncertainty quantification has led to robust frameworks capable of operating in partially observable and memory-based environments. For instance, a “Lyapunov-based uncertainty-aware safe reinforcement learning” model combines Lyapunov functions with Transformer architectures to process long time horizons, ensuring safety even under out-of-distribution observations by estimating the probability of constraint violations [372]. These approaches effectively transform complex trajectory-based safety constraints into local conditions that can be efficiently verified and enforced during the learning process.

To address the inherent uncertainties in dynamical models where exact reachable sets are intractable, probabilistic safety constraints and distributional perspectives have gained prominence. “Safe Model-Based Reinforcement Learning with an Uncertainty-Aware Reachability Certificate” proposes a distributional reachability certificate (DRC) that characterizes robustly persistently safe states, addressing the lack of safety robustness against model uncertainties in training [373]. By leveraging a line search method and a shield policy, this framework achieves comparable returns with significantly fewer constraint violations. Additionally, “Safe Distributional Reinforcement Learning” formalizes safety within a distributional RL setting, accepting various definitions such as bounds on Conditional Value at Risk (CVaR), variance, or the probability of reaching bad states [374]. This perspective allows for more nuanced risk management, moving beyond simple expectation constraints to capture the tail risks that are critical in high-stakes robotic applications.

Planning algorithms operating under dynamic risk budgets further enhance the adaptability of safe RL systems, enabling them to navigate the trade-off between exploration and safety. “Safe Reinforcement Learning via Hierarchical Adaptive Chance-Constraint Safeguards” introduces Adaptive Chance-constrained Safeguards (ACS), which use the safety recovery rate as a surrogate chance constraint to iteratively ensure safety during exploration and post-convergence [375]. This method theoretically guarantees forward invariance to the safe set through relaxed probabilistic constraints, demonstrating nearly zero violations in real-world stochastic settings. Moreover, “Sampling-based Safe Reinforcement Learning for Nonlinear Dynamical Systems” develops a single-stage, sampling-based approach that learns controllers with classical convergence guarantees while satisfying hard safety constraints throughout training and deployment, effectively bridging the gap between control-theoretic guarantees and RL convergence [376].

The synthesis of these methodologies highlights a growing trend toward unifying formal verification tools with data-driven learning to create robust safety shields. Techniques such as “Provably Safe Reinforcement Learning via Action Projection using Reachability Analysis and Polynomial Zonotopes” employ reachability analysis to create safety shields that project unsafe actions onto the closest safe action, handling nonlinear effects and dynamic obstacles with robust guarantees [377]. Similarly, “Safety-Guided Deep Reinforcement Learning via Online Gaussian Process Estimation” utilizes online Gaussian Process estimation to approximate the derivative of a safety cost function, treating it as a candidate Lyapunov function to guide exploration in unknown environments [378]. Collectively, these frameworks represent a significant leap forward in enabling RL agents to operate safely under uncertainty, balancing the dual objectives of performance optimization and rigorous probabilistic safety adherence. While these methods provide strong guarantees for individual agents, the challenge escalates when scaling to multi-agent systems; the following section addresses this by exploring compositional design and decentralized safety mechanisms that extend these risk-aware

principles to large-scale, interacting robotic fleets.

7.6 Compositional Design, Contracts, and Decentralized Safety for Multi-Agent Systems

7.6 Compositional Design, Contracts, and Decentralized Safety for Multi-Agent Systems

While the risk-aware planning and probabilistic safety mechanisms discussed in the preceding section provide robust guarantees for individual agents operating under uncertainty, the challenge escalates dramatically when scaling these principles to large-scale, interacting robotic fleets. The deployment of distributed sensor networks, autonomous vehicle swarms, and cooperative legged robots introduces formidable hurdles due to the curse of dimensionality, where traditional centralized synthesis methods often fail to scale gracefully. As the number of interacting subsystems increases, monolithic approaches become computationally intractable for real-world applications involving thousands of degrees of freedom. To address this scalability bottleneck while preserving the rigorous safety guarantees established in Section 7.5, recent research has pivotally shifted towards compositional design frameworks. This paradigm decomposes global safety specifications into manageable local subproblems, relying on the synthesis of robust set-invariance controllers and the rigorous definition of assume-guarantee contracts to enable decentralized optimization without sacrificing formal guarantees for the aggregate system.

At the core of compositional safety is the concept of assume-guarantee contracts, which serve as formal interfaces between interconnected subsystems. These contracts characterize the admissible sets of states and controls for individual agents, ensuring that if each subsystem satisfies its local contract under the assumption that its neighbors satisfy theirs, the global safety specification is met. A seminal approach in this domain involves the compositional synthesis of decentralized robust set-invariance controllers for large-scale linear systems. By utilizing an optimization-based framework that iteratively computes subsystem-level assume-guarantee contracts in the form of robust control invariant sets, researchers have demonstrated the ability to design correct-by-construction controllers that require only local state information for online implementation. This methodology has been successfully validated on systems with dimensions reaching into the thousands, proving its efficacy in managing complex coupled dynamics without centralized coordination [379].

Further advancing this theoretical foundation, novel parameterizations of assume-guarantee contracts have been developed to ensure that the set of parameters composing correctly forms a convex set. This convexity is critical, as it allows the verification and synthesis problems for the aggregate system to be broken down into solving small, independent convex programs for individual subsystems. The introduction of a potential function that quantifies the distance of contracts from a correct composition enables subsystems to negotiate contract parameters in a distributed manner using gradient information derived from local synthesis problems. This approach significantly enhances scalability, having been applied to distributed Model Predictive Control (MPC) problems in power systems with tens of thousands of dimensions [380]. The utility of these contracts extends beyond linear time-invariant systems to time-varying dynamics, where they facilitate the handling of finite-time reach-avoid and infinite-time invariance specifications through a rich parameterization that preserves compositional correctness [381].

In the context of heterogeneous and nonlinear multi-agent systems, such as cooperative legged robots, compositional safety is increasingly integrated with Control Barrier Functions (CBFs) to bridge the gap between abstract contracts and physical dynamics. For instance, safety-critical coor-

dination of legged robots has been achieved by leveraging CBFs combined with forward reachable sets within a layered control architecture. In this framework, a top-level planner generates safety-ensured trajectories for each agent by accounting for dynamic constraints and inter-agent coupling, while middle and low-level controllers track these trajectories and manage whole-body motions. This layered approach guarantees safe coordination during locomotion even in the presence of fixed holonomic constraints and physical environmental limitations [382]. Similarly, for cooperative locomotion involving interconnected agents, CBFs are employed to maintain desired formations and avoid obstacles, utilizing interconnected single rigid body dynamics to generate optimal ground reaction forces that respect interconnection dynamics [383].

To handle the complexities of nonlinear interconnected systems where traditional Sum-of-Squares (SOS) techniques struggle with dimensionality, compositional methods have been adapted to synthesize barrier functions via assume-guarantee contracts. By formulating the synthesis problem as a set of small-size problems that construct local contracts, a negotiation scheme among subsystems can be established at the contract level. This strategy effectively mitigates the scalability issues inherent in monolithic SOS approaches, as demonstrated in applications ranging from vehicle platooning to room temperature regulation [384]. Furthermore, the integration of signal temporal logic (pSTL) as a template for contracts allows for the formal description of subsystem behaviors, enabling the composition of robust control invariant sets (RCIs) for the entire network. An epigraph algorithm with linear complexity for sparse networks can then solve for valid contracts, ensuring that the state of each subsystem remains within safe sets even under severe contingencies like topology changes [385].

Decentralized optimization also plays a pivotal role in enforcing coupling constraints without central oversight, extending these compositional principles to active navigation and stochastic environments. In multi-robot navigation, distributed controller synthesis frameworks leverage CBFs to formulate collision avoidance as constraints on control inputs within a state-dependent network optimization problem. This approach ensures that the resulting controller is distributed, satisfies safety constraints at all times, and achieves asymptotic optimality for general nonlinear dynamics, as validated in hardware experiments with teams of differential-drive robots [386]. Additionally, for stochastic multi-robot systems, Distributed Model Predictive Covariance Steering (DMPCS) blends covariance steering theory with distributed optimization to steer state distributions to desired targets while satisfying probabilistic safety constraints. By employing a decentralized consensus-based algorithm using the Alternating Direction Method of Multipliers (ADMM), DMPCS achieves safety and scalability in problems involving hundreds of robots, outperforming standard stochastic MPC approaches [387].

The extension of these principles to learning-based control further underscores their versatility, allowing for the safe integration of data-driven policies within a compositional framework. Distributed Model Predictive Safety Certification (DMPSC) schemes have been developed to ensure state and input constraint satisfaction when applying arbitrary learning-based control algorithms to uncertain distributed linear systems. By utilizing a distributed tube-based MPC concept where subsystems negotiate local tube sizes, these frameworks allow for performance improvements over robust distributed MPC while rigorously guaranteeing safety [388]. Moreover, the synthesis of separable controlled invariant sets allows for modular local control design, reducing correct-by-construction synthesis problems to subproblems of modest dimension through optimization over linear matrix inequalities. This method facilitates the compositional solution of local control synthesis problems, addressing the curse of dimensionality in dynamically coupled linear subsystems [389].

Finally, the application of assume-guarantee reasoning extends to autonomous systems incorporating deep learning components, where contracts encode the input-output behavior of neural network modules. This compositional verification approach allows designers to model learning-enabled components working alongside traditional controllers, providing a pathway to scalable formal verification for complex autonomous vehicles [390]. Collectively, these advancements in compositional design, contract-based synthesis, and decentralized optimization represent a critical frontier in ensuring the safe deployment of large-scale, heterogeneous, and distributed robotic systems. However, while these frameworks offer rigorous theoretical guarantees for multi-agent coordination, their practical efficacy ultimately hinges on validation within complex, high-dimensional environments where analytical models may not fully capture real-world stochasticity. The following section addresses this necessity by exploring simulation-based validation, falsification, and chaos engineering techniques designed to stress-test these compositional guarantees before real-world deployment.

7.7 Simulation-Based Validation, Falsification, and Chaos Engineering

7.7 Simulation-Based Validation, Falsification, and Chaos Engineering

While the compositional frameworks and decentralized safety contracts discussed in the preceding section provide rigorous theoretical guarantees for multi-agent coordination, their practical efficacy ultimately hinges on validation within complex, high-dimensional environments where analytical models may not fully capture real-world stochasticity. The deployment of autonomous robotic systems in safety-critical settings necessitates rigorous validation protocols that extend beyond standard performance metrics. Given the prohibitive costs and inherent dangers of exhaustive real-world testing, simulation-based validation has emerged as a cornerstone for ensuring system reliability. However, the efficacy of such validation is often constrained by the fidelity of the simulator and the completeness of the test scenarios. To address these challenges, the research community has developed sophisticated black-box safety validation techniques that integrate joint falsification and fidelity optimization, automated red-teaming, and chaos engineering principles to systematically probe system resilience, thereby bridging the gap between formal contract satisfaction and operational robustness.

A primary challenge in simulation-based validation is the trade-off between computational efficiency and environmental accuracy. High-fidelity simulators accurately model real-world dynamics but are computationally expensive, limiting the scale of exhaustive testing, whereas low-fidelity simulators offer speed but risk yielding false conclusions due to modeling inaccuracies. To navigate this dichotomy, recent frameworks propose a joint falsification and fidelity optimization approach. This methodology mathematically combines counterexample searches with dynamic simulator fidelity improvements, facilitating a more efficient exploration of critical environmental configurations. By theoretically analyzing counterexample sensitivity and sample complexity, these approaches enable a targeted testing process that optimizes computational resources while enhancing confidence in safety validation outcomes [391]. Furthermore, multi-fidelity Bayesian optimization frameworks have been developed to automatically determine the appropriate level of fidelity for specific safety evaluations. These methods intelligently switch between inexpensive, inaccurate information from low-fidelity simulators and expensive, accurate data from high-fidelity environments, achieving falsification performance comparable to single-fidelity methods but at a significantly reduced computational cost [392]. Extending this concept, algorithms utilizing transfer learning have been introduced to improve the efficiency of iterative safety validation. By encoding knowledge from previous validation tasks into action value functions and attention weights, these methods can accelerate the discovery of failures in related systems, even when failure modes differ substantially

[393].

Beyond fidelity management, the core objective of safety validation is falsification: the systematic search for input sequences or environmental disturbances that cause the system to violate safety specifications. In the context of black-box systems where internal dynamics are unknown, stochastic optimization and reinforcement learning (RL) have become predominant tools. Adaptive Stress Testing (AST) formulates the search for the most likely path to a failure event as a Markov Decision Process, utilizing RL to optimize the discovery of rare failure scenarios in partially observable and continuous-valued systems [394]. To further enhance the efficiency of finding failures in high-fidelity simulations, hybrid approaches have been proposed that initially identify failures in low-fidelity environments and then adapt them to high-fidelity settings using backward algorithms trained on expert demonstrations. This strategy significantly reduces the number of expensive high-fidelity simulation steps required to validate complex autonomous vehicle systems [395]. Additionally, dimensionality reduction techniques have been employed to prioritize the exploration of effective policy parameters, thereby coping with the scalability issues of black-box optimization in high-dimensional robotic control spaces [396].

Complementing falsification efforts, automated red-teaming has gained prominence as a method for evaluating the robustness of AI-driven systems, particularly those integrating large language models or complex decision-making modules. Red-teaming involves generating adversarial test cases to expose hidden vulnerabilities and failure modes. For instance, automated safety scenario red-teaming frameworks employ methods such as semantically aligned augmentation and adversarial knowledge injection to algorithmically generate diverse test suites that cover semantic equivalence and adversarial settings. These evaluations have revealed statistically significant performance discrepancies in state-of-the-art models, highlighting critical concerns for physical safety in deployed systems [397]. In black-box settings where query access is limited, Bayesian red-teaming methods utilize Bayesian optimization to iteratively identify diverse positive test cases leading to model failures, offering a query-efficient alternative to brute-force approaches [398]. Furthermore, parametric red-teaming through “unalignment” has been proposed to expose hidden biases and harms by tuning model parameters rather than just crafting input prompts, demonstrating high success rates in bypassing safety guardrails in both proprietary and open-source models [399].

Finally, the application of chaos engineering principles provides a paradigm shift from passive testing to active resilience verification. Originally developed for distributed software systems, chaos engineering involves subjecting systems to unexpected conditions and perturbations to evaluate their ability to self-heal and maintain functionality. In the realm of cyber-physical systems (CPS), this methodology is adapted to introduce runtime uncertainties, such as hardware faults, cyberattacks, or environmental disturbances, to assess system resilience in real-time operations. By applying chaos engineering analyses to industrial CPS, researchers can predict environmental changes and apply mitigation measures that limit the blast radius of adverse events, thereby taming systemic uncertainty [340]. Frameworks like CHeSS have been proposed to systematically evaluate self-adaptive and self-healing systems by leveraging chaos engineering to subject them to unanticipated failures, providing crucial insights into their operational limits and recovery capabilities [400]. Moreover, simulation-based falsification techniques are increasingly being viewed through the lens of chaos engineering, where the goal is not merely to find a single bug but to characterize the decision manifolds between desired and undesired behaviors, offering a more detailed understanding of system reliability under dynamic conditions [401]. Collectively, these advanced simulation-based validation, falsification, and chaos engineering techniques form a comprehensive toolkit for ensuring the safety and robustness of next-generation autonomous robotics before real-world deployment. The

insights gained from these stress tests—regarding failure modes, resource utilization under duress, and resilience limits—directly inform the subsequent phase of development: the precise estimation of computational resources and reliability metrics required to meet functional safety specifications in embedded environments.

7.8 Resource Estimation and Reliability from Functional Safety Specifications

7.8 Resource Estimation and Reliability from Functional Safety Specifications

The insights gained from the stress tests described in the preceding section—specifically regarding failure modes, resource utilization under duress, and resilience limits—directly inform the subsequent phase of development: the precise estimation of computational resources and reliability metrics required to meet functional safety specifications. The deployment of autonomous robotic systems in safety-critical environments necessitates a rigorous alignment between these high-level functional safety requirements, derived from extensive validation, and the underlying computational resource constraints. Traditionally, reliability and fault tolerance in embedded cyber-physical systems (CPS) have been treated as platform-specific implementation details, often addressed only after the functional design has been finalized. However, this decoupled approach frequently leads to suboptimal designs where safety-critical functionalities are either over-provisioned with expensive hardware resources or, conversely, fail to meet strict safety-integrity levels due to unanticipated computational bottlenecks exposed during simulation-based falsification. A paradigm shift is required to address reliability at the functional safety specification level, thereby widening the scope for resource optimization by targeting only those functionalities that are genuinely safety-critical rather than applying blanket redundancy across the entire platform [402]. By formally modeling functional reliability annotations overlaid on safety-critical specifications, it becomes possible to extract constraint satisfaction problems that determine the optimal set of physical resources and computation bandwidth required to meet specific reliability targets early in the design phase [402].

The complexity of modern robotic systems, which integrate heterogeneous smart technologies, poses significant challenges to verifying that physical and random aspects of various modules adhere to strict safety and performance requirements identified during validation [403]. To overcome these challenges, researchers have proposed generic frameworks for reliability modeling that encompass the environment, communication, and computational layers of the system architecture. These frameworks allow for the formalization of reliability models for architectural components, ensuring that the derived reliability metrics cover almost all possible failure modes required for general cyber-physical systems [404]. Such holistic modeling is essential because the reliability of a complex system is the most critical requirement; an unreliable system can lead to service disruptions, property damage, financial losses, and even fatalities. Consequently, the ability to estimate resource needs based on formal functional safety specifications enables designers to balance the trade-offs between temporal redundancies and the replication of physical resources, a capability that is increasingly vital for software-based control functionalities [402].

Furthermore, the integration of machine learning components into CPS introduces new uncertainties that complicate traditional reliability estimation methods, particularly given the black-box nature of many controllers validated in Section 7.7. Dependability assurance for systems embedding machine learning components, often referred to as learning-enabled systems, requires a quantitative notion of assurance that goes beyond binary pass/fail metrics. This involves characterizing assurance measures as a probabilistic quantification of confidence that the system possesses the necessary functional capabilities and dependability attributes [405]. To support this, advanced methodologies

utilize Satisfiability Modulo Theories (SMT) and Integer Linear Programming (ILP) solvers to solve the optimization problems derived from reliability annotations. These solvers facilitate the determination of optimal resource sets, as demonstrated in case studies involving Satellite Launch Vehicle Navigation, Guidance, and Control systems, where early estimation proved feasible and critical for mission success [402].

In addition to static resource estimation, dynamic risk management and runtime verification are crucial for maintaining safety integrity under distribution shifts and operational uncertainties. Robotic systems designed and validated in simulators often behave differently when deployed in the real world, necessitating predictive runtime verification algorithms that can account for distribution shifts between design-time and deployment-time data distributions. Techniques based on robust conformal prediction allow for the construction of interpretable algorithms that quantify the relationship between calibration data, desired confidence levels, and permissible distribution shifts, providing statistically valid guarantees for stochastic CPS [406]. This is particularly important when dealing with the “black-box” nature of many learning-enabled controllers, where formal verification of the internal logic is intractable, and reliance must be placed on statistical guarantees derived from functional specifications and runtime monitoring.

The interplay between safety and security further complicates resource estimation, as failures caused by cyber attacks—often simulated via chaos engineering—must be considered alongside physical faults. Traditional risk assessment methods often overlook security failures, leading to an incomplete picture of system reliability. Integrated approaches, such as the Cyber Layer Of Protection Analysis (CLOPA), extend existing frameworks to include failures caused by cyber attacks, providing a rigorous mathematical formulation that expresses the quantitative trade-off between designing a highly reliable versus a highly secure CPS [407]. Ignoring these security-induced reliability losses can result in architectures that appear robust against random hardware failures but are vulnerable to targeted adversarial exploits. Therefore, early-stage resource estimation must incorporate threat modeling and risk assessment to ensure that the allocated computational resources are sufficient to handle both accidental faults and malicious intrusions [408].

Moreover, the timing predictability of these systems on modern multi-core platforms is a decisive factor in meeting safety specifications. Temporal interference caused by contention on shared resources such as caches and memory buses can significantly increase task execution times, potentially violating real-time constraints essential for safety [409]. Analytical techniques that bound this temporal interference are therefore integral to the resource estimation process, ensuring that the theoretical safety guarantees derived from functional specifications hold true during actual execution. By combining formal functional safety specifications with detailed reliability annotations and timing analysis, engineers can create a comprehensive resource estimation methodology. This approach not only optimizes the hardware footprint and energy consumption of robotic systems but also ensures that they meet the stringent safety-integrity levels required for deployment in unstructured and hazardous environments. Ultimately, bridging the gap between abstract safety goals validated through simulation and concrete resource allocation is fundamental to the development of trustworthy, accountable, and legally compliant autonomous systems.

8 Applications, Benchmarking Ecosystems, and Evaluation Protocols

8.1 Real-World Manipulation Benchmarks and Datasets

8.1 Real-World Manipulation Benchmarks and Datasets

While simulation provides a safe sanctuary for initial policy training, the ultimate validation of offline and preference-based reinforcement learning lies in unstructured real-world environments. The transition from controlled laboratory settings to practical deployment is critically dependent on the availability of rigorous, reproducible benchmarks and high-quality datasets. Historically, progress in this domain was hindered by a reliance on simulation-only evaluations or proprietary setups that prevented direct algorithmic comparison. To bridge this gap, the research community has cultivated a diverse ecosystem of benchmarks targeting specific manipulation capabilities, ranging from dexterous grasping and bi-manual assembly to complex long-horizon tasks in obstructed environments. These initiatives provide the standardized metrics, open-source hardware designs, and large-scale data collections essential for validating the efficacy of learning-based methods before they are subjected to the dynamic complexities of mobile platforms discussed in subsequent sections.

A primary focus of recent benchmarking efforts is dexterous manipulation, which demands that robots handle objects with high degrees of freedom using multi-fingered hands. The [410] benchmark addresses the limitations of parallel grippers by evaluating policies on diverse articulated objects within a physical simulator, emphasizing generalization to unseen object categories through 3D representation learning. Complementing this, the [411] project offers a comprehensive simulator for bimanual dexterous hands, featuring thousands of target objects and tasks calibrated to human motor skill levels, although it highlights the current struggles of existing algorithms in multi-task and few-shot settings. For real-world validation, the [412] and its associated benchmarking study [413] provide access to TriFinger robotic platforms via the cloud, facilitating fair comparisons of structured policies that combine classical robotics with modern optimization to achieve sample efficiency and reliability on real hardware. Furthermore, the [414] provides a massive synthetic dataset of stable grasps, which has been shown to significantly outperform previous datasets when training dexterous grasp synthesis algorithms, bridging the gap between simulation data generation and real-world application.

Beyond single-arm dexterity, bi-manual manipulation introduces unique challenges regarding coordination, collision avoidance, and force control. The [415] tackles long-horizon complexity by focusing on real-world furniture assembly, providing over 5,000 demonstrations and a realistic simulator to test offline RL and imitation learning algorithms on tasks that require sequencing multiple skills. Similarly, the [416] study demonstrates the feasibility of transferring deep RL policies trained in simulation to real dual-arm systems for magnetic block assembly, achieving significant success rates without real-world fine-tuning by carefully addressing action delays and perception noise. For tasks requiring tactile feedback, the [417] system integrates low-cost high-resolution tactile sensors with dual arms, offering a suite of tasks like bi-reorienting and bi-gathering that highlight the necessity of touch for robust manipulation. Additionally, the [418] framework introduces a role-assignment strategy where one arm stabilizes the environment while the other acts, demonstrating superior performance in complex tasks like zipping jackets with minimal demonstration data.

The challenge of operating in cluttered or obstructed environments represents another critical frontier addressed by recent benchmarks. The [419] dataset stands out by providing over 110,000 contact-rich manipulation sequences captured in the real world, featuring multi-modal perception

including vision, force, audio, and language descriptions, which is vital for training agents to generalize across hundreds of diverse skills. In the domain of warehouse automation, the [420] offers a large-scale dataset from an Amazon warehouse context, focusing on object singulation from containers with heterogeneous contents to evaluate perception and manipulation under realistic clutter. For more constrained scenarios, the [421] provides dynamic scenes and real time-series data for block stacking, revealing that models trained on less constrained datasets often fail when workspace restrictions are introduced. Furthermore, the [422] paper proposes a modular framework combining object-centric generative models with motion planning and RL to navigate obstructed spaces, demonstrating the need for integrated approaches in such settings.

To lower the barrier to entry and ensure reproducibility, standardized evaluation platforms have also emerged. The [423] introduces a low-cost, self-contained hardware stack designed to make benchmarking accessible to a wider range of researchers, complete with a grasping benchmark and dataset. The [424] provides a flexible simulation environment with 31 manipulation tasks and baseline algorithms to encourage direct comparisons. The [425] focuses on industrial assembly tasks using 3D-printed parts and digital twins to assess planning and fault recovery capabilities. Moreover, the [426] study critically evaluates offline RL methods using data collected from safe operations of related tasks, arguing that such realistic data sources are more practical for real-world deployment than contrived simulation buffers. Finally, the [427] facilitates the collection of high-quality demonstration data across various robot morphologies using affordable VR headsets, supporting the data-hungry nature of modern learning algorithms. Collectively, these benchmarks and datasets form the backbone of current research, enabling the community to systematically address the sample inefficiency and generalization gaps inherent in deploying learning-based manipulation policies. While these frameworks rigorously test stationary manipulators, the following section extends this evaluation paradigm to the coupled challenges of locomotion and mobile manipulation, where stability and navigation add further layers of complexity.

8.2 Locomotion and Mobile Manipulation Evaluation Frameworks

8.2 Locomotion and Mobile Manipulation Evaluation Frameworks

Building upon the foundation of stationary manipulation benchmarks discussed previously, the deployment of autonomous systems in unstructured environments necessitates a paradigm shift toward evaluating coupled locomotion and mobile manipulation capabilities. While Section 8.1 established rigorous standards for dexterous grasping and assembly in fixed workspaces, mobile platforms introduce compounded complexities involving dynamic terrain negotiation, whole-body coordination, and long-horizon navigation. Traditional evaluations relying on isolated control objectives or simplified toy tasks often fail to capture these multifaceted challenges. To address this, recent research has cultivated specialized ecosystems designed to facilitate robust algorithm comparison, ranging from comprehensive imitation learning suites to hierarchical sim-to-real transfer protocols. These frameworks are critical for assessing not only raw task performance but also generalization capabilities, data efficiency, and resilience to the distributional shifts inherent in mobile deployment, serving as a vital precursor to the social interaction metrics explored in the subsequent section.

A cornerstone in the evaluation of locomotion policies is the emergence of benchmarks specifically tailored for imitation learning, which address the limitations of reinforcement learning in sparse reward settings. The [428] framework represents a significant stride in this direction by providing a diverse set of environments encompassing quadrupeds, bipeds, and musculoskeletal human models.

Unlike previous benchmarks that focused on narrow domains, [428] offers datasets ranging from real noisy motion capture data to ground-truth expert and sub-optimal demonstrations, enabling researchers to evaluate algorithms across a spectrum of difficulty levels. Furthermore, it integrates interfaces for dynamics randomization and partially observable tasks, ensuring that learned agents are tested against the uncertainties inherent in physical systems. This comprehensive approach allows for a more nuanced assessment of policy robustness, moving beyond simple success rates to include metrics for stability and adaptability under varying environmental conditions.

Parallel to locomotion-specific benchmarks, the field of mobile manipulation requires evaluation frameworks that can handle the coupled complexity of navigation and interaction. The [429] framework addresses this by lifting the action space from low-level joint controls to high-level subgoals managed by a motion generator. By benchmarking interactive navigation and mobile manipulation tasks in photo-realistic simulations, [429] demonstrates that hierarchical approaches significantly outperform standard reinforcement learning baselines in long-horizon scenarios. This framework highlights the importance of evaluating not just the final task success, but also the efficiency of exploration and the ability to alternate seamlessly between navigation and manipulation phases. The superior transferability observed in [429] suggests that future evaluation protocols must prioritize architectures that decouple high-level planning from low-level execution to ensure practical deployability on real hardware.

Beyond structured benchmarks, the efficacy of evaluation frameworks is increasingly being tested against simple yet powerful baselines that challenge the necessity of complex deep learning architectures. The study presented in [430] serves as a critical reality check for the community, proposing a model-free open-loop strategy using simple oscillators. The findings indicate that such baselines can achieve respectable performance with a fraction of the parameters required by deep reinforcement learning algorithms, while exhibiting greater robustness to sensor noise and failure modes. This underscores a vital gap in current evaluation practices: the need to compare sophisticated learning-based methods against minimalistic, physics-informed baselines to ensure that added complexity yields genuine improvements in reliability and sim-to-real transfer. Consequently, modern frameworks must incorporate these open-loop controls as standard reference points to prevent the overestimation of learning algorithm capabilities.

The challenge of transferring policies from simulation to reality remains a central theme in evaluation, prompting the development of hierarchical sim-to-real strategies. Frameworks like those discussed in [431] utilize domain randomization and hierarchical policy structures to enable zero-shot transfer of coordinated behaviors. By training low-level goal-reaching skills in simulation and employing them as action spaces for high-level controllers, these methods provide a modular paradigm for evaluating increasingly complex multi-agent behaviors. Similarly, the [432] initiative introduces an obstacle course inspired by dog agility competitions to quantify agility through time-based scoring. This benchmark encourages the development of controllers that balance speed with controllability, utilizing both specialist skills and generalist transformer-based policies. The inclusion of such dynamic, whole-body tasks in evaluation frameworks ensures that policies are assessed on their ability to handle the continuous, high-frequency demands of real-world locomotion.

Furthermore, the evaluation of loco-manipulation capabilities is expanding to include non-prehensile interactions and whole-body coordination. Research such as [433] and [434] pushes the boundaries by evaluating robots on tasks that require using legs for manipulation, such as climbing walls or pressing buttons, without dedicated arms. These studies emphasize the need for benchmarks that assess the emergence of novel behaviors and the robustness of policies against external perturbations. The integration of visual feedback and adaptive control, as seen in [435], further necessitates

evaluation protocols that measure performance in visually complex and dynamically changing environments.

In conclusion, the landscape of evaluation frameworks for locomotion and mobile manipulation is evolving from isolated task success metrics to holistic ecosystems that stress-test generalization, robustness, and transferability. By leveraging comprehensive datasets like those in [428], adopting hierarchical abstractions as seen in [429], and maintaining rigorous baselines as suggested by [430], the community can better gauge the true readiness of robotic systems for real-world deployment. These advancements in physical robustness and navigational agility provide the essential substrate for the next frontier: operating safely and effectively within human-populated spaces, where social compliance and interaction protocols become the primary measures of success.

8.3 Human-Robot Interaction and Social Navigation Protocols

8.3 Human-Robot Interaction and Social Navigation Protocols

While the preceding section detailed frameworks for assessing locomotion and mobile manipulation in unstructured physical terrains, the deployment of autonomous robots in human-populated environments necessitates a distinct shift in evaluation priorities. Here, metrics must extend beyond path efficiency and collision avoidance to capture the nuanced dynamics of social compliance, safety, and subjective human perception. As robots transition from isolated industrial settings to dynamic domestic and public spaces, the research community has developed specialized benchmarking ecosystems designed to quantify complex human-robot interaction (HRI) attributes such as comfort, trust, and fairness—dimensions that traditional robotic performance indicators often fail to address. This focus on social and perceptual validity serves as a critical bridge between the physical robustness discussed previously and the rigorous sim-to-real predictivity metrics explored in the following section.

A primary focus of modern evaluation protocols is the establishment of standardized simulation environments that enable the reproducible testing of social navigation algorithms. One such framework is [436], which provides a lightweight, extensible 2D simulation environment specifically designed to abstract away perception challenges and focus explicitly on social navigation behaviors. This platform enables researchers to compare diverse learning approaches, including Learning from Demonstration (LfD) and Reinforcement Learning (RL), against a suite of benchmark scenarios. By facilitating the comparison of human-engineered policies with model-based learning approaches, [436] helps ground future research in data efficiency and task performance metrics that are directly transferable to real-world autonomy. Similarly, [437] offers an advanced simulation environment capable of generating a wide variety of social scenarios. It supports the integration of both hand-crafted and data-driven reward signals, allowing agents to be trained from scratch to navigate complex social situations. Studies utilizing [437] have demonstrated that agents trained with data-driven rewards exhibit superior social compliance compared to those relying on heuristic-based functions, highlighting the importance of flexible benchmarking tools in advancing policy learning.

Beyond pure simulation, there is a growing emphasis on grounded simulation testing frameworks that leverage real-world pedestrian data to ensure ecological validity, addressing the distributional shifts inherent in human behavior. [438] addresses the fragmentation in experimental procedures by providing a simulator with photo-realistic capabilities and curated scenarios derived from actual pedestrian interactions. This framework includes a comprehensive suite of metrics to quantify performance trade-offs, enabling consistent and interpretable comparisons across disparate social navigation methods. The move toward standardized metrics is further supported by [439], which

proposes a unified definition of a socially navigating robot based on principles such as safety, comfort, legibility, politeness, and responsiveness. This work advocates for a structured approach to developing benchmarks, datasets, and simulators, aiming to create a common language for evaluating social navigation that transcends specific hardware or algorithmic implementations.

In the realm of physical Human-Robot Collaboration (HRC), where the consequences of failure are immediate, safety guarantees become non-negotiable. [440] introduces a modular simulation framework featuring eight challenging HRC tasks, distinguished by the inclusion of a “safety shield” that provably guarantees human safety during the training of RL agents. This innovation bridges the gap between theoretical RL research and real-world deployment by ensuring that agents adhere to strict safety specifications while learning complex collaborative tasks. The efficacy of such safety mechanisms is critical, as traditional reactive planners often fail to account for the nuanced, game-theoretic nature of human-robot interactions. For instance, [441] demonstrates that leveraging non-cooperative game theory can produce trajectories that are indistinguishable from human paths in terms of social acceptability, a metric validated through large-scale user surveys. This underscores the necessity of evaluation protocols that incorporate human subjective feedback, such as Turing-test-style questionnaires, to measure true social integration.

Furthermore, evaluation protocols must increasingly address the ethical dimensions of robot behavior, including fairness and bias, which are unique to social domains. [442] argues that models trained solely on human observations may inadvertently replicate societal unfairness, such as discrimination or segregation. This work proposes a “relearning” component to detect and correct harmful outcomes, emphasizing that benchmarking must include metrics for equity and social justice, not just efficiency. Complementing this, [443] suggests computational models for detecting microaggressions and social exclusion, advocating for regulatory mechanisms that promote fairness in mixed human-robot groups.

The assessment of human perception remains a cornerstone of these protocols, linking objective robot actions to subjective human experiences. [444] utilizes the Robotic Social Attributes Scale (RoSAS) to measure social compatibility, validating that metrics like extra time ratio and extra distance ratio correlate strongly with human perceptions of warmth and competence. Similarly, [445] investigates the multidimensional nature of perceived safety, linking it to comfort, predictability, and trust through both subjective questionnaires and objective physiological data. These studies highlight that effective evaluation protocols must integrate psychological constructs with robotic performance data. Finally, [446] offers a methodology for defining canonical social scenarios and using in-situ questionnaires to evaluate performance, ensuring that benchmarks are realistic, scalable, and repeatable across different physical spaces. Together, these frameworks and protocols form a robust ecosystem for advancing safe, socially compliant, and ethically aligned robotics. By establishing rigorous standards for human-centric evaluation, this section sets the stage for the subsequent discussion on quantifying the fidelity of these simulations and predicting their transferability to physical reality.

8.4 Sim-to-Real Predictivity and Transfer Metrics

8.4 Sim-to-Real Predictivity and Transfer Metrics

Building upon the human-centric evaluation protocols and social compliance metrics established in the preceding section, the deployment of autonomous robots ultimately hinges on a more fundamental challenge: ensuring that policies trained in simulation perform reliably on physical hardware. While Section 8.3 addressed the *what* of evaluation—defining social safety, comfort, and fairness—

this subsection addresses the *how* of validation, focusing on rigorous methodologies to quantify the “sim-to-real” gap before costly and risky real-world deployment occurs. The central objective here is to assess “sim-to-real predictivity,” a property determining whether performance improvements observed in simulation reliably translate to progress in reality. Moving beyond simple success rates, this domain employs nuanced statistical correlations, probabilistic models, and adaptive strategies to bridge the divide between virtual training and physical execution, thereby setting the stage for the standardized multi-agent ecosystems discussed in the following section.

A cornerstone of quantifying this transferability is the Sim-vs-Real Correlation Coefficient (SRCC), a metric designed to measure the persistence of performance rankings from simulation to physical deployment. Research utilizing the Habitat-PyRobot Bridge revealed that standard simulation benchmarks often suffer from low predictivity, with an SRCC as low as 0.18 for navigation success metrics [447]. This discrepancy arises when agents exploit simulator imperfections, such as non-physical collision dynamics, that do not exist in the real world. Crucially, however, this study demonstrated that by tuning simulation parameters to better align with physical reality, the SRCC could be improved dramatically to over 0.84 [447]. This finding underscores a paradigm shift: the simulator must be treated not as a fixed truth, but as a tunable proxy whose fidelity requires continuous validation against real-world data.

Complementing correlation-based approaches, probabilistic dynamics models have emerged as efficient tools for predicting transfer performance without requiring extensive real-world rollouts. By training a probabilistic model alongside a policy and evaluating it against a fixed set of real-world trajectories, researchers can derive transfer metrics that are highly correlated with actual policy performance [448]. This approach streamlines the selection of training setups, hyperparameters, and randomization strategies, effectively acting as a filter to identify high-potential policies. Similarly, frameworks leveraging out-of-distribution (OOD) detection techniques have been proposed to rank policies generated through domain randomization. The VSDR (Validate on Sim, Detect on Real) score, for instance, combines OOD detection with simulation evaluation to significantly enhance policy ranking accuracy without necessitating additional real-world execution data [449]. These methods represent a critical evolution from brute-force real-world testing toward data-efficient predictive validation.

The role of domain randomization (DR) in bridging the reality gap is multifaceted, serving both as a primary training strategy and a subject of metric evaluation. While DR is widely adopted to expose models to diverse scenarios, theoretical frameworks suggest its success relies on mild conditions and the use of history-dependent policies to bound the sim-to-real gap [205]. However, the efficacy of DR is heavily contingent on the choice of randomization distributions. Benchmarking studies indicate that design choices, such as rendering quality and the specific type of randomization (e.g., textures versus distractors), significantly impact generalization to novel environments [450]. Furthermore, the assumption that higher-fidelity simulations invariably lead to better transfer has been challenged; systematic evaluations suggest that lower-fidelity simulations can sometimes yield superior transfer performance by preventing overfitting to inaccurate physics and enabling faster, large-scale learning [451].

To address the complexities of selecting optimal randomization distributions, several adaptive and offline methods have been developed and benchmarked. Comparative analyses of offline and on-line adaptive domain randomization methods, including SimOpt, BayRn, DROID, and DROPO, highlight distinct trade-offs: online methods are often constrained by the quality of the current policy, whereas offline methods may struggle with open-loop trajectory replay [452]. Specifically, methods like DROPO utilize limited offline datasets to explicitly estimate domain randomization

distributions, modeling parameter uncertainty to match real data [453]. Conversely, frameworks like AdaptSim focus on task-driven adaptation, optimizing simulation parameters specifically for target task performance rather than global dynamics matching, thereby achieving superior data efficiency [213].

Furthermore, evaluating sensor-specific discrepancies is critical for autonomous systems reliant on GPS and IMU data. Traditional direct comparisons of raw sensor data often fail to effectively quantify the gap; instead, employing a state-of-the-art state-estimation package as a “judge” allows for the isolation of sim-to-real discrepancies stemming specifically from sensor simulations [454]. In visual domains, generative label-to-image synthesis has been employed to bypass domain shift, utilizing transferability measures such as correlation analysis of Intersection over Union (IoU) to validate simulation-based testing results against real-life data [455].

Despite these methodological advances, significant limitations remain. In domains such as precision agriculture, current sim-to-real techniques are often insufficient due to the extreme complexity of required dynamics and visual fidelity, highlighting that simulation accuracy remains a bottleneck in specific high-stakes applications [456]. Consequently, the community is increasingly moving toward unified ecosystems and standardized metrics that assess full system performance in unmodified real-world settings, rather than relying on isolated simulation successes. The integration of causal inference perspectives, where domain randomization is viewed as an intervention to encourage invariance to irrelevant features, further refines our understanding of what makes a simulation predictive [457]. Ultimately, robust sim-to-real transfer demands a holistic evaluation strategy combining statistical correlation metrics, probabilistic modeling, and adaptive randomization. This rigorous approach to validating simulation fidelity provides the necessary foundation for the subsequent discussion on standardized ecosystems for multi-agent and modular robotics, where the scale and complexity of deployment make accurate predictivity even more critical.

8.5 Standardized Ecosystems for Multi-Agent and Modular Robotics

8.5 Standardized Ecosystems for Multi-Agent and Modular Robotics

While the preceding discussion on sim-to-real predictivity establishes the critical need for accurate metrics to bridge the reality gap, these metrics are only as effective as the environments in which they are applied. The transition of multi-robot reinforcement learning (MRRL) and modular robotics from theoretical simulations to robust real-world deployment is currently hindered by a fragmentation in evaluation ecosystems that often lacks the fidelity and standardization required for meaningful comparison. While single-agent benchmarks have matured significantly, the landscape for coordinated multi-robot systems and reconfigurable modular architectures remains disjointed, frequently relying on custom-built simulators that fail to capture complex physical dynamics or support reproducible hardware deployment. To address these challenges, the research community is developing standardized ecosystems that bridge the gap between virtual training and physical validation. These platforms emphasize unified interfaces, realistic dynamics, and scalable hardware testbeds, providing the necessary infrastructure to rigorously assess algorithms under the complex constraints of communication latency, heterogeneous sensing, and inter-robot collision avoidance that define real-world multi-agent operations.

A paramount example of such standardization is the emergence of platforms designed to evaluate MRRL algorithms on physical hardware with high reproducibility, directly addressing the predictivity concerns raised in previous sections. Traditional multi-agent benchmarks frequently overlook the nuances of robotic kinematics and the risks associated with sim-to-real transfer. In response,

[458] introduces a comprehensive framework that marries the Georgia Tech Robotarium with the OpenAI Gym interface. This integration allows researchers to train policies in a standardized software environment while deploying them on a physical fleet of robots, thereby addressing the critical deficit of platforms that support both realistic dynamics and actual sim-to-real performance evaluation. [458] provides a highly controllable environment featuring barrier certificate-based obstacle avoidance, ensuring safety during the evaluation of cooperative and competitive scenarios. By offering five novel scenarios inspired by common multi-robot system challenges, this platform facilitates a direct comparison of algorithmic suitability for physical deployment, moving beyond the idealized conditions of purely virtual agents.

Parallel to fleet-based standardization, there is a pressing need for modular frameworks that accommodate the diverse hardware configurations inherent in autonomous navigation and modular robotics. The complexity of integrating deep reinforcement learning with standard robotics middleware has led to the development of [459]. This framework exemplifies the shift toward modularity by fully integrating training and testing within ROS2 and Gazebo, allowing researchers to easily customize simulations by selecting new platforms, sensors, or models. [459] demonstrates the efficacy of this approach by benchmarking policies across various indoor and outdoor navigation tasks, providing a complete set of metrics that serve as a baseline for future autonomous navigation research. Such modular architectures are vital for modular robotics, where the ability to rapidly prototype and evaluate different morphologies without rewriting core control logic is a prerequisite for innovation.

The necessity for standardized ecosystems extends further into the realm of heterogeneous and self-assembling systems, where coordination strategies must adapt to dynamic changes in robot composition. Research into decentralized exploration by swarms of miniature robots highlights the importance of scalable and robust architectures that can handle heterogeneous units, such as ground-based mappers and wall-climbing supporters, without relying on a centralized global map [460]. To support the development of such complex interactions, physical testbeds like [461] provide low-cost, standardized hardware tiles and diverse mobile manipulators. These platforms validate the usefulness of modular approaches through exemplary applications, enabling the systematic testing of scalability and flexibility in unstructured environments. Similarly, for systems relying on programmable self-assembly, [462] offers a universal testbed capable of demonstrating fully autonomous and distributed stochastic self-assembly, emulating the performance of various existing modular systems through a hybrid cube model.

Furthermore, the evaluation of multi-robot coordination often requires environments that capture complex spatial-temporal interactions and social norms, adding another layer of complexity to the sim-to-real gap. The [463] study introduces SAMARL, a novel benchmark that employs a spatial-temporal transformer to capture intricate interactions between robots and humans, addressing the simulation-to-reality gap often found in social navigation tasks. Complementing this, [464] presents a specialized platform for multi-agent quadruped robots, emphasizing hierarchical policy structures and challenging evaluation scenarios that reflect real-world collaborative and competitive tasks. These environments are crucial for benchmarking state-of-the-art MARL algorithms, revealing that while hierarchical reinforcement learning can simplify task acquisition, advanced algorithms are still required to handle the intricate dynamics of multi-agent interactions.

The push for standardization also encompasses the software infrastructure required to scale these evaluations, ensuring that computational constraints do not become a bottleneck for system-level assessment. Frameworks like [465] provide a reproducible and scalable stack for distributed RL, abstracting away hardware complexities and enabling algorithms to scale across thousands of CPU

cores and hundreds of GPUs. This level of computational standardization is mirrored in simulation tools like [466], which interfaces with Ignition Gazebo to offer modular architecture, multiple physics engines, and distributed simulation capabilities. Such tools ensure that environments are agnostic to specific runtimes, allowing for execution in real-time settings on actual robotic platforms driven by different middlewares. Additionally, [467] contributes to this ecosystem by offering a unified task interface across diverse domains, including multi-agent and hierarchical learning, supported by expert demonstrations and baseline results that establish a standard for benchmarking.

In conclusion, the development of standardized ecosystems like [458], [459], and various modular testbeds represents a paradigm shift in how multi-robot and modular systems are evaluated. These platforms emphasize the necessity of reproducible physical testbeds and unified interfaces to overcome the limitations of fragmented, simulation-only research. By providing robust environments that account for realistic dynamics, heterogeneous capabilities, and complex coordination constraints, these ecosystems enable the rigorous validation of learning algorithms, accelerating the deployment of reliable multi-robot systems in real-world applications. However, the existence of these diverse platforms also underscores a recurring challenge: the lack of a single, unified metric system to compare performance across them. This fragmentation sets the stage for the subsequent discussion on the urgent need for unified metrics and end-to-end system evaluation, which aims to synthesize these disparate efforts into a cohesive framework for assessing full system performance in unmodified real-world settings.

8.6 The Need for Unified Metrics and End-to-End System Evaluation

8.6 The Need for Unified Metrics and End-to-End System Evaluation

Building upon the diverse standardized ecosystems and physical testbeds discussed in the previous section, a critical paradox emerges: while platforms like MARBLER, PIC4rl-gym, and HeROS provide the necessary infrastructure for reproducible experimentation, the community lacks a unified, metrics-driven framework to synthesize these disparate efforts into a cohesive assessment of full system performance. The rapid advancement of offline and preference-based reinforcement learning for robotics has precipitated a fragmented landscape of evaluation protocols, where success is often measured in isolated simulation environments or platform-specific benchmarks rather than through holistic, real-world deployment. This fragmentation creates a dangerous illusion of progress, where algorithms demonstrate superior performance in controlled simulators or specific hardware configurations but fail catastrophically when subjected to the stochasticity and complexity of unmodified physical reality. The core issue lies in the disconnect between the metrics optimized during training—often simplified reward scores or simulation-specific success rates—and the actual requirements for safe, robust, and efficient operation in dynamic environments. To bridge this gap, it is imperative to move beyond the component-level validation facilitated by current ecosystems and adopt end-to-end evaluation strategies that treat the robot, its perception stack, and its control policy as a singular, integrated cyber-physical system.

A primary limitation of current evaluation paradigms, even within these advanced ecosystems, is the over-reliance on simulation fidelity without rigorous quantification of the simulation-to-reality gap. Many studies report impressive results based on synthetic data or idealized physics engines, yet these findings often do not translate to physical hardware due to unmodeled dynamics, sensor noise, and environmental shifts. While the modular frameworks mentioned earlier address the

scalability of networked system evaluations, they highlight the broader difficulty of achieving end-to-end evaluation coverage when bottlenecks exist in software or hardware components that are not perfectly modeled [468]. Similarly, insights from systems engineering emphasize that accurate timing and synchronization are critical for credible evaluation, demonstrating the necessity of modular simulation to combine diverse components like processors and networks into virtual testbeds [469]. These findings underscore a critical lesson for robotics: without a cohesive architecture that integrates perception, planning, and control into a single evaluative loop across different platforms, metrics derived from partial simulations or isolated sub-tasks remain poor predictors of real-world utility.

Furthermore, the absence of unified metrics complicates the assessment of robustness and safety, particularly when dealing with distributional shifts and adversarial conditions across heterogeneous systems. In the realm of Cyber-Physical Systems (CPS), formal verification techniques have been proposed to ensure safety specifications are met, yet these methods often struggle with the computational prohibitions of exhaustively simulating complex operational scenarios [470]. The difficulty is compounded by the fact that current tools cannot easily optimize simulation campaigns using existing datasets of collected scenarios, nor can they accurately predict the time required to complete verification processes [471]. This inability to leverage historical data for efficient validation mirrors the challenges in offline reinforcement learning, where policies must be evaluated without further environment interaction. Without standardized protocols for off-policy evaluation that account for noisy labels and heterogeneous data sources across different robotic platforms, claims of algorithmic superiority remain anecdotal and unreproducible.

The necessity for a unified metric system is further highlighted by the challenges in benchmarking optimization approaches and handling the heterogeneous data generated by diverse ecosystems. In fields ranging from systems biology to power grid operations, researchers have noted that insufficient performance of optimization approaches often stems from a lack of unbiased benchmarking guidelines and the inability to reproduce results under realistic scenarios [472; 473]. In robotics, this manifests as a proliferation of custom datasets and task definitions that prevent fair comparison between offline RL algorithms and preference-based fine-tuning methods, even when using standard testbeds. The community urgently requires a “pFL-Bench” equivalent for robotics—a comprehensive benchmark that unifies data partitioning, heterogeneous settings, and systematic evaluation metrics to facilitate rapid and reproducible research [474]. Such a framework would not only standardize dataset variants but also enforce rigorous evaluation of generalization, fairness, and system overhead, moving the field away from cherry-picked success stories on specific platforms toward statistically significant performance gains across the board.

Moreover, the evaluation of robotic systems must account for the resource constraints and heterogeneity of real-world deployment environments, extending beyond the physical capabilities of the robots to their computational substrates. Just as heterogeneity measures like HEET have been developed to quantify the impact of diverse computing resources on system performance in distributed systems, robotic evaluation must incorporate metrics that reflect computational latency, energy consumption, and hardware variability [475]. Ignoring these factors leads to policies that are theoretically optimal but practically infeasible on edge devices or mobile platforms. The concept of “specification overfitting,” where systems optimize for specific proxy metrics at the expense of high-level requirements, poses a significant risk in autonomous robotics [476]. If evaluation protocols focus solely on task completion rates in simulation or specific benchmark scenarios while neglecting safety constraints, energy efficiency, or social compliance, the resulting agents may be brittle and unsafe. Therefore, future evaluation frameworks must integrate multi-objective metrics

that balance performance with robustness, ensuring that improvements in one domain do not come at the cost of catastrophic failures in another.

Finally, the path toward trustworthy autonomous robotics requires a shift from isolated component testing within standardized ecosystems to full system validation in unmodified real-world settings. This involves adopting methodologies that can propagate uncertainty bounds through composite systems, allowing researchers to derive valid upper bounds on failure probabilities even when simulations are imperfect [477]. It also necessitates the development of cost-effective test selection mechanisms that can identify informative scenarios from vast datasets, minimizing the computational and physical costs of validation while maximizing fault detection [478]. By synthesizing these approaches, the robotics community can establish a rigorous, unified evaluation culture that prioritizes end-to-end system reliability over isolated metric optimization. Only through such comprehensive and standardized assessment can we ensure that the transition from static data and human preferences to deployed robotic agents is both safe and effective, ultimately fostering trust in autonomous systems operating alongside humans.

9 Open Challenges and Future Research Directions

9.1 Scaling Foundation Models for High-Dimensional Sensory Inputs and Real-Time Control

9.1 Scaling Foundation Models for High-Dimensional Sensory Inputs and Real-Time Control

The integration of large foundation models into robotic systems promises a paradigm shift towards general-purpose agents capable of zero-shot reasoning and complex task execution. However, realizing this potential in physical domains introduces a critical bottleneck: the immense computational demand of processing high-dimensional sensory inputs and generating control signals in real-time often conflicts with the strict Size, Weight, Area, and Power (SWAP) constraints inherent to mobile and embedded platforms. While cloud-offloading strategies offer a temporary reprieve for non-latency-critical tasks, they frequently fail to meet the stringent timing requirements of closed-loop control systems, where milliseconds dictate stability and safety. Consequently, a primary research frontier involves developing strategies to scale down these massive models or optimize their execution without sacrificing the emergent capabilities that make them valuable. This challenge necessitates a holistic approach encompassing model distillation, efficient architecture design, dynamic resource allocation, and intelligent online model selection, laying the computational groundwork for the reliable decision-making discussed in subsequent sections.

One of the most promising avenues for addressing the computational burden is knowledge distillation, particularly when adapted for the unique constraints of robotics. While standard distillation techniques have proven effective in static domains, their application to robotic foundation models requires careful consideration of the “small data pitfall,” where methods optimized on small-scale datasets underestimate the potential of vanilla distillation on larger scales. Recent investigations suggest that employing stronger data augmentation and leveraging large-scale datasets can significantly close the performance gap between distilled student models and their teacher counterparts, even for complex architectures like Vision Transformers [479]. Furthermore, for multimodal foundation models which are increasingly common in robotics for grounding language in vision, module-wise adaptive distillation offers a sophisticated refinement. By treating different architectural components as arms in a multi-armed bandit problem, systems can dynamically identify and

distill the most contributive modules, thereby maximizing cumulative performance gains while minimizing the computational footprint required for deployment [480]. This is crucial for robots that must operate on edge devices where every megabyte of memory and milliwatt of power counts.

Beyond distillation, the design of efficient architectures tailored for sequential decision-making is paramount. The standard Transformer architecture, despite its dominance, suffers from quadratic scaling with sequence length, making it ill-suited for the long-horizon contexts often required in robotic planning. Emerging alternatives, such as state space models like Mamba, offer linear-time sequence modeling with selective state spaces, providing a hardware-aware parallel algorithm that achieves significantly higher throughput than traditional Transformers while maintaining state-of-the-art performance across modalities [481]. For resource-constrained platforms specifically, such as tiny quadruped robots used in search-and-rescue, casting imitation learning as a conditional sequence modeling task via Decision Transformers, followed by aggressive software optimizations like quantization and pruning, has demonstrated the feasibility of deploying generative policies on ultra-low-cost hardware. Empirical results indicate that quantizing models down to 4 bits and applying structured pruning can reduce model size by approximately 30% while preserving competitive reward signals, enabling natural gaits on devices with severe memory limitations [482].

However, static optimization is often insufficient given the dynamic nature of real-world environments and the heterogeneity of available computing resources. This has led to the emergence of online model selection and elastic collaborative computing as critical strategies. Robots often face a trade-off between utilizing powerful but costly closed-source models and free but less capable open-source alternatives. To navigate this, user-centric online model selection frameworks have been proposed, which combine open-source encoders to extract context with online learning algorithms that maximize a composite reward function balancing performance, execution time, and cost. Such approaches have been shown to improve task success rates significantly in language-based robotic tasks by dynamically selecting the optimal model for the current context [483]. Extending this concept to the system level, elastic collaborative computing frameworks like ElasticROS enable algorithm-level adaptation where tasks are dynamically distributed between the robot and cloud servers based on real-time conditions. By employing dynamic action decision algorithms that update parameters to adapt to changing network and computational loads, these systems ensure stable elasticity and sublinear regret, effectively managing the latency and power consumption bottlenecks associated with large model inference [484].

Furthermore, the challenge of real-time control extends to the efficient management of memory and batch sizes during on-device training and adaptation. Autonomous systems requiring continuous adaptation, such as search and rescue robots, must co-optimize batch size and replay buffer size to balance timing constraints with algorithmic performance under tight memory budgets. Holistic solutions like R³ employ deadline-driven feedback loops and dynamic batch sizing to ensure timing predictability and prevent out-of-memory errors, demonstrating that efficient memory management is as critical as model architecture design for real-time deep reinforcement learning on autonomous robots [485]. Additionally, the deployment of large diffusion models for vision-based control on mobile devices has been accelerated through GPU-aware optimizations, achieving inference latencies under 12 seconds on consumer smartphones, thereby broadening the applicability of generative AI in offline, privacy-preserving robotic scenarios [486].

In conclusion, scaling foundation models for robotics is not merely a matter of compressing weights but involves a synergistic combination of architectural innovation, adaptive distillation, and dynamic system-level orchestration. These computational optimizations provide the necessary infrastructure to deploy intelligent agents on real-world hardware; however, efficiency alone does

not guarantee safety. As these scaled-down models begin to drive high-level planning, new risks emerge regarding the fidelity and reliability of their outputs. Future research must therefore not only continue to bridge the gap between theoretical capabilities and embedded deployment—through hardware-specific optimizations like FPGAs [487] and robust local-cloud switching frameworks—but also address the qualitative failures that persist even in optimized systems. Addressing these computational bottlenecks is a prerequisite for, yet distinct from, the critical challenge of mitigating hallucinations and ensuring the trustworthiness of language-guided decision making, which forms the focus of the following discussion.

9.2 Mitigating Hallucinations and Ensuring Reliability in Language-Guided Decision Making

9.2 Mitigating Hallucinations and Ensuring Reliability in Language-Guided Decision Making

While Section 9.1 established the computational infrastructure necessary to deploy scaled-down foundation models on real-world hardware, efficiency alone does not guarantee safety. As these optimized agents begin to drive high-level planning using Large Language Models (LLMs) and Vision-Language Models (VLMs), the phenomenon of hallucination emerges as a paramount barrier to safe deployment. In physical domains, unlike purely digital text generation, a hallucinated action plan can lead to catastrophic hardware failures, safety violations, or irreversible environmental damage. The core challenge lies in the disconnect between the statistical nature of language models, which predict plausible token sequences, and the deterministic constraints of physical laws and safety requirements. Recent surveys highlight that while foundation models offer emergent reasoning capabilities, their propensity to generate content that contradicts established world knowledge or user inputs poses substantial risks to reliability [488]. Furthermore, the application of these models in decision-making tasks requires a rigorous re-evaluation of what constitutes a hallucination in an embodied context, moving beyond factual inaccuracies to include physically infeasible or unsafe actions [489].

To address the unpredictability of LLM-generated plans, a promising research direction involves the development of layered architectures that harmonize high-level intelligence with low-level “robot instinct.” This paradigm proposes decoupling the generative capabilities of LLMs from the execution layer by introducing innate or learned response mechanisms dedicated to survival-essential tasks such as obstacle avoidance and stability maintenance [490]. By defining robot instinct as a set of prioritized responses that operate independently of the language model’s output, systems can ensure that safety-critical constraints are met even when the LLM confidently proposes a hallucinated or dangerous trajectory. This approach effectively creates a safety filter that prunes unsafe actions before they reach the controller, aligning with frameworks that enforce constraints via queryable safety modules based on linear temporal logic [491]. Such modules allow for the natural language encoding of prohibitions and the verification of compliance, ensuring that the “don’ts” of robotic operation are strictly adhered to alongside the “dos.”

Complementing architectural safeguards is the critical need for robust uncertainty quantification (UQ) to enable robots to recognize when they lack sufficient knowledge to proceed safely. A significant portion of hallucinations stems from the model’s overconfidence in ambiguous situations or out-of-distribution scenarios. Research indicates that LLMs often fail to express their internal uncertainty during generation, leading to factual hallucinations even when the model possesses the

correct internal knowledge state [492]. To mitigate this, future systems must integrate mechanisms that allow agents to abstain from answering or executing plans when uncertainty exceeds a predefined threshold. Techniques such as conformal prediction have been successfully applied to provide statistical guarantees on task completion, enabling planners to know when to ask for human assistance rather than executing a potentially flawed plan [493]. By aligning the model’s confidence scores with the actual probability of success, often through the use of scene affordance scores, robots can significantly reduce the rate of harmful hallucinations and the frequency of required human interventions [494].

Moreover, verification mechanisms that operate post-generation or iteratively during the planning process are essential for catching undesirable behaviors that simple task-success metrics might miss. Traditional verification often checks only if a goal condition is met, ignoring subtleties such as the manner of execution or social norms. Recent work suggests utilizing large Vision-Language Models as scalable “behavior critics” to evaluate video trajectories for undesirable attributes, providing a more holistic assessment of agent behavior [495]. Additionally, frameworks like Chain-of-Verification (CoVe) encourage models to deliberate on their own responses by planning verification questions and answering them independently, thereby reducing hallucinations in complex reasoning tasks [496]. In the context of robotics, this self-reflection can be extended to include physical grounding, where the model verifies the feasibility of its proposed actions against a grounded model of the environment before execution [497].

Despite these advances, significant gaps remain in ensuring the reliability of language-guided decision making. The illusion of Theory of Mind in LLMs, for instance, can lead to misaligned expectations in Human-Robot Interaction, where models fail to maintain invariance to irrelevant context perturbations [498]. Furthermore, the integration of retrieval-augmented generation (RAG) and other external knowledge sources, while helpful, does not fully eliminate the risk of hallucination when prompts contradict pre-trained biases [499]. Future research must therefore focus on developing hybrid systems that combine the semantic breadth of foundation models with the precision of formal verification, probabilistic safety certificates, and adaptive uncertainty-aware policies. Only by tightly coupling language-based planning with rigorous, real-time validation layers can we transition from experimental prototypes to trustworthy autonomous agents. Establishing this foundational safety is a prerequisite for addressing the next critical bottleneck: the prohibitive cost and scarcity of high-quality human-annotated data required to train these robust systems, which forms the focus of the following discussion on reducing annotation burdens via autonomous data collection.

9.3 Reducing Human Annotation Burden via Autonomous Data Collection and Synthetic Continual Learning

9.3 Reducing Human Annotation Burden via Autonomous Data Collection and Synthetic Continual Learning

While Sections 9.2 established the necessity of mitigating hallucinations to ensure the safety of language-guided planning, and Section 9.4 will address the need for interpretability to foster human trust, a fundamental practical barrier remains: the prohibitive cost and scarcity of high-quality human-annotated data required to train these robust systems. Traditional reinforcement learning and supervised learning paradigms often rely on extensive human feedback to define reward functions or label trajectories, creating a significant bottleneck for scalability. This dependency is particularly acute in domains such as healthcare, finance, and personal assistance, where data col-

lection is frequently constrained by strict privacy regulations, preventing the aggregation of the diverse experiences necessary for robust generalization. To bridge the gap between the theoretical safety mechanisms discussed previously and the trustworthy deployment envisioned next, future research must pivot towards integrating foundation models for autonomous instruction generation, leveraging synthetic data creation techniques, and developing privacy-preserving continual learning frameworks.

A promising direction for alleviating the human annotation burden lies in utilizing foundation models to autonomously generate instructions and synthetic training environments. Large Language Models (LLMs) and generative models possess the capacity to synthesize plausible task descriptions and environmental contexts without direct human intervention, effectively acting as autonomous instructors. By fine-tuning pretrained generative language models with differential privacy, it is possible to generate useful synthetic text that maintains strong privacy protection while providing diverse training signals [500]. This approach can be extended to robotics, where LLMs generate varied task goals and constraint specifications that guide robotic agents through self-supervised exploration. Moreover, the integration of label-guided synthetic content has shown potential in low-resource settings, such as speech understanding, where models trained on synthetic samples provide effective initialization points, thereby reducing the need for massive collections of user data [501]. In robotics, similar strategies could enable agents to practice complex manipulation or navigation tasks in simulated environments populated with synthetically generated objects and scenarios, significantly lowering the reliance on physical human demonstration.

However, the generation of synthetic data must be balanced with rigorous privacy guarantees to prevent the leakage of sensitive information from the underlying real-world distributions used to train the generative models. Generative models, particularly deep neural networks, are prone to memorizing details of individual training records, which poses severe risks when dealing with sensitive data [502]. To mitigate this, future robotic learning systems should incorporate differentially private synthetic data generation mechanisms. Techniques such as Differentially Private Tabular Latent Diffusion Models (DP-TLDM) have demonstrated the ability to enhance the utility of synthetic data while preserving meaningful theoretical privacy guarantees, outperforming traditional adversarial approaches in data resemblance and discriminability [503]. Similarly, the application of diffusion paradigms to tabular data synthesis, as seen in TableDiffusion, offers a smoother training process that avoids mode collapse and achieves state-of-the-art performance in privatised data synthesis [504]. Adapting these diffusion-based methods to generate high-dimensional robotic state-action trajectories could provide a rich, privacy-compliant corpus for offline reinforcement learning, allowing robots to learn from diverse behaviors without compromising user privacy.

Beyond static synthetic datasets, the dynamic nature of robotic environments necessitates continual learning strategies that can adapt to new tasks and shifting distributions over time. Yet, continual learning often exacerbates privacy risks as models accumulate knowledge from sequential data streams. Research into continual learning with differential privacy has introduced mechanisms to bound the sensitivity of data records across tasks, ensuring formal privacy guarantees while maintaining model utility [505]. For robotic systems, this implies the development of algorithms that can incrementally update policies based on new synthetic or real-world interactions without forgetting previous skills or leaking sensitive historical data. The concept of “Draw and Discard” machine learning offers a novel framework for client-distributed learning, utilizing random sampling and model averaging to achieve differential privacy in asynchronous settings, which is highly relevant for fleets of robots learning collaboratively [506].

Furthermore, the synergy between synthetic data and active learning presents a viable pathway

to optimize the annotation process. By fine-tuning differentially private models through active learning on public or synthetic data, researchers have achieved state-of-the-art accuracy while retaining strict privacy guarantees [507]. In a robotic context, agents could employ active learning strategies to identify the most informative regions of the state space, requesting human feedback only when necessary, while filling the rest of the experience buffer with high-fidelity synthetic data. This hybrid approach reduces the total volume of required human annotations. Additionally, frameworks that leverage representation learning via autoencoders to generate privacy-preserving embedded data allow organizations to share data representations for collaborative training without exposing raw sensory inputs [508]. Such methods are crucial for multi-robot systems where data sharing is essential for collective improvement but is restricted by confidentiality concerns.

Finally, the evaluation of these synthetic and privacy-preserving methods must go beyond simple utility metrics to include comprehensive assessments of fairness and bias, ensuring that the data used to train safe and interpretable systems does not introduce new vulnerabilities. Studies have shown that marginal-based synthetic data generators can train models that achieve utility and fairness characteristics close to those obtained with real data, highlighting the potential for synthetic data to not only preserve privacy but also enhance equitable outcomes [509]. Future research in robotics must therefore establish rigorous benchmarks to evaluate the long-term stability, safety, and fairness of policies learned from autonomous synthetic data streams. By combining the generative power of foundation models with robust differential privacy mechanisms and efficient continual learning algorithms, the robotics community can overcome the current limitations of data scarcity and privacy. This foundation of abundant, private, and diverse data is essential for realizing the reliable decision-making discussed in Section 9.2 and enabling the transparent, human-centered metrics explored in Section 9.4, ultimately paving the way for truly autonomous and trustworthy robotic agents capable of learning continuously in complex, human-centric environments.

9.4 Enhancing Interpretability and Trust through Explainable AI and Human-Centered Metrics

9.4 Enhancing Interpretability and Trust through Explainable AI and Human-Centered Metrics

While Section 9.3 addressed the critical challenge of acquiring diverse, privacy-preserving data to overcome annotation bottlenecks, the subsequent deployment of robots trained on such datasets introduces a new imperative: ensuring that their decision-making processes are transparent and trustworthy to human collaborators. The opacity of deep reinforcement learning models, particularly those trained on vast synthetic or offline corpora, presents a formidable barrier to adoption in safety-critical environments. The fundamental challenge lies in bridging the chasm between the high-dimensional, non-linear decision boundaries of black-box policies and the human need for transparent, justifiable actions. Current research indicates that while post-hoc explanation methods are frequently proposed to mitigate this opacity, they often fail to provide consistent fidelity across different user groups or scenarios. For instance, audits of explanation quality have revealed that the approximation fidelity of surrogate models can differ significantly between protected subgroups, raising serious concerns about the fairness and reliability of these explanations in diverse operational contexts [510]. This variability suggests that a single explanatory approach is rarely sufficient, necessitating a shift toward more robust, causal, and human-centered frameworks for evaluating and designing interpretable robotic agents.

A critical avenue for future research involves moving beyond associative attributions toward causal

interpretability to ensure that explanations reflect true mechanistic understanding rather than spurious correlations found in training data. Traditional explanation methods often rely on feature importance scores that identify correlations but fail to capture the underlying causal mechanisms driving a robot’s behavior. This limitation is particularly dangerous in robotics, where understanding “why” an action was taken is essential for predicting failure modes and ensuring safety. Recent advancements propose principled causality-based approaches, such as probabilistic contrastive counterfactuals, which quantify the direct and indirect influences of specific variables on decisions. These methods offer actionable recourse and can generate explanations at local, global, and contextual levels without requiring access to the model’s internal gradients [511]. Furthermore, integrating causal concepts into the explanation process allows for the construction of structural causal models that capture relationships between semantic concepts, thereby providing explanations that are both concept-based and causally sound [512]. By adopting such causal lenses, future robotic systems can provide explanations that are not only descriptive but also predictive of how changes in the environment would alter the policy’s output, a capability essential for validating behaviors before the physical realities discussed in Section 9.5 are encountered.

However, the efficacy of any explanation system is ultimately determined by its ability to foster appropriate human trust, distinguishing between mere transparency and genuine reliability. There is a growing consensus that interpretability alone is neither necessary nor sufficient for trust; rather, trust is built through the ability to verify model behavior via rigorous behavioral certificates and interactive validation [513]. The phenomenon of “fairwashing,” where unfair models are rationalized by high-fidelity but misleading explanations, underscores the danger of relying solely on superficial transparency metrics [514]. Consequently, future research must prioritize the development of standardized, quantitative metrics that assess explanation quality beyond mere fidelity. Proposed frameworks suggest evaluating explanations based on consistency, sufficiency, and the information transfer rate required for humans to replicate model predictions [515; 516]. Additionally, the “Turing Test for Transparency” offers a novel metric where the goal is for human interrogators to be unable to distinguish between machine-generated and human-generated explanations, serving as a benchmark for the naturalness and comprehensibility of robotic rationales [517].

The diversity of stakeholders in robotic deployment further complicates the landscape of interpretability, requiring systems that adapt to varying levels of technical literacy. Different users, ranging from field operators to system regulators, possess distinct explanation needs. Research demonstrates that individuals with and without AI backgrounds perceive explanations differently, often placing unwarranted faith in numerical outputs or misinterpreting visual cues based on their cognitive heuristics [518]. Therefore, a one-size-fits-all explanation strategy is inadequate. Future systems must incorporate interactive and adaptive explanation mechanisms that allow users to probe the model through “what-if” queries and adjust the level of abstraction according to their expertise [519]. Interactive interfaces that visualize causal relationships and allow users to explore the decision space have shown promise in enhancing understanding for both expert and non-expert users [520].

Moreover, resolving the trade-off between model performance and interpretability remains a contentious yet vital issue for deploying high-capacity models trained on the rich data sources described in Section 9.3. While some argue for the exclusive use of inherently interpretable models in high-stakes decisions to avoid the pitfalls of explaining black boxes [521], others advocate for a synergistic approach where accurate black-box models are paired with robust explanation layers. The future likely lies in “interpretable-by-design” neural architectures that do not sacrifice accuracy for transparency, utilizing techniques like adaptive routing and diagnostic benchmarks to maintain

high performance while ensuring intrinsic explainability [522]. Finally, establishing ethical compliance requires metrics that evaluate not just the technical correctness of an explanation, but its impact on perceived fairness and the user’s willingness to rely on the system. Studies utilizing response shift paradigms to quantify how explanations sway human decisions provide a pathway to measuring “appropriate trust,” ensuring that humans neither over-rely on nor unjustifiably reject autonomous recommendations [523]. By integrating causal reasoning, interactive customization, and rigorous human-centered metrics, the next generation of robotic learning systems can achieve the transparency required for safe deployment. This foundation of trust and interpretability is the prerequisite for addressing the subsequent challenge: ensuring that these transparent policies remain robust when transitioning from controlled simulations to the chaotic physical world, as explored in Section 9.5.

9.5 Bridging the Sim-to-Real Gap with Photorealistic Simulation and Open-World Perception

9.5 Bridging the Sim-to-Real Gap with Photorealistic Simulation and Open-World Perception

While the preceding discussion on explainability addresses the *interpretability* of robotic policies, the practical deployment of these systems hinges on their *robustness* when transitioning from controlled training environments to the chaotic physical world. The “sim-to-real” gap remains a formidable barrier, where policies that excel in simulation often fail catastrophically upon deployment due to discrepancies in physical dynamics, sensor noise, and visual complexity. Addressing this challenge requires a paradigm shift that complements the causal reasoning frameworks discussed earlier with hyper-realistic virtual worlds, comprehensive open-world perception datasets, and task-driven scene understanding. Only by grounding agents in high-fidelity simulations and open-ended perceptual capabilities can we ensure that the transparent and trustworthy behaviors designed in Section 9.4 remain reliable when faced with the unstructured realities required for the general-purpose robotics outlined in Section 9.6.

The first critical frontier is the enhancement of photorealism in synthetic data generation to minimize the visual domain gap. Traditional simulators have historically struggled to render lighting, textures, and material properties with sufficient fidelity, leading to brittle deep learning models. However, recent advances leverage powerful game engines to create environments visually indistinguishable from reality. For instance, [524] introduces an environment built on Unreal Engine 4 specifically designed to reduce the reality gap by leveraging hyper-realistic indoor scenes where robot agents interact with objects in a visually plausible manner. Similarly, [525] demonstrates that synthetic datasets created using photorealistic rendering techniques can achieve state-of-the-art results for training and validation, effectively serving as a proxy for real-world street scenes. The efficacy of such approaches is further validated by studies showing that pre-training on virtual data can significantly improve performance on real-world tasks, as seen in [526], which provides quantitative evidence that modern deep learning algorithms behave similarly in real and virtual worlds when the visual fidelity is high. Furthermore, [527] emphasizes that more realistic rendering is worth the computational cost, showing that pretraining with large-scale physically-based rendered images improves results in surface normal prediction and semantic segmentation beyond current state-of-the-art limits. These high-fidelity environments provide the necessary substrate for generating the diverse data required to train robust, interpretable policies.

Beyond visual fidelity, the semantic richness and scope of training data must expand from closed-set

recognition to open-world perception to support the adaptability needed for general-purpose agents. Most existing benchmarks rely on a fixed taxonomy of object classes, rendering robots incapable of recognizing or interacting with novel objects encountered in dynamic environments—a limitation that directly impedes the compositional reasoning capabilities discussed in the following section. To overcome this, the community must prioritize the creation of open-world panoptic segmentation datasets that support the identification of unseen categories. [528] addresses this by providing a novel benchmark for open-world panoptic segmentation in crowded human settings, featuring diverse object classes and comprehensive 2D and 3D synchronized data modalities essential for accurate navigation and decision-making. Complementing this, [529] tackles the scarcity of 3D-text pairs by harnessing pre-trained vision-language foundation models to generate captions for multi-view images, enabling the localization and recognition of unseen object categories in 3D space. The push for holistic understanding is further articulated in [530], which proposes “holistic segmentation” to identify and separate objects of unseen, unknown categories without prior knowledge, a crucial capability for safety-critical applications where failing to detect an outlier can have dangerous consequences. Additionally, [531] extends urban scene understanding by introducing diverse LiDAR point cloud annotations, addressing the lack of diversity in existing datasets that hinders the learning of robust tracking methods.

Finally, bridging the sim-to-real gap requires moving beyond passive perception to active, task-driven scene understanding, ensuring that robots can reason about which elements of the scene are relevant to their current objectives. This aligns with the need for agents that do not merely perceive but actively plan and execute complex behaviors. [532] argues that the granularity of object representation should be intrinsically task-dependent, proposing a framework where robots select the subset of objects and scene structures necessary to complete natural language instructions. This approach aligns with the need for flexible simulation environments that bridge scene understanding and task execution, as highlighted in [533], which presents TESSE as a simulator developed specifically to support metric-semantic mapping and dynamic scene graph generation for interactive task learning. Moreover, the validation of these systems requires rigorous benchmarking protocols that quantify the discrepancy between simulation and reality. [534] proposes a framework for studying sim-to-real algorithms with consistent metrics, while [454] demonstrates that direct comparison of raw data is insufficient, advocating for the use of state-estimation packages as judges to isolate sensor simulation discrepancies. Further efforts like [535] and [536] underscore the necessity of generating high-resolution, diverse, and domain-specific synthetic data with full ground truth to foster progress in generalization.

In conclusion, closing the sim-to-real gap demands a concerted effort to integrate hyper-realistic rendering pipelines, develop open-world datasets that challenge models to recognize the unknown, and formulate scene understanding as a task-driven, adaptive process. By leveraging insights from [537] regarding domain transfer and [538] on unifying diverse taxonomies, future research can create robotic agents that are not only proficient in simulated benchmarks but also robust, adaptable, and safe in the unpredictable real world. Establishing this foundation of perceptual robustness and realistic simulation is the prerequisite for the next evolutionary step: the development of general-purpose agents capable of compositional reasoning and long-horizon planning in truly unstructured environments.

9.6 Toward General-Purpose Robotics: Compositional Reasoning, Long-Horizon Planning, and Multi-Agent Coordination

9.6 Toward General-Purpose Robotics: Compositional Reasoning, Long-Horizon Planning, and Multi-Agent Coordination

Building upon the foundation of perceptual robustness and open-world scene understanding established in Section 9.5, the next evolutionary leap in robotic artificial intelligence is the realization of general-purpose agents capable of operating seamlessly in unstructured, dynamic environments. While high-fidelity simulation and task-driven perception provide the necessary sensory grounding, moving beyond the current paradigm of task-specific policies trained in isolation requires a fundamental shift toward systems that exhibit compositional reasoning, robust long-horizon planning, and scalable multi-agent coordination. Although recent advancements in foundation models have provided the semantic backbone for such agents, significant challenges remain in grounding these capabilities in physical reality, ensuring reliability over extended temporal horizons, and orchestrating collaborative behaviors among heterogeneous robot teams. Future research must therefore address the integration of symbolic abstraction with continuous control, the mitigation of planning hallucinations, and the development of decentralized communication protocols that scale with team size, thereby transforming robust perception into adaptive action.

A primary hurdle in achieving general-purpose robotics is the ability to perform compositional reasoning, where agents synthesize novel behaviors by recombining learned skills rather than learning each task from scratch. Current approaches often struggle with the combinatorial explosion of possible task sequences and the lack of transferability between domains. To overcome this, future systems must leverage hierarchical frameworks that decouple high-level abstraction from low-level execution. For instance, methods that learn temporally extended skills as symbolic actions allow agents to utilize forward models in a discrete state abstraction for efficient planning, effectively bridging the gap between continuous control and long-horizon reasoning [257]. Furthermore, the concept of compositional plan vectors offers a promising direction, enabling policies to perform arithmetic operations on task representations to execute complex sequences without additional supervision [539]. By treating trajectories as sums of subtasks, agents can generalize to unseen compositions, a capability essential for adapting to the infinite variability of real-world demands. However, realizing this potential requires architectures that can dynamically select and parameterize skills based on contextual cues, moving beyond static libraries to fluid, generative skill synthesis [71].

Long-horizon planning presents another critical frontier, particularly when dealing with tasks that require hundreds of steps and involve complex dependencies. Traditional reinforcement learning agents often fail in these settings due to sparse rewards and the difficulty of credit assignment over time. The integration of Large Language Models (LLMs) and Vision-Language Models (VLMs) has emerged as a potent solution, providing agents with the commonsense reasoning necessary to decompose abstract goals into executable sub-goals. Systems like ViLa demonstrate that grounding planners in visual perception allows for a deeper understanding of spatial layouts and object attributes, significantly outperforming text-only planners in open-world manipulation tasks [540]. Similarly, frameworks that leverage scene graphs to ground LLM inputs enable more precise translation of human requests into executable plans, reducing the incidence of infeasible actions caused by hallucinations [541]. Despite these advances, the challenge of “hallucination” remains potent; LLMs may generate syntactically correct but physically impossible plans. Future research must therefore focus on robust verification mechanisms, such as conformal prediction, to quantify uncer-

tainty and ensure that generated plans adhere to safety and feasibility constraints before execution [542]. Additionally, the development of self-improving loops, where agents use their own successful trajectories to refine future planning, will be crucial for autonomous adaptation [543].

The transition from single-agent intelligence to coordinated multi-robot systems introduces a new layer of complexity involving communication efficiency and decentralized decision-making. As the number of agents increases, centralized planning becomes computationally prohibitive and vulnerable to single points of failure. The future of multi-robot coordination lies in hybrid architectures that balance global oversight with local autonomy. Recent studies indicate that hybrid communication frameworks, which combine centralized strategy generation with decentralized execution, achieve superior task success rates and scalability compared to purely centralized or decentralized approaches [294]. Moreover, hierarchical architectures utilizing cloud-edge end structures can leverage multiple specialized LLMs to handle complex task decomposition and motion control, allowing heterogeneous robot clusters to collaborate effectively in open scenarios [544]. A key research direction involves optimizing token efficiency in LLM-based coordination to prevent context window saturation while maintaining coherent group behavior. Furthermore, enabling agents to simulate human-like social interactions and task-oriented coordination through collaborative generative agents could unlock new potentials in service and industrial robotics [545].

Ultimately, the path toward general-purpose robotics requires a holistic synthesis of these elements: compositional skill libraries that enable zero-shot generalization, vision-language grounding that ensures physical feasibility, and decentralized coordination protocols that support scalable teamwork. The vision of an “Agent Foundation Model” suggests a move away from reductionist, task-specific training toward cohesive systems that integrate perception, reasoning, and action into a unified framework [546]. Achieving this will necessitate not only algorithmic innovations but also the creation of comprehensive benchmarks that evaluate agents on long-horizon, multi-task, and multi-agent criteria, such as those proposed in the LIBERO and CALVIN suites [547; 548]. By addressing these challenges, the community can transition from brittle, narrow AI systems to resilient, generalist agents capable of thriving in the complexity of the real world. However, the deployment of such powerful, autonomous agents introduces profound societal and legal implications. As these systems gain the capacity for independent decision-making and complex coordination, the need for rigorous governance becomes paramount, setting the stage for the discussion in Section 9.7 on establishing roadmaps for trustworthy, accountable, and legally compliant autonomous systems.

9.7 Establishing Roadmaps for Trustworthy, Accountable, and Legally Compliant Autonomous Systems

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While the preceding section outlines the architectural for general-purpose robotics through compositional reasoning and multi-agent coordination, the realization of such advanced autonomous systems demands a parallel evolution in governance. The deployment of robots trained via offline and preference-based reinforcement learning necessitates a critical transition from abstract ethical principles to concrete, actionable engineering roadmaps. Despite significant strides in algorithmic efficiency, the broader ecosystem currently lacks standardized mechanisms to ensure these agents operate within legal, ethical, and safety boundaries throughout their lifecycle. A foundational step in this roadmap is the development of comprehensive accountability catalogues that bridge the gap

between high-level governance goals and technical implementation. Current frameworks often suffer from a “principles-to-practices” gap, where ethical guidelines remain too abstract for practitioners to operationalize effectively [549]. To address this, future research must focus on creating granular metrics catalogues that define specific process, resource, and product metrics for AI accountability, ensuring that responsible AI becomes the norm rather than an exception [550]. These catalogues should serve as dynamic references that evolve alongside regulatory landscapes, providing clear instructions for developers on how to embed accountability directly into the architecture of robotic systems [551].

Furthermore, establishing legally compliant autonomous systems requires rigorous alignment with emerging global regulations, such as the EU AI Act and various liability directives. The legal landscape is shifting towards holding developers and deployers accountable for harms caused by AI, yet current liability frameworks are often described as half-hearted or insufficiently detailed for complex autonomous behaviors [552]. A robust roadmap must therefore integrate legal compliance directly into the software engineering lifecycle, moving beyond post-hoc audits to “Responsible AI by Design” approaches. This involves embedding traceability mechanisms that record not only the system’s outputs but also the provenance of its training data, the logic of its preference models, and the decision-making processes underlying its actions [553]. Such traceability is essential for satisfying regulatory requirements for transparency and for enabling effective redress when systems fail. Additionally, the role of compute providers and infrastructure hosts is becoming increasingly significant in the regulatory ecosystem, suggesting that future compliance roadmaps must consider the entire supply chain, including the verification of customer activities and the safeguarding of critical infrastructure [554].

Privacy and data governance represent another pivotal pillar of this roadmap, particularly for robots learning from human preferences which may contain sensitive personal information. The integration of privacy-preserving techniques, such as federated learning and differential privacy, must be standard practice rather than an optional add-on [555]. Research indicates that datasets used for training often contain inherent biases and privacy vulnerabilities that can propagate into the deployed system, necessitating a shift towards “data-centric governance” where governance requirements are systematized via datasets and algorithmic evaluations [556]. Future directions must include the development of standardized protocols for evaluating the fairness, privacy, and regulatory compliance of datasets prior to their use in offline reinforcement learning, ensuring that no dataset is deployed without a thorough responsible rubric assessment [557]. Moreover, the tension between different human values, such as privacy versus transparency, must be explicitly managed through sociotechnical frameworks that recognize these trade-offs rather than treating them as isolated technical problems [558].

Finally, fostering community standards for reproducible research and rigorous evaluation is paramount for building trust in the autonomous systems described in previous sections. The current state of AI evaluation is fragmented, with many studies failing to provide holistic assessments of safety, robustness, and societal impact [559]. There is an urgent need for unified evaluation ecosystems that employ end-to-end auditing frameworks, allowing organizations to assess their systems internally before deployment and continuously monitor them thereafter [560]. These evaluation protocols should be grounded in human rights and democratic values, utilizing assurance frameworks that integrate context-based risk analysis with stakeholder engagement [561]. By adopting a sociotechnical approach that bridges the gap between technical systems and the social contexts in which they are deployed, the research community can develop decentralized computational infrastructures that support responsible behavior and curb irresponsible use [562].

Ultimately, the path forward requires a concerted effort to define clear accountabilities for all stakeholders, from data collectors to system deployers, ensuring that the rapid advancements in robotic learning are matched by equally robust frameworks for trust, safety, and legal compliance [563]. Only through such comprehensive and interconnected roadmaps can we ensure that the general-purpose agents of the future serve humanity reliably and ethically.

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